

Harshit-Kumar-Containerization

Lab Experiments – Containerization & DevOps

This section contains all the **laboratory experiments** performed as part of the **Containerization and DevOps** coursework.

Each experiment is designed to provide hands-on exposure to core DevOps concepts, tools, and practices through practical implementation.

The experiments are organized sequentially and focus on building a strong foundation in containerization, automation, and deployment workflows.

List of Experiments

The table below provides an overview of the lab experiments included in this repository:

Experiment No.	Title	Result
Experiment-1	VM vs Container - Utilization	The experiment demonstrates that containers are significantly more lightweight and resource-efficient compared to virtual machines
Experiment-2	Docker Installation, Configuration, and Running Images	Docker images were successfully pulled, containers executed, and lifecycle commands performed.
Experiment-3	Docker Deploy NGINX Using Different Base Images and Comparing Image Layers	NGINX was deployed using different base images. The Alpine image was smaller and had fewer layers than Ubuntu, showing that lightweight base images improve efficiency and reduce storage usage.
Experiment-4	Containerizing Applications with Dockerfile	Containerizing prevents from dependency issue or version outages
Experiment-5	Docker - Volumes, Environment Variables, Monitoring & Networks	Essential Docker features for building, configuring, and managing production-ready containerized applications.

Experiment No.	Title	Result
Experiment-6	Docker Run vs Docker-Compose	Compose is an easy way to create containers
Experiment-7	CI/CD via Jenkins	Automate Build via Jenkins
Experiment-9	Ansible	Automate Infrastructure Orchestration & Updates
Experiment-10	Working with SonarQube	Generate Code Quality Report via SonarQube
Experiment-11	Orchestration with Docker Swarm	Compose defines the application. Swarm runs it reliably.
Experiment-12	Kubernetes	Kubernetes automatically decides where to run your app

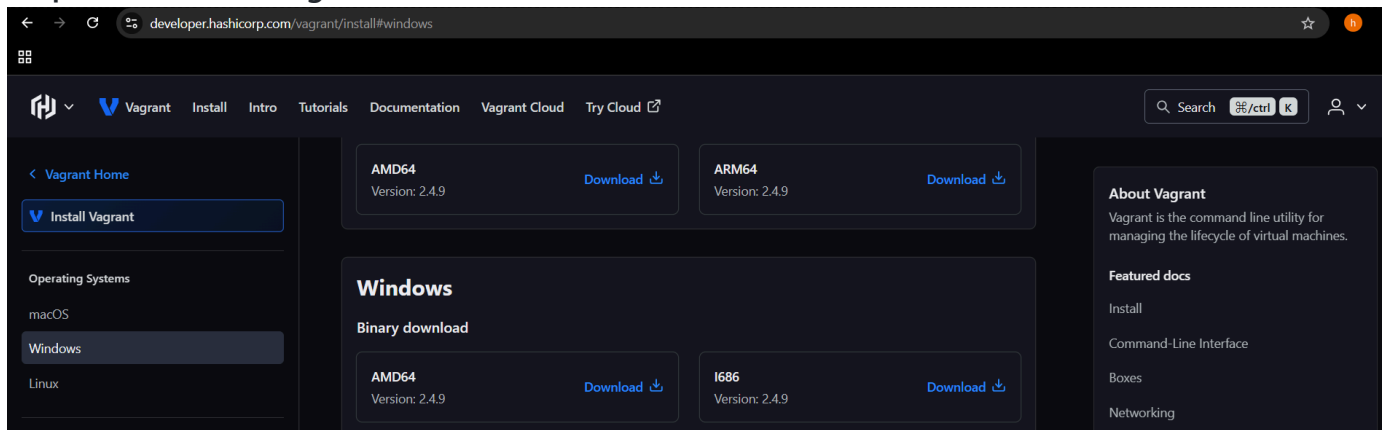
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Experiment-1 : VM vs Container

PART-A:- Virtual Machine (Windows)

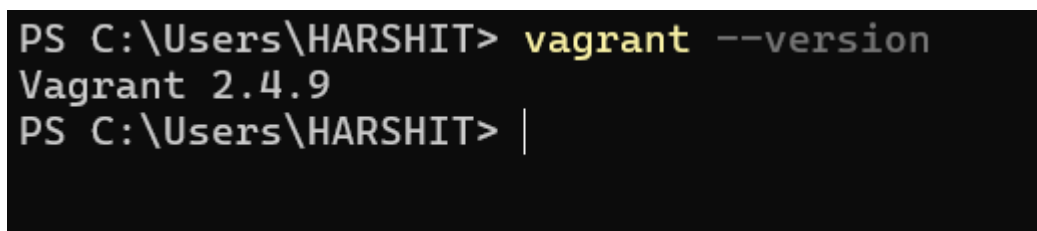
Step 1: Download Virtual Box from [here](#)

Step 2: Download Vagrant from [here](#)



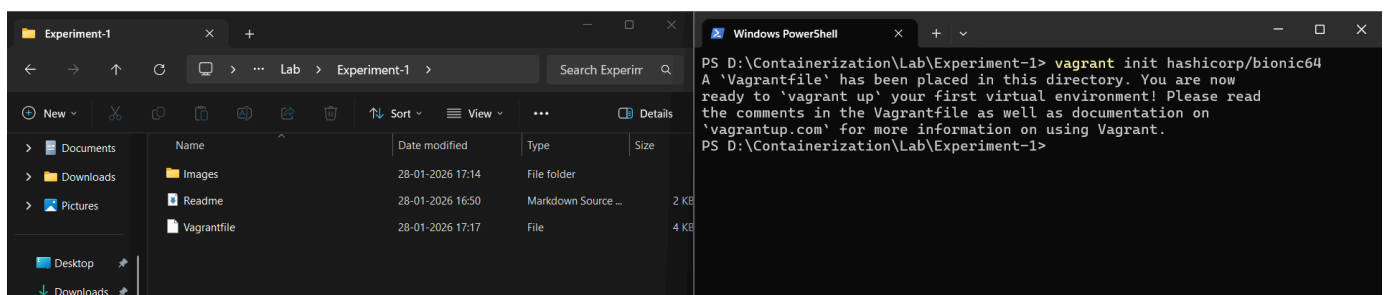
Step 3: To verify the installation we will check the version via following command

```
vagrant --version
```



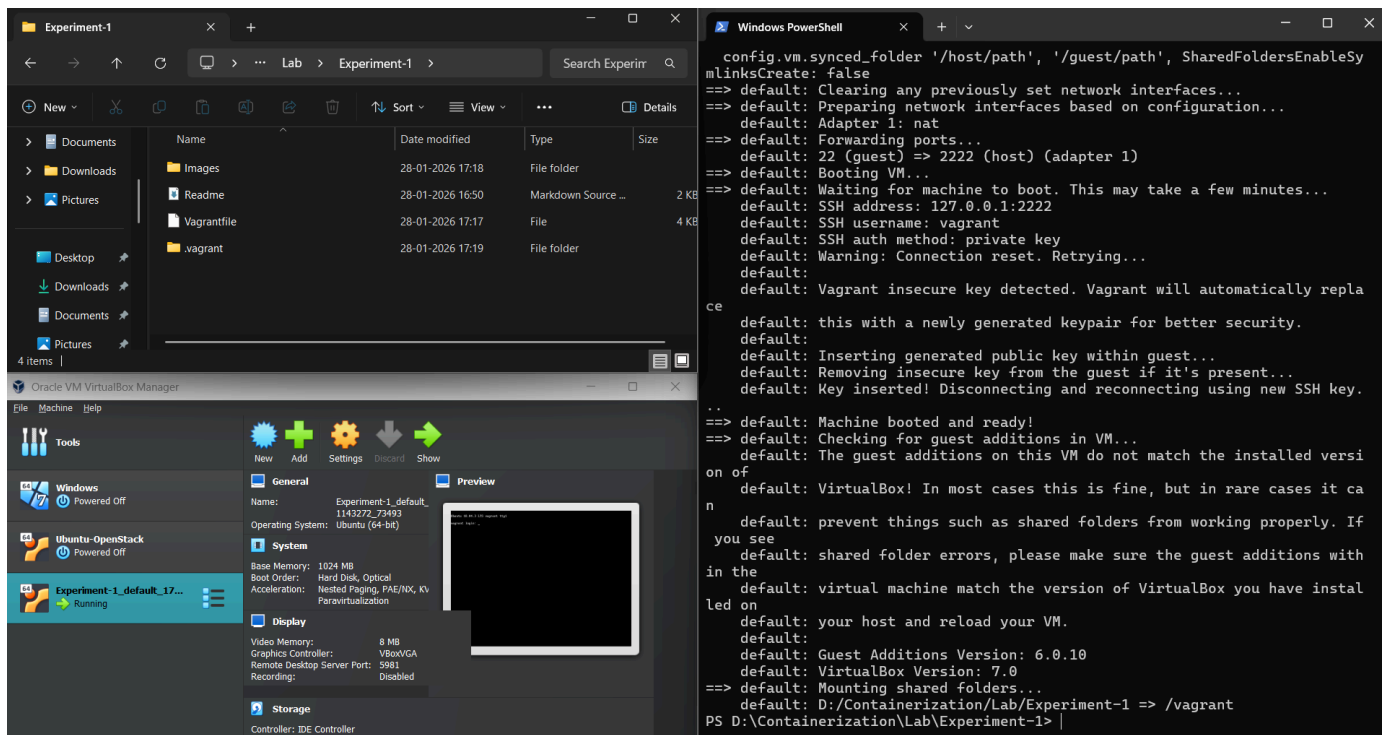
Step 4: Initialize Vagrant with Ubuntu box:

```
vagrant init hashicorp/bionic64
```



Step 5: Start the VM:

```
vagrant up
```



Step 6: Access the VM:

```
vagrant ssh
```



```
vagrant@vagrant: ~
PS D:\Containerization\Lab\Experiment-1> vagrant ssh
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-58-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Wed Jan 28 11:55:12 UTC 2026

System load:  0.37               Processes:            92
Usage of /:   2.5% of 61.80GB    Users logged in:     0
Memory usage: 11%               IP address for eth0: 10.0.2.15
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8
S just raised the bar for easy, resilient and secure K8s cluster deployment
.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

0 packages can be updated.
0 updates are security updates.

vagrant@vagrant:~$
```

Step 7: Install Nginx inside VM

```
sudo apt update
sudo apt install -y nginx
sudo systemctl start nginx
```

Step 8: Verify Nginx

```
curl localhost
```

```

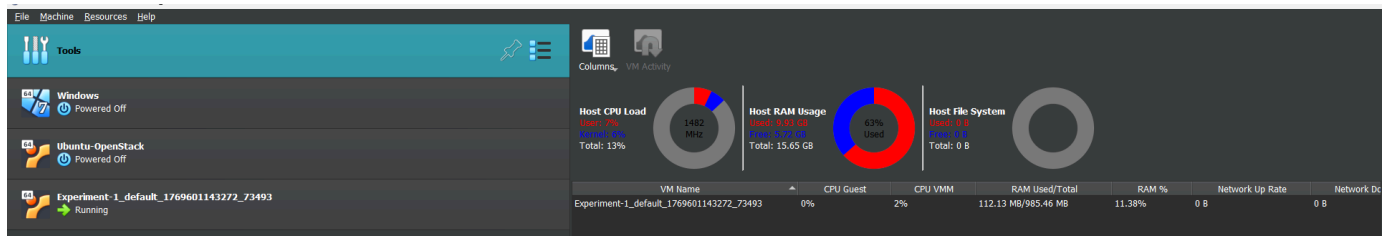
vagrant@vagrant: ~$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
vagrant@vagrant: ~$

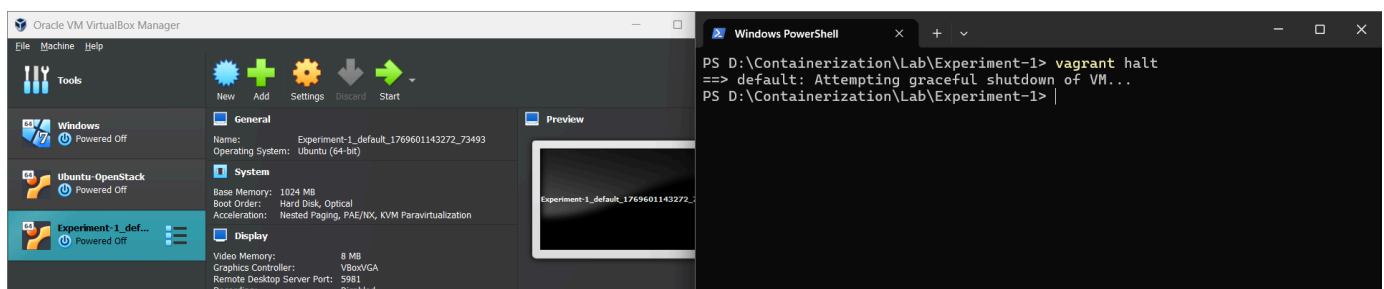
```

Step 9: Utilization Matrix In Running State

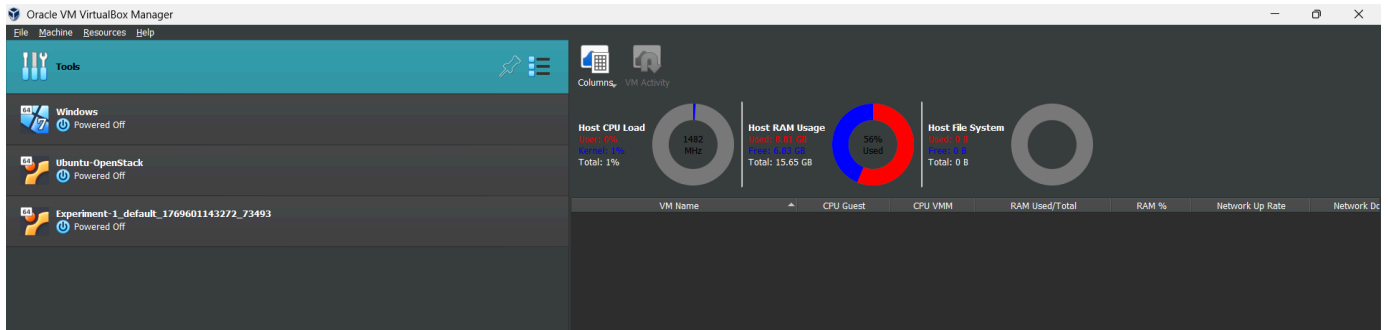


Step 10: Stop VM

```
vagrant halt
```

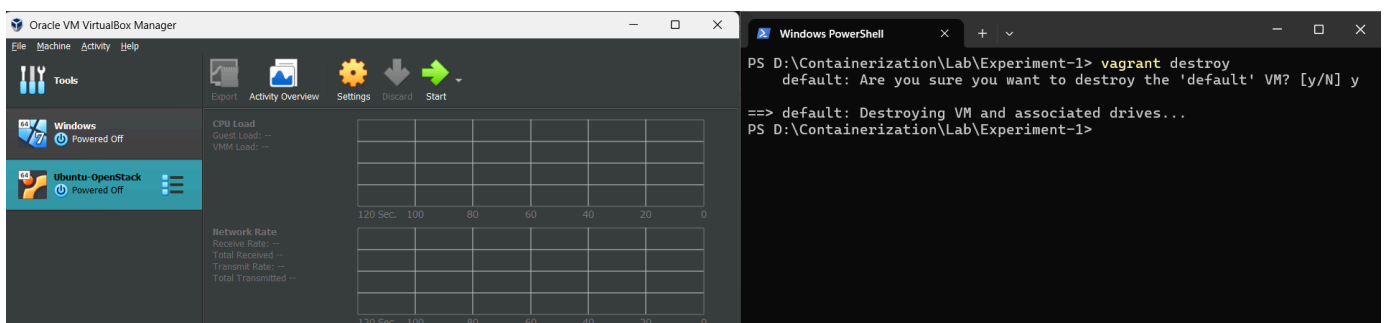


Step 11: Utilization Matrix In Stop State



Step 12: Remove VM

vagrant destroy



PART-B:- Containers using WSL (Windows)

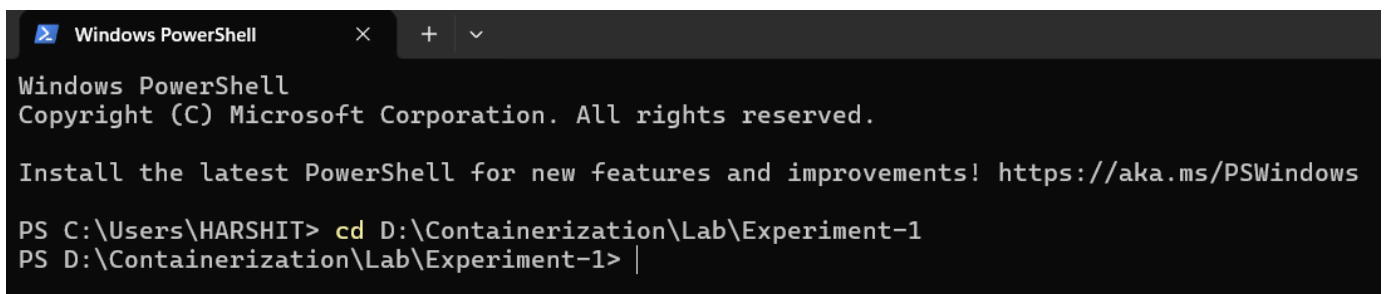
Step 1: Install WSL

```
wsl --install
```

Reboot the system after installation.

Step 2: Install Ubuntu on WSL

```
wsl --install -d Ubuntu
```



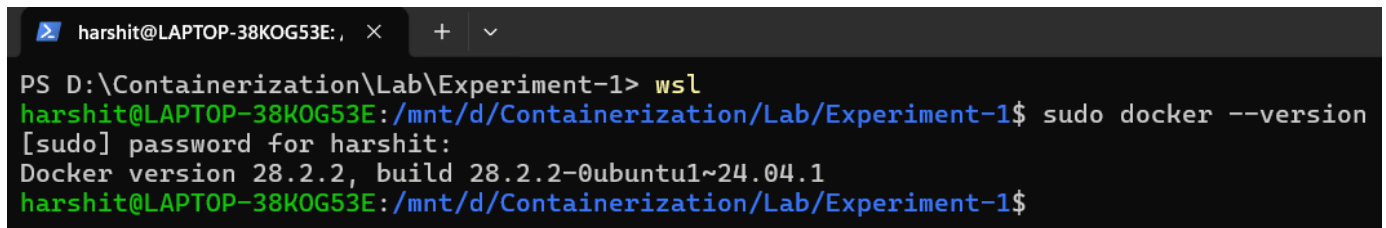
Step 3: Install Docker Engine inside WSL

```
sudo apt update
sudo apt install -y docker.io
sudo systemctl start docker
sudo usermod -aG docker $USER
```

Logout and login again to apply group changes.

Step 4: Verify Docker Installation

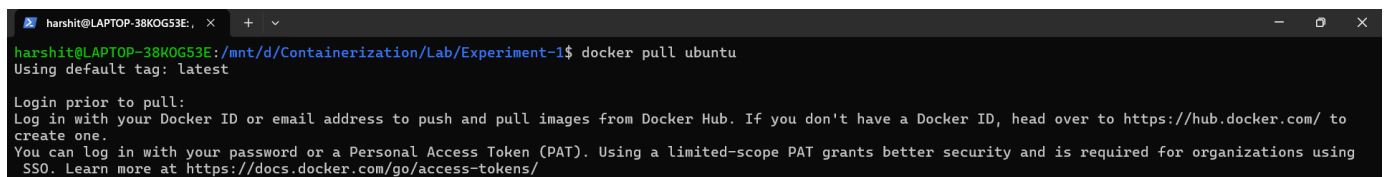
```
docker --version
```



```
PS D:\Containerization\Lab\Experiment-1> wsl
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-1$ sudo docker --version
[sudo] password for harshit:
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-1$
```

Step 5: Pull Ubuntu Image

```
docker pull ubuntu
```

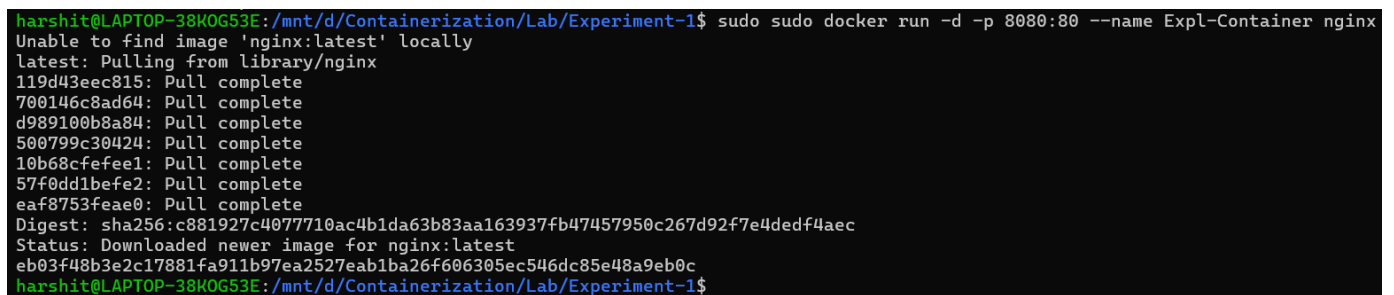


```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-1$ docker pull ubuntu
Using default tag: latest

Login prior to pull:
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/
```

Step 6: Run Ubuntu Container with Nginx

```
docker run -d -p 8080:80 --name nginx-container nginx
```



```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-1$ sudo docker run -d -p 8080:80 --name Expl-Container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfeee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
eb03f48b3e2c17881fa911b97ea2527eab1ba26f606305ec546dc85e48a9eb0c
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-1$
```

Step 7: Verify Nginx in Container

```
curl localhost:8080
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-1$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-1$

```

Step 8: Container Observation Commands

```

docker stats
free -h

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-1$ docker stats
CONTAINER ID   NAME          CPU %          MEM USAGE / LIMIT   MEM %           NET I/O       BLOCK I/O      PIDS
eb03f48b3e2c   Expl-Container  --             -- / --             --              --            --             --

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-1$ free -h
Mem:            7.6Gi       559Mi       7.0Gi          3.6Mi       240Mi       7.0Gi
Swap:           2.0Gi          0B         2.0Gi
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-1$

```

Parameters to Compare

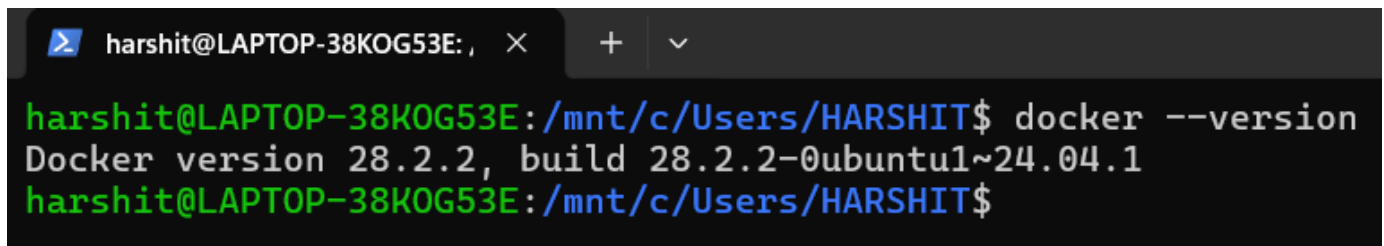
Parameter	Virtual Machine	Container
Boot Time	High	Very Low
RAM Usage	High	Low
CPU Overhead	Higher	Minimal
Disk Usage	Larger	Smaller
Isolation	Strong	Moderate

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Experiment 2:- Docker Installation, Configuration, and Running Images

Step 1: Pull Image

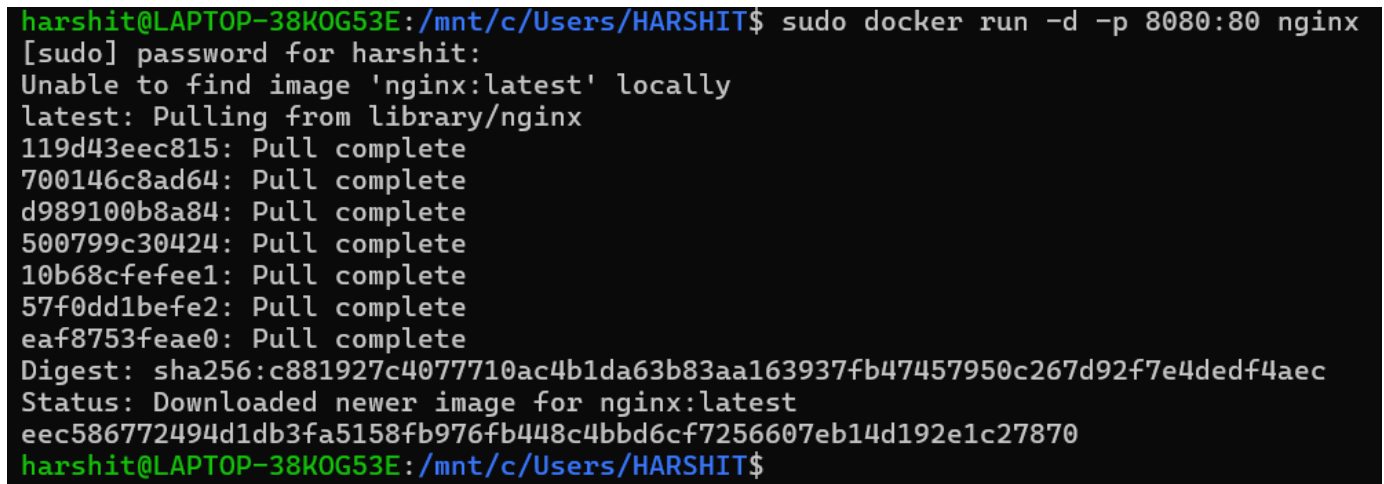
```
docker --version
```



```
harshit@LAPTOP-38KOG53E: , × + v
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$
```

Step 2: Run Container with Port Mapping

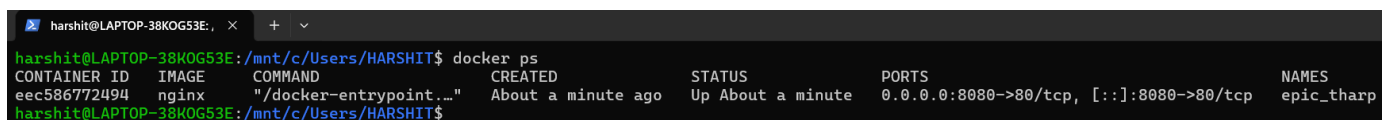
```
docker run -d -p 8080:80 nginx
```



```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ sudo docker run -d -p 8080:80 nginx
[sudo] password for harshit:
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
eec586772494d1db3fa5158fb976fb448c4bbd6cf7256607eb14d192e1c27870
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$
```

Step 3: Verify Running Containers

```
docker ps
```



```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
eec586772494   nginx    "/docker-entrypoint...." About a minute ago Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   epic_tharp
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$
```

Step 4: Stop Container

```
docker stop <container_id>
```

```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker stop eec
eec
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
71dc434cecaf	nginx	"/docker-entrypoint..."	5 minutes ago	Created		bold_liskov
eec586772494	nginx	"/docker-entrypoint..."	5 minutes ago	Exited (0) 38 seconds ago		epic_tharp
50718c8d6516	java-app:1.0	"bash"	27 hours ago	Exited (0) 26 hours ago		happy_borg
804c18b86fb3	java-app:1.0	"java Hello"	27 hours ago	Exited (0) 27 hours ago		happy_mccarthy
4caf5a088f14	hello-world	"/hello"	10 days ago	Exited (0) 10 days ago		epic_mendel
91afd5288876	hello-world	"/hello"	10 days ago	Exited (0) 10 days ago		festive_mcnulty

```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$
```

Step 5: Remove Container

```
docker rm <container_id>
```

```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker rm eec
eec
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
71dc434cecaf	nginx	"/docker-entrypoint..."	17 minutes ago	Created		bold_liskov
50718c8d6516	java-app:1.0	"bash"	27 hours ago	Exited (0) 27 hours ago		happy_borg
804c18b86fb3	java-app:1.0	"java Hello"	27 hours ago	Exited (0) 27 hours ago		happy_mccarthy
4caf5a088f14	hello-world	"/hello"	10 days ago	Exited (0) 10 days ago		epic_mendel
91afd5288876	hello-world	"/hello"	10 days ago	Exited (0) 10 days ago		festive_mcnulty

Step 5: Remove Image

```
docker rmi nginx
```

```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Deleted: sha256:4af177a024eb8a1e43f4fb6c66735bb8260115cb5925a64f51673219bd97c144
Deleted: sha256:74821a6b71dfac14eaeb80c7c9e48130391306557f75228278054afe105cde75
Deleted: sha256:ae46c5037a8cc17f658f78b0e6523c9363ca800bf51f91f58cbc3aad8ffe53b5
Deleted: sha256:ec81248cd6b602bb24b1c32d4f59194a4a42aab3c81a42631efa1af7cba4f69
Deleted: sha256:774cc13f29089d7d0ebaaa291c5ac2d12841f2d70ab07a21c470d3cbdc11c059
Deleted: sha256:f1155703f3479bf6b295f5e17684156f82f815ca0798af8a2d08cf994411a055
Deleted: sha256:a455653294af11c547d0b3d2d769fd184f357cd5c0decabd1f89ffe9b380985aa
Deleted: sha256:e50a58335e1366e2581fe61794c1651afe2fe04e881e795aa166f24f4fc78d92
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
javaapp	2.0	0045139ba152	27 hours ago	773MB
java-app	1.0	3c923bd3aefd	3 days ago	773MB
ubuntu	22.04	65c77cbc27c2	3 weeks ago	77.9MB
hello-world	latest	1b44b5a3e06a	5 months ago	10.1kB

```
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT$
```

Result Docker images were successfully pulled, containers executed, and lifecycle commands performed.

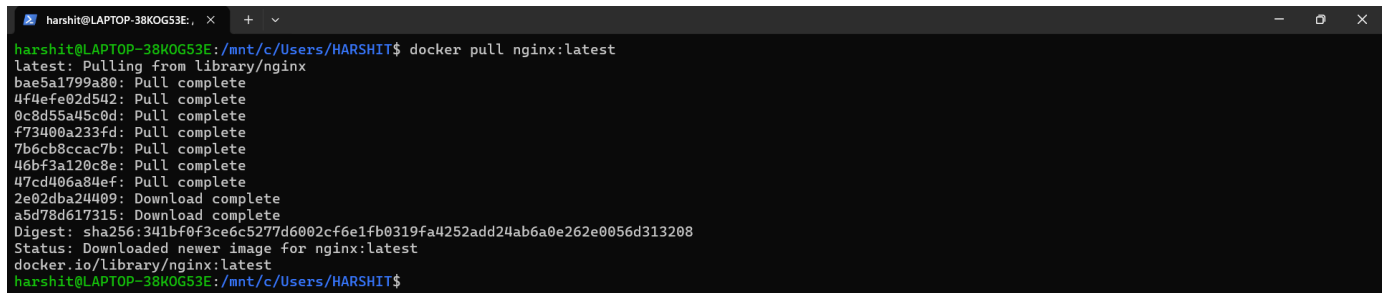
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Experiment 3:- Docker Deploy NGINX Using Different Base Images and Comparing Image Layers

Part 1: Deploy NGINX Using Official Image (Recommended Approach)

Step 1: Pull the Image

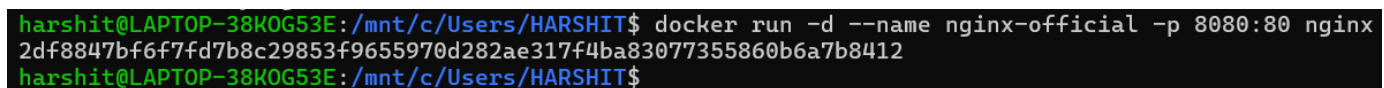
```
docker pull nginx:latest
```



```
harshit@LAPTOP-38KOG53E: /mnt/c/Users/HARSHIT$ docker pull nginx:latest
latest: Pulling from library/nginx
bae5a1799a80: Pull complete
4f4efe02d542: Pull complete
0c8d55a45c0d: Pull complete
f73400a233fd: Pull complete
7b6cb8ccac7b: Pull complete
46bf3a120c8e: Pull complete
47cd406a84ef: Pull complete
2e02dba24409: Download complete
a5d78d617315: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
harshit@LAPTOP-38KOG53E: /mnt/c/Users/HARSHIT$
```

Step 2: Run the Container

```
docker run -d --name nginx-official -p 8080:80 nginx
```



```
harshit@LAPTOP-38KOG53E: /mnt/c/Users/HARSHIT$ docker run -d --name nginx-official -p 8080:80 nginx
2df8847bf6f7fd7b8c29853f9655970d282ae317f4ba83077355860b6a7b8412
harshit@LAPTOP-38KOG53E: /mnt/c/Users/HARSHIT$
```

Step 3: Verify

```
curl http://localhost:8080
```



```

harshit@LAPTOP-38K0G53E:/mnt/c/Users/HARSHIT$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
harshit@LAPTOP-38K0G53E:/mnt/c/Users/HARSHIT$

```

You should see the NGINX welcome page.

Key Observations

docker images nginx

```

harshit@LAPTOP-38K0G53E:/mnt/c/Users/HARSHIT$ docker images nginx
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    341bf0f3ce6c   2 days ago    240MB
harshit@LAPTOP-38K0G53E:/mnt/c/Users/HARSHIT$

```

- Image is pre-optimized
- Minimal configuration required
- Uses Debian-based OS internally

Part 2: Custom NGINX Using Ubuntu Base Image

Step 1: Create Dockerfile

```
FROM ubuntu:22.04
```

```

RUN apt-get update && \
    apt-get install -y nginx && \

```

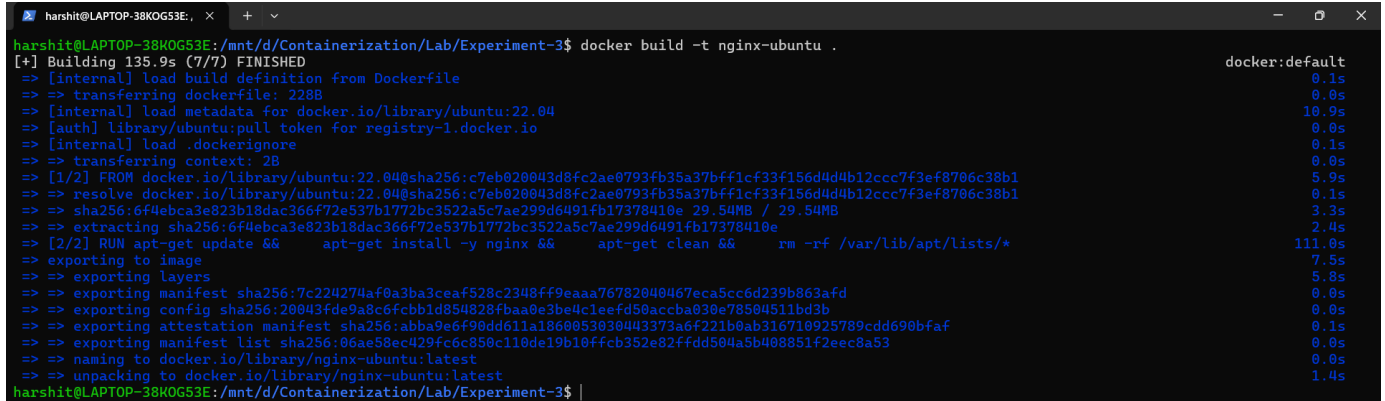
```
apt-get clean && \
rm -rf /var/lib/apt/lists/*
```

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

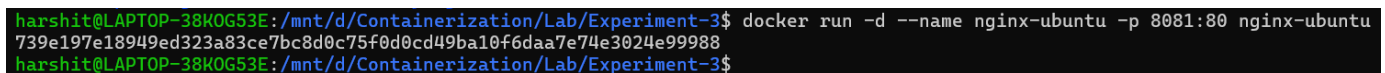
Step 2: Build Image

```
docker build -t nginx-ubuntu .
```



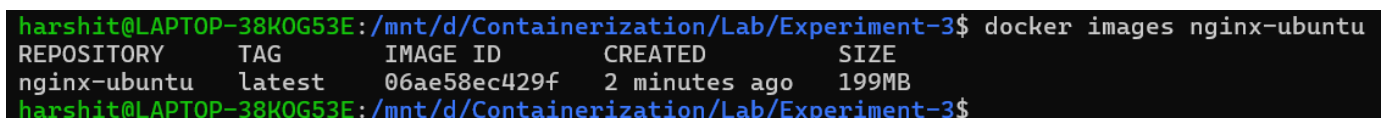
Step 3: Run Container

```
docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu
```



Observations

```
docker images nginx-ubuntu
```



- Much larger image size
- More layers
- Full OS utilities available

Part 3: Custom NGINX Using Alpine Base Image

Step 1: Create Dockerfile

```
FROM alpine:latest

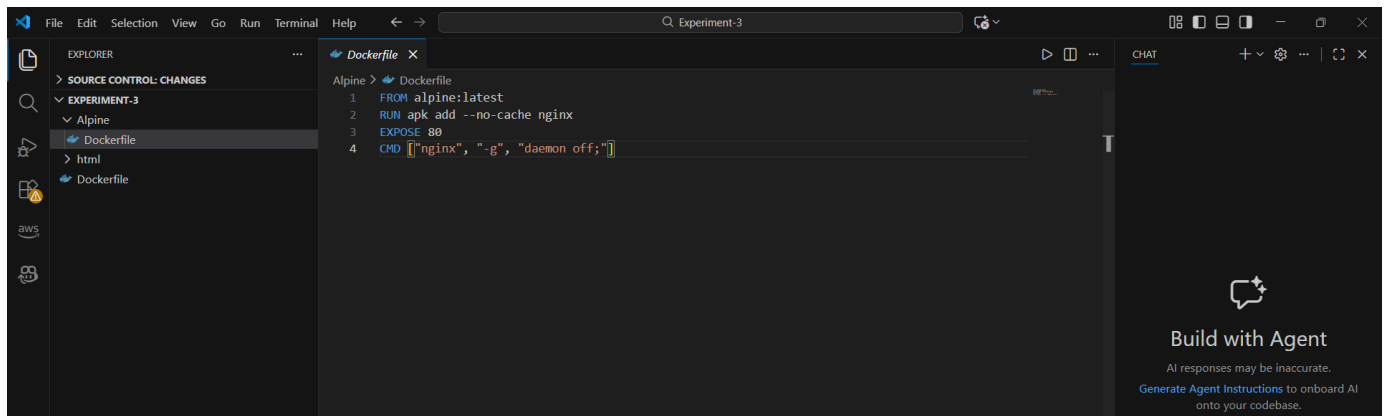
RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Step 2: Build Image

```
docker build -t nginx-alpine .
```



Step 3: Run Container

```
docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
```

```
harshit@LAPTOP-38K0G53E:/mnt/d/Containerization/Lab/Experiment-3/Alpine$ docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
ccc5445853cd6f2eaa582a551f13831f59e8f4b6e7cb26fbf78e2eecd8bffd96
harshit@LAPTOP-38K0G53E:/mnt/d/Containerization/Lab/Experiment-3/Alpine$
```

Observations

```
docker images nginx-alpine
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$ docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
ccc5445853cd6f2eaa582a551f13831f59e8f4b6e7cb26fbf78e2eecd8bffd96
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$ docker images nginx-alpine
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx-alpine   latest    afbaba358e5c   58 seconds ago 16MB
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$
```

- Extremely small image
- Fewer packages
- Faster pull and startup time

Part 4: Image Size and Layer Comparison

Compare Sizes

```
docker images | grep nginx
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3$ docker images | grep nginx
nginx-alpine   latest    afbaba358e5c   25 minutes ago 16MB
nginx-ubuntu   latest    06ae58ec429f   34 minutes ago 199MB
nginx          latest    341bf0f3ce6c   2 days ago     240MB
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3$
```

Typical result (approx):

Image Type	Size
nginx:latest	240 MB
nginx-ubuntu	199 MB
nginx-alpine	16 MB

Inspect Layers

```
docker history nginx
docker history nginx-ubuntu
docker history nginx-alpine
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$ docker history nginx
IMAGE          CREATED        CREATED BY          SIZE      COMMENT
341bf0f3ce6c   2 days ago    CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.v0
<missing>      2 days ago    STOPSIGNAL SIGQUIT 0B        buildkit.dockerfile.v0
<missing>      2 days ago    EXPOSE map[80/tcp:{}] 0B        buildkit.dockerfile.v0
<missing>      2 days ago    ENTRYPOINT ["/docker-entrypoint.sh"] 0B        buildkit.dockerfile.v0
<missing>      2 days ago    COPY 30-tune-worker-processes.sh /docker-ent... 16.4kB    buildkit.dockerfile.v0
<missing>      2 days ago    COPY 20-envsubst-on-templates.sh /docker-ent... 12.3kB    buildkit.dockerfile.v0
<missing>      2 days ago    COPY 15-local-resolvers.envsh /docker-entryp... 12.3kB    buildkit.dockerfile.v0
<missing>      2 days ago    COPY 10-listen-on-ipv6-by-default.sh /docker... 12.3kB    buildkit.dockerfile.v0
<missing>      2 days ago    COPY docker-entrypoint.sh / # buildkit 8.19kB    buildkit.dockerfile.v0
<missing>      2 days ago    RUN /bin/sh -c set -x && groupadd --syst... 86.6MB    buildkit.dockerfile.v0
<missing>      2 days ago    ENV DYNPKG_RELEASE=1~trixie 0B        buildkit.dockerfile.v0
<missing>      2 days ago    ENV PKG_RELEASE=1~trixie 0B        buildkit.dockerfile.v0
<missing>      2 days ago    ENV ACME_VERSION=0.3.1 0B        buildkit.dockerfile.v0
<missing>      2 days ago    ENV NJS_RELEASE=1~trixie 0B        buildkit.dockerfile.v0
<missing>      2 days ago    ENV NJS_VERSION=0.9.5 0B        buildkit.dockerfile.v0
<missing>      2 days ago    ENV NGINX_VERSION=1.29.5 0B        buildkit.dockerfile.v0
<missing>      2 days ago    LABEL maintainer=NGINX Docker Maintainers <d... 0B        buildkit.dockerfile.v0
<missing>      5 days ago    # debian.sh --arch 'amd64' out/ 'trixie' '01... 87.4MB    debuerreotype 0.17
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$ docker history nginx-ubuntu
IMAGE          CREATED        CREATED BY          SIZE      COMMENT
06ae58ec429f   10 minutes ago CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.v0
<missing>      10 minutes ago EXPOSE [80/tcp] 0B        buildkit.dockerfile.v0
<missing>      10 minutes ago RUN /bin/sh -c apt-get update && apt-get... 58.6MB    buildkit.dockerfile.v0
<missing>      4 weeks ago    /bin/sh -c #(nop) CMD ["/bin/bash"] 0B        buildkit.dockerfile.v0
<missing>      4 weeks ago    /bin/sh -c #(nop) ADD file:b499000226bd9a7c5... 87.5MB    buildkit.dockerfile.v0
<missing>      4 weeks ago    /bin/sh -c #(nop) LABEL org.opencontainers.... 0B        buildkit.dockerfile.v0
<missing>      4 weeks ago    /bin/sh -c #(nop) LABEL org.opencontainers.... 0B        buildkit.dockerfile.v0
<missing>      4 weeks ago    /bin/sh -c #(nop) ARG LAUNCHPAD_BUILD_ARCH 0B        buildkit.dockerfile.v0
<missing>      4 weeks ago    /bin/sh -c #(nop) ARG RELEASE 0B        buildkit.dockerfile.v0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$ docker history nginx-alpine
IMAGE          CREATED        CREATED BY          SIZE      COMMENT
afbaba358e5c   About a minute ago CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.v0
<missing>      About a minute ago EXPOSE [80/tcp] 0B        buildkit.dockerfile.v0
<missing>      About a minute ago RUN /bin/sh -c apk add --no-cache nginx # bu... 2.2MB    buildkit.dockerfile.v0
<missing>      10 days ago    CMD ["/bin/sh"] 0B        buildkit.dockerfile.v0
<missing>      10 days ago    ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /... 9.11MB    buildkit.dockerfile.v0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3/Alpine$

```

Observations:

- Ubuntu has many filesystem layers
- Alpine has minimal layers
- Official NGINX image is optimized but heavier than Alpine

Part 5: Functional Tasks Using NGINX

Task 1: Serve Custom HTML Page

```

mkdir html
echo "<h1>Harshit Kumar - 500122407</h1>" > html/index.html

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3$ mkdir html
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3$ echo "<h1>Harshit Kumar - 500122407</h1>" > html/index.html
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-3$

```

Run:

```

docker run -d \
-p 8083:80 \

```

```
-v $(pwd)/html:/usr/share/nginx/html \
nginx
```

```
harshit@LAPTOP-38K0G53E:/mnt/d/Containerization/Lab/Experiment-3$ docker run -d \
-p 8083:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx
ec25b53c4141b6d1ce89368f25331f7e3fbac6e65aa1a37ecdec3a111e487e65
harshit@LAPTOP-38K0G53E:/mnt/d/Containerization/Lab/Experiment-3$
```



Harshit Kumar - 500122407

Task 2: Reverse Proxy (Conceptual)

NGINX can:

- Forward traffic to backend services
- Load balance multiple containers
- Terminate SSL

Example use cases:

- Frontend for microservices
- API gateway
- Static file server

Part 6: Comparison Summary

Image Comparison Table

Feature	Official NGINX	Ubuntu + NGINX	Alpine + NGINX
Image Size	Medium	Large	Very Small
Ease of Use	Very Easy	Medium	Medium
Startup Time	Fast	Slow	Very Fast
Debugging Tools	Limited	Excellent	Minimal
Security Surface	Medium	Large	Small
Production Ready	Yes	Rarely	Yes

Part 7: When to Use What

Use Official NGINX Image When:

- Production deployment
- Standard web hosting
- Reverse proxy / load balancer

Use Ubuntu-Based Image When:

- Learning Linux + NGINX internals
- Heavy debugging needed
- Custom system-level dependencies

Use Alpine-Based Image When:

- Microservices
- CI/CD pipelines
- Cloud and Kubernetes workloads

[< Previous](#) [Home](#) [Next >](#)

Harshit-Kumar-Containerization

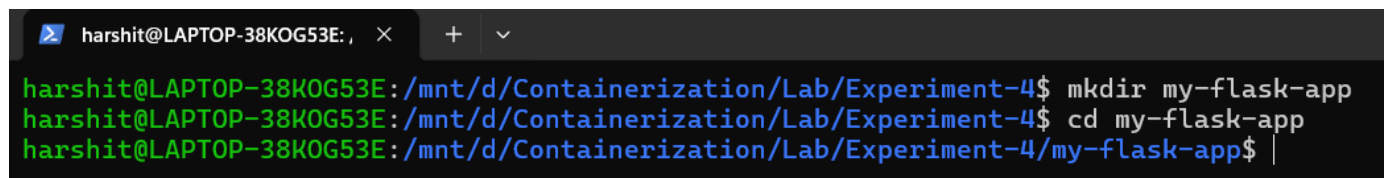
Experiment 4: Docker Essentials

Dockerfile .dockerignore tagging publishing

Containerizing Applications with Dockerfile

Step-1: Create a Directory for Python application

```
mkdir my-flask-app  
cd my-flask-app
```

A terminal window with a dark background. The title bar shows 'harshit@LAPTOP-38KOG53E:'. The terminal shows three lines of commands and their outputs: 1. 'mkdir my-flask-app' is executed, and the prompt changes to the new directory. 2. 'cd my-flask-app' is executed, and the prompt changes to the full path of the directory. 3. The prompt is shown again at the end of the third line.

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4$ mkdir my-flask-app  
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4$ cd my-flask-app  
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Step-2:- Create an **** app.py ****

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route('/')  
def hello():  
    return "Hello from Docker!"  
  
@app.route('/health')  
def health():  
    return "OK"  
  
if __name__ == '__main__':  
    app.run(host='0.0.0.0', port=5000)
```



```

harshit@LAPTOP-38KOG53E: ~
GNU nano 7.2 app.py
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from Docker!"

@app.route('/health')
def health():
    return "OK"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)

```

Step-3:- Create **requirements.txt** :

```
Flask==2.3.3
```

```

harshit@LAPTOP-38KOG53E: ~
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ cat requirements.txt
Flask==2.3.3
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$

```

Step-4: Create **Dockerfile**

```

# Use Python base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY app.py .

# Expose port
EXPOSE 5000

```

Run the application

CMD ["python", "app.py"]

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ nano Dockerfile
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ cat Dockerfile
# Use Python base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY app.py .

# Expose port
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Step-5:- Create a **.dockerignore**

```
# Python files
__pycache__/
*.pyc
*.pyo
*.pyd

# Environment files
.env
.venv
env/
venv/

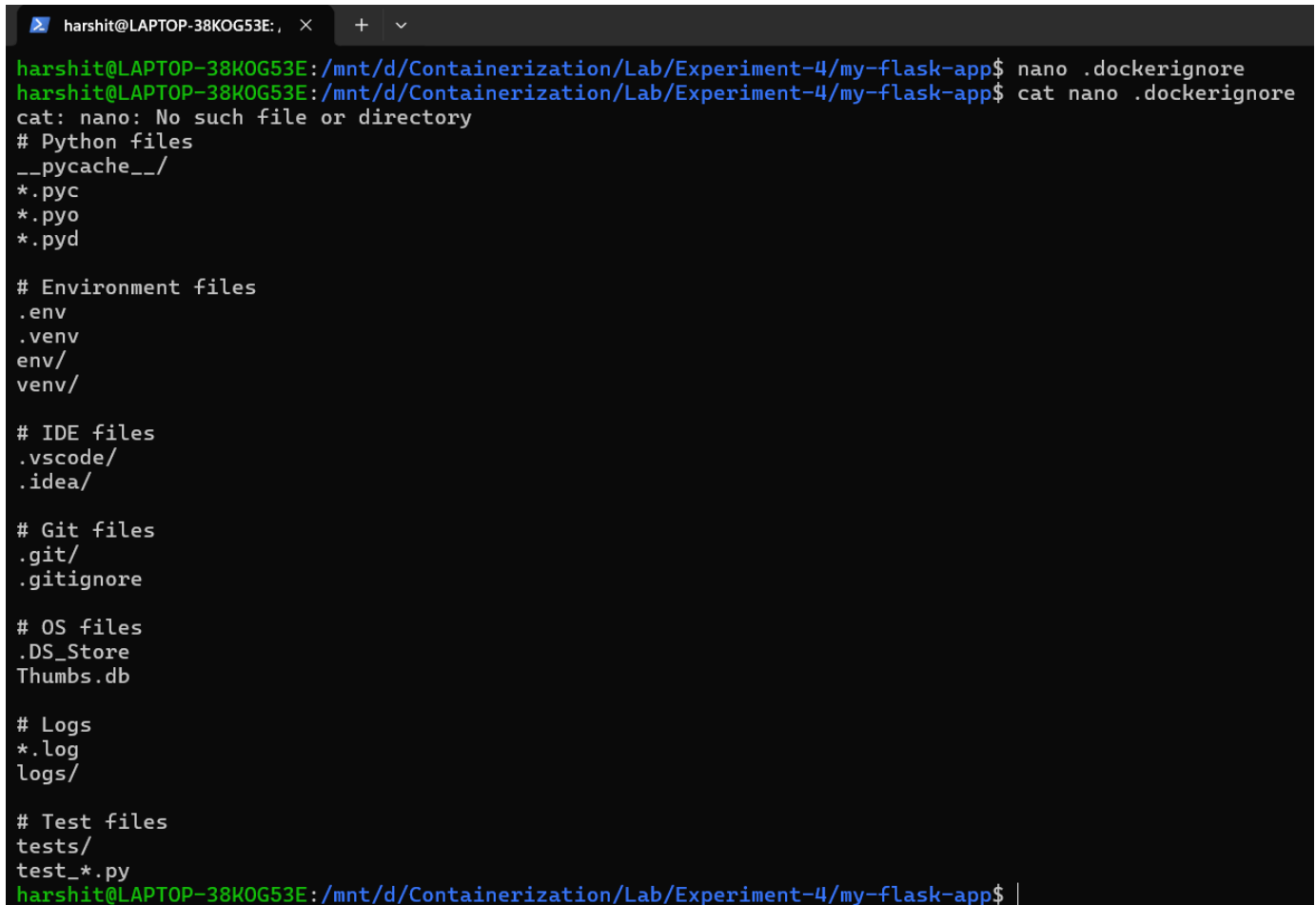
# IDE files
.vscode/
.idea/

# Git files
.git/
.gitignore

# OS files
.DS_Store
Thumbs.db

# Logs
*.log
logs/
```

```
# Test files
tests/
test_*.py
```

A terminal window with a dark background. The prompt is 'harshit@LAPTOP-38KOG53E:'. The user runs 'nano .dockerignore' and then 'cat nano .dockerignore'. The output shows the content of the file: '# Python files', '__pycache__/', '*.pyc', '*.pyo', '*.pyd', '# Environment files', '.env', '.venv', 'env/', 'venv/', '# IDE files', '.vscode/', '.idea/', '# Git files', '.git/', '.gitignore', '# OS files', '.DS_Store', 'Thumbs.db', '# Logs', '*.log', 'logs/', '# Test files', 'tests/', 'test_*.py'. The prompt at the bottom is 'harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app\$ |'.

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ nano .dockerignore
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ cat nano .dockerignore
cat: nano: No such file or directory
# Python files
__pycache__/
*.pyc
*.pyo
*.pyd

# Environment files
.env
.venv
env/
venv/

# IDE files
.vscode/
.idea/

# Git files
.git/
.gitignore

# OS files
.DS_Store
Thumbs.db

# Logs
*.log
logs/

# Test files
tests/
test_*.py
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Why `.dockerignore` is Important

- Prevents unnecessary files from being copied
- Reduces image size
- Improves build speed
- Increases security

Step-6: Build image from Dockerfile

```
docker build -t my-flask-app .
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker build -t my-flask-app .
[+] Building 37.4s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 361B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 273B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> => sha256:ea56f685404df81680322f152d2cfec62115b30dda481c2c458078315beb508 251B / 251B
=> => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB
=> => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB
=> => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 2.3s
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.3s
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 1.4s
=> => extracting sha256:ea56f685404df81680322f152d2cfec62115b30dda481c2c458078315beb508 0.0s
=> [internal] load build context
=> => transferring context: 321B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY app.py .
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:1000d936b946975e16972fcb3cb5303528b45ac5cf0cd2bb67c113a66654317
=> => exporting config sha256:4a7048f3c831ddc1046bb351892b4f4ccd70a1f15be4f25d1415bd48feca887
=> => exporting attestation manifest sha256:c41b96361492f7c660192fd65efc71e92eb63bbb97bbe7fb6f842638ef265ced
=> => exporting manifest list sha256:f49a1ae3b7f04e26dfd7d3a5b46cd7841a55ab06b3acb4ac733be92dbf51d66
=> => naming to docker.io/library/my-flask-app:latest
=> => unpacking to docker.io/library/my-flask-app:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$

```

Step-7:- Verify Image

docker images

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
my-flask-app        latest             f49a1ae3b7f0       23 seconds ago     200MB
singlestage         2.0               0772d72ef207       10 days ago        1.29MB
singlestage         1.0               c9497eb2ed74       10 days ago        504MB
nginx-alpine        latest            afbaba358e5c       2 weeks ago        16MB
nginx-ubuntu        latest            06ae58ec429f       2 weeks ago        199MB
mysql               latest            db32c8ec843c       2 weeks ago        1.27GB
sapid-checker       500122407         baebf693c5b5       2 weeks ago        242MB
nginx               latest            341bf0f3ce6c       2 weeks ago        240MB
python              3.11.14-alpine3.23 6ce68f8bfbb4       2 weeks ago        84.4MB
alpine              latest            25109184c71b       3 weeks ago        13.1MB
myrepo/test-app     latest            aa8b582800c6       3 weeks ago        217MB
ubuntu              latest            cd1dba651b30       5 weeks ago        119MB
hello-world         latest            05813aedc15f       6 months ago       25.9kB
busybox             latest            b3255e7dfbcd       17 months ago      6.77MB
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$

```

Step-8: Add Tag in Image

docker build -t my-flask-app:1.0 .

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker build -t my-flask-app:1.0 .
[+] Building 2.9s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 361B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 273B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 63B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY app.py .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:1000d936b946975e16972fcb3cb5303528b45ac5cf0cd2bb67c113a666654317
=> => exporting config sha256:4a7048fb3c831ddc1046bb351892b4f4ccd70a1f15be4f25d1415bd48feca887
=> => exporting attestation manifest sha256:d5ed2b31fd32c46b62ccad3f2272611fa03993a032389265bfb144021244919
=> => exporting manifest list sha256:23440825d2284bf6c6302848889a6f4c6e4aae5c5f3a2254f14ad7c19cc3c5132
=> => naming to docker.io/library/my-flask-app:1.0
=> => unpacking to docker.io/library/my-flask-app:1.0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-9:- Add multiple tags

```
docker build -t my-flask-app:latest -t my-flask-app:1.0 .
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker build -t my-flask-app:latest -t my-flask-app:1.0 .
[+] Building 1.9s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 361B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 273B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 63B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY app.py .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:1000d936b946975e16972fcb3cb5303528b45ac5cf0cd2bb67c113a666654317
=> => exporting config sha256:4a7048fb3c831ddc1046bb351892b4f4ccd70a1f15be4f25d1415bd48feca887
=> => exporting attestation manifest sha256:e2ba1dc12eb44779452060edc0ab27b1079634b3dd1ee1a6314da7d78514eaff
=> => exporting manifest list sha256:ea042c20cb9b6d3cba9faa96595c1b95ac2ee18faba40a0e511fcc169c173b6f
=> => naming to docker.io/library/my-flask-app:latest
=> => unpacking to docker.io/library/my-flask-app:latest
=> => naming to docker.io/library/my-flask-app:1.0
=> => unpacking to docker.io/library/my-flask-app:1.0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-10:- Add Tag with custom registry

```
docker build -t username/my-flask-app:1.0 .
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker build -t username/my-flask-app:1.0 .
[+] Building 2.0s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 361B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 273B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 63B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY app.py .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:1000d936b946975e16972fcb3cb5303528b45ac5cf0cd2bb67c113a666654317
=> => exporting config sha256:4a7048fb3c831ddc1046bb351892b4f4ccd70a1f15be4f25d1415bd48feca887
=> => exporting attestation manifest sha256:bc6128bd0d48f7fc4e2292a354240c1513491e575562d99ca865eff970cb6871
=> => exporting manifest list sha256:b5deb2bd63ecd013d2d50fe2d3b01278777fda056f02fafd8474fa48241dab
=> => naming to docker.io/username/my-flask-app:1.0
=> => unpacking to docker.io/username/my-flask-app:1.0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-11:- Tag existing image

```
docker tag my-flask-app:latest my-flask-app:v1.0
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker tag my-flask-app:latest my-flask-app:v1.0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-12: View Image Details

docker images

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
username/my-flask-app	1.0	b5deb2bdb63e	2 minutes ago	200MB
my-flask-app	1.0	ea042c20cb9b	2 minutes ago	200MB
my-flask-app	latest	ea042c20cb9b	2 minutes ago	200MB
my-flask-app	v1.0	ea042c20cb9b	2 minutes ago	200MB
singlestage	2.0	0772d72ef207	10 days ago	1.29MB
singlestage	1.0	c9497eb2ed74	10 days ago	504MB
nginx-alpine	latest	afbaba358e5c	2 weeks ago	16MB
nginx-ubuntu	latest	06ae58ec429f	2 weeks ago	199MB
mysql	latest	db32c8ec843c	2 weeks ago	1.27GB
sapid-checker	500122407	baebf693c5b5	2 weeks ago	242MB
nginx	latest	341bf0f3ce6c	2 weeks ago	240MB
python	3.11.14-alpine3.23	6ce68f8bfbb4	2 weeks ago	84.4MB
alpine	latest	25109184c71b	3 weeks ago	13.1MB
myrepo/test-app	latest	aa8b582800c6	3 weeks ago	217MB
ubuntu	latest	cd1dba651b30	5 weeks ago	119MB
hello-world	latest	05813aedc15f	6 months ago	25.9kB
busybox	latest	b3255e7dfbcd	17 months ago	6.77MB

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-13:- Show image history

docker history my-flask-app

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker history my-flask-app
```

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
ea042c20cb9b	2 minutes ago	CMD ["python" "app.py"]	0B	buildkit.dockerfile.v0
<missing>	2 minutes ago	EXPOSE [5000/tcp]	0B	buildkit.dockerfile.v0
<missing>	2 minutes ago	COPY app.py . # buildkit	12.3kB	buildkit.dockerfile.v0
<missing>	2 minutes ago	RUN /bin/sh -c pip install --no-cache-dir -r...	13.4MB	buildkit.dockerfile.v0
<missing>	2 minutes ago	COPY requirements.txt . # buildkit	12.3kB	buildkit.dockerfile.v0
<missing>	2 minutes ago	WORKDIR /app	8.19kB	buildkit.dockerfile.v0
<missing>	3 months ago	CMD ["python3"]	0B	buildkit.dockerfile.v0
<missing>	3 months ago	RUN /bin/sh -c set -eux; for src in idle3 p...	16.4kB	buildkit.dockerfile.v0
<missing>	3 months ago	RUN /bin/sh -c set -eux; savedAptMark="\$(a...	45.7MB	buildkit.dockerfile.v0
<missing>	3 months ago	ENV PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee8...	0B	buildkit.dockerfile.v0
<missing>	3 months ago	ENV PYTHON_VERSION=3.9.25	0B	buildkit.dockerfile.v0
<missing>	3 months ago	ENV GPG_KEY=E3FF2839C048B25C084DEBE9B26995E3...	0B	buildkit.dockerfile.v0
<missing>	3 months ago	RUN /bin/sh -c set -eux; apt-get update; a...	4.94MB	buildkit.dockerfile.v0
<missing>	3 months ago	ENV LANG=C.UTF-8	0B	buildkit.dockerfile.v0
<missing>	3 months ago	ENV PATH=/usr/local/bin:/usr/local/sbin:/usr...	0B	buildkit.dockerfile.v0
<missing>	4 months ago	# debian.sh --arch 'amd64' out/ 'trixie' '@1...	87.4MB	debuerreotype 0.16

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-14:- Inspect image details

docker inspect my-flask-app

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker inspect my-flask-app
[
  {
    "Architecture": "amd64",
    "Author": "",
    "Comment": "buildkit.dockerfile.v0",
    "Config": {
      "ArgsEscaped": true,
      "Cmd": [
        "python",
        "app.py"
      ],
      "Entrypoint": null,
      "Env": [
        "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
        "PYTHON_VERSION=3.9.25",
        "PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee84d2a40dae404a99303e3f97701c10966c91834"
      ],
      "ExposedPorts": {
        "5000/tcp": {}
      },
      "Labels": null,
      "OnBuild": null,
      "User": "",
      "Volumes": null,
      "WorkingDir": "/app"
    },
    "Created": "2026-02-21T09:04:20.945303564Z",
    "Descriptor": {
      "digest": "sha256:ea042c20cb9b6d3cba9faa96595c1b95ac2ee10faba40a0e511fcc169c173b6f",
      "mediaType": "application/vnd.oci.image.index.v1+json",
      "size": 856
    },
    "DockerVersion": "",
    "GraphDriver": {
      "Data": null,
      "Name": "overlayfs"
    },
    "Id": "sha256:ea042c20cb9b6d3cba9faa96595c1b95ac2ee10faba40a0e511fcc169c173b6f",
  }
]

```

Step-15: Run Container

```
docker run -d -p 5000:5000 --name flask-container my-flask-app
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
a7b2f0f68e42818b797a5d3a0e62cf536db435401a497bf0d79cb77a71ef3e76
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |

```

Step-16:- Test the application

```
curl http://localhost:5000
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ curl http://localhost:5000
Hello from Docker!harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |

```

Step-17:- View running containers

```
docker ps
```



```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
a7b2f0f68e42   my-flask-app   "python app.py"         59 seconds ago Up 57 seconds 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp flask-container
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-18:- View container logs

```
docker logs flask-container
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker logs flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [21/Feb/2026 09:09:16] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [21/Feb/2026 09:09:29] "GET / HTTP/1.1" 200 -
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-19:- Stop container

```
docker stop flask-container
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker stop flask-container
flask-container
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-20:- Start stopped container

```
docker start flask-container
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker start flask-container
flask-container
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-21:- Remove container

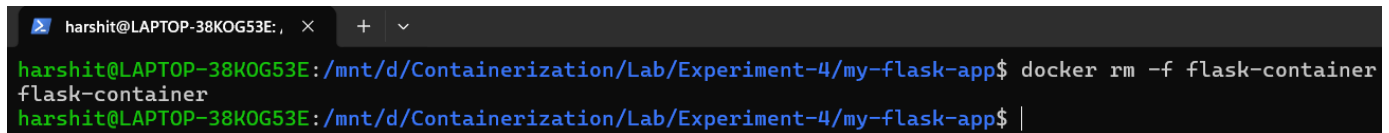
```
docker rm flask-container
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker rm flask-container
Error response from daemon: cannot remove container "flask-container": container is running: stop the container before removing or force remove
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-22:- Remove container forcefully

Since Container is in Running State, so we will remove it forcefully...


```
docker rm -f flask-container
```



```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker rm -f flask-container
flask-container
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Multi-stage Builds

Why Multi-stage Builds?

- Smaller final image size
- Better security (remove build tools)
- Separate build and runtime environments

Step-23: Create a Simple **** Dockerfile.multistage : ****

```
# STAGE 1: Builder stage
FROM python:3.9-slim AS builder

WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage
FROM python:3.9-slim

WORKDIR /app

# Copy virtual environment from builder
COPY --from=builder /opt/venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"

# Copy application code
COPY app.py .

# Create non-root user
RUN useradd -m -u 1000 appuser
USER appuser

# Expose port
EXPOSE 5000
```

```
# Run application
CMD ["python", "app.py"]
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4$ nano Dockerfile.multistage
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4$ cat Dockerfile.multistage
# STAGE 1: Builder stage
FROM python:3.9-slim AS builder

WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage
FROM python:3.9-slim

WORKDIR /app

# Copy virtual environment from builder
COPY --from=builder /opt/venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"

# Copy application code
COPY app.py .

# Create non-root user
RUN useradd -m -u 1000 appuser
USER appuser

# Expose port
EXPOSE 5000

# Run application
CMD ["python", "app.py"]
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4$ |
```

Step-24:- Build regular image

```
docker build -t flask-regular .
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker build -t flask-regular .
[+] Building 4.1s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 361B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 273B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd4e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd4e13cf1731b1b
=> [internal] load build context
=> => transferring context: 63B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY app.py .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:1000d936b946975e16972fcb3cb5303528b45ac5cf0cd2bb67c113a666654317
=> => exporting config sha256:4a7048fb3c831ddc1046bb351892b4f4ccd70a1f15be4f25d1415bd48feca887
=> => exporting attestation manifest sha256:6244fb462322dc2f50f3ce57a69eb2fc42bfed6eb3f76eaeccbbe6c020637175
=> => exporting manifest list sha256:fc07ac8ab08c4b55200b539cb98e2839bbacad4274c43fc9e2864a438069919f
=> => naming to docker.io/library/flask-regular:latest
=> => unpacking to docker.io/library/flask-regular:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Step-25:- Build multi-stage image

```
docker build -f Dockerfile.multistage -t flask-multistage .
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker build -f Dockerfile.multistage -t flask-multistage .
[+] Building 30.5s (13/13) FINISHED
=> [internal] load build definition from Dockerfile.multistage
=> => transferring dockerfile: 675B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 273B
=> [internal] load build context
=> => transferring context: 63B
=> [builder 1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b
=> CACHED [builder 2/5] WORKDIR /app
=> CACHED [builder 3/5] COPY requirements.txt
=> [builder 4/5] RUN python -m venv /opt/venv
=> [builder 5/5] RUN pip install --no-cache-dir -r requirements.txt
=> [stage-1 3/5] COPY --from=builder /opt/venv /opt/venv
=> [stage-1 4/5] COPY app.py
=> [stage-1 5/5] RUN useradd -m -u 1000 appuser
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:3e7159860d8123fb457f346584110d0a3374be5b9527892ef689cdc13a3609fe
=> => exporting config sha256:9f732ae1631ac2c732a1688aabe8062187df81ad1a794c70315b7566b76706f2
=> => exporting attestation manifest sha256:71dd07ed7789becb5f69d24cd10692fdc4936e50bfcc14c18fd759f683886bc1
=> => exporting manifest list sha256:407f3393d2a0f6d169edd0c61ee286493ab13ba2a694ae4435d72683f4fbbf9
=> => naming to docker.io/library/flask-multistage:latest
=> => unpacking to docker.io/library/flask-multistage:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Step-26:- Compare sizes

```
docker images | grep flask-
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker images | grep flask-
flask-multistage      latest      407f3393d2a0  30 seconds ago  219MB
flask-regular         latest      fc07ac8ab08c  20 minutes ago  200MB
my-flask-app          1.0        ea042c20cb9b  20 minutes ago  200MB
my-flask-app          latest      ea042c20cb9b  20 minutes ago  200MB
my-flask-app          v1.0       ea042c20cb9b  20 minutes ago  200MB
username/my-flask-app 1.0        b5deb2bdb63e  20 minutes ago  200MB
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Publishing to Docker Hub

Step-27:- Login to Docker Hub

```
docker login
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker login
Authenticating with existing credentials... [Username: harshitkumar22]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ |
```

Step-28:- Tag image for Docker Hub

```
docker tag my-flask-app:latest username/my-flask-app:1.0
docker tag my-flask-app:latest username/my-flask-app:latest
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker tag my-flask-app:latest harshitkumar22/my-flask-app:1.0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker tag my-flask-app:latest harshitkumar22/my-flask-app:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-29:- Push to Docker Hub

```
docker push username/my-flask-app:1.0
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker push harshitkumar22/my-flask-app:1.0
The push refers to repository [docker.io/harshitkumar22/my-flask-app]
463edab584aa: Pushed
38513bd72563: Pushed
ea56f685404a: Pushed
73ec3de0cc64: Pushed
fc7443084902: Pushed
02ec0bbe9c4c: Pushed
9ed960885e60: Pushed
d734a13928ac: Pushed
b3ec39b36ae8: Pushed
1.0: digest: sha256:ea042c20cb9b6d3cba9faa96595c1b95ac2ee10faba40a0e511fcc169c173b6f size: 856
```

Step-30:- Push Latest Image

```
docker push username/my-flask-app:latest
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker push harshitkumar22/my-flask-app:latest
The push refers to repository [docker.io/harshitkumar22/my-flask-app]
463edab584aa: Layer already exists
38513bd72563: Layer already exists
73ec3de0cc64: Layer already exists
fc7443084902: Layer already exists
02ec0bbe9c4c: Layer already exists
ea56f685404a: Layer already exists
9ed960885e60: Layer already exists
d734a13928ac: Layer already exists
b3ec39b36ae8: Layer already exists
latest: digest: sha256:ea042c20cb9b6d3cba9faa96595c1b95ac2ee10faba40a0e511fcc169c173b6f size: 856
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-31:- Pull from Docker Hub (on another machine)

```
docker pull username/my-flask-app:latest
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker pull harshitkumar22/my-flask-app:latest
latest: Pulling from harshitkumar22/my-flask-app
Digest: sha256:ea042c20cb9b6d3cba9faa96595c1b95ac2ee10faba40a0e511fcc169c173b6f
Status: Image is up to date for harshitkumar22/my-flask-app:latest
docker.io/harshitkumar22/my-flask-app:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Step-32:- Run the pulled image

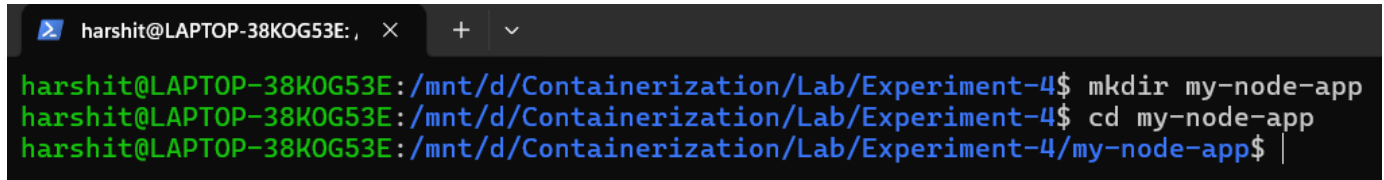
```
docker run -d -p 5000:5000 username/my-flask-app:latest
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$ docker run -d -p 5000:5000 harshitkumar22/my-flask-app:latest
7c87358eddc59a1ba9d834af9154acceca4efb5fe4be644ae79c6816510079
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-flask-app$
```

Node.js Application Example (Quick Version)**

Step-33:- Create a directory Node.js Application

```
mkdir my-node-app
cd my-node-app
```



```
harshit@LAPTOP-38KOG53E: ~$ mkdir my-node-app
harshit@LAPTOP-38KOG53E: ~$ cd my-node-app
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$
```

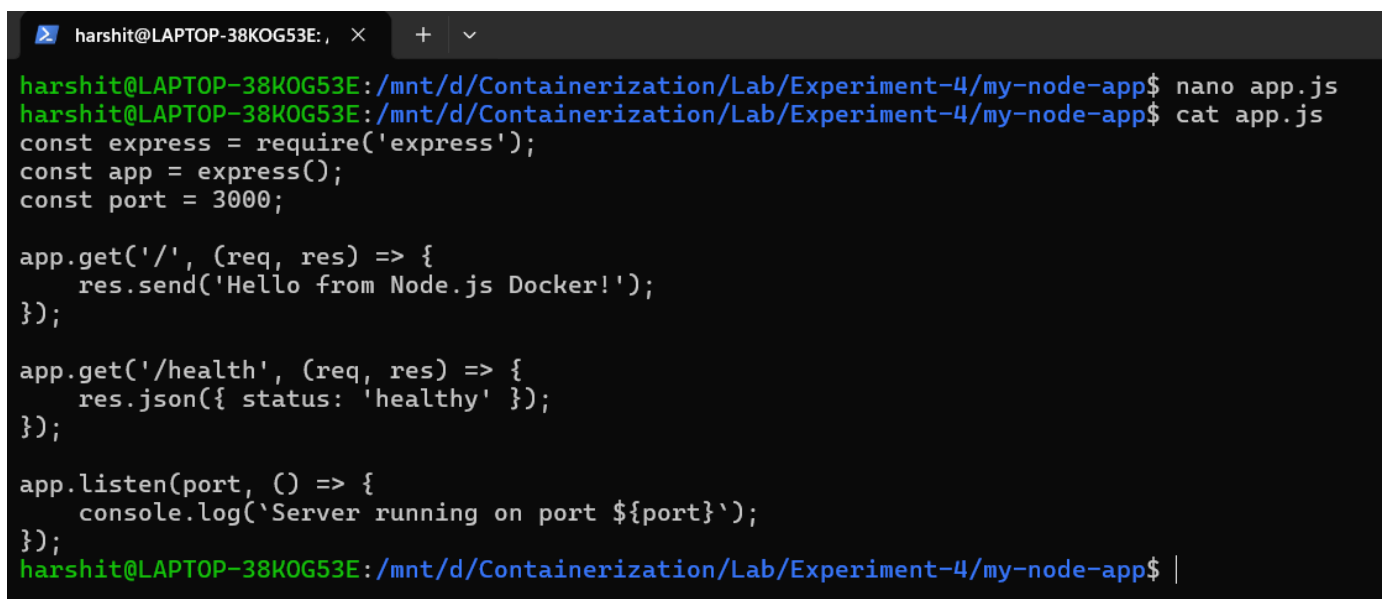
Step-34:- Create a ** app.js **:~

```
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req, res) => {
  res.send('Hello from Node.js Docker!');
});

app.get('/health', (req, res) => {
  res.json({ status: 'healthy' });
});

app.listen(port, () => {
  console.log(`Server running on port ${port}`);
});
```



```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ nano app.js
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ cat app.js
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req, res) => {
  res.send('Hello from Node.js Docker!');
});

app.get('/health', (req, res) => {
  res.json({ status: 'healthy' });
});

app.listen(port, () => {
  console.log(`Server running on port ${port}`);
});
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$
```

Step-35:- Create a ** package.json **:~

```
{
  "name": "node-docker-app",
  "version": "1.0.0",
  "main": "app.js",
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-node-app$ nano package.json
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-node-app$ cat package.json
{
  "name": "node-docker-app",
  "version": "1.0.0",
  "main": "app.js",
  "dependencies": {
    "express": "^4.18.2"
  }
}
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-node-app$ |
```

Step-36:- Create a Node.js Dockerfile

```
FROM node:18-alpine

WORKDIR /app

COPY package*.json ./
RUN npm install --only=production

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-node-app$ nano Dockerfile
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-node-app$ cat Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY package*.json ./
RUN npm install --only=production

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/Lab/Experiment-4/my-node-app$ |
```

Step-37:- Build image

```
docker build -t my-node-app .
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ docker build -t my-node-app .
[+] Building 36.6s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 182B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> => sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f8d26702d6e3 1.26MB / 1.26MB
=> => sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 446B / 446B
=> => sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feea161b2b5a3d0e 40.01MB / 40.01MB
=> => sha256:f18232174bc91741fdcf3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB
=> => extracting sha256:f18232174bc91741fdcf3da96d85011092101a032a93a388b79e99e69c2d5c870 0.45s
=> => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feea161b2b5a3d0e 2.75s
=> => extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f8d26702d6e3 0.11s
=> => extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 0.08s
=> [internal] load build context
=> => transferring context: 514B
=> [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install --only=production
=> [5/5] COPY app.js .
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:956ef6b6e59e98f8c21b6dc7296cab880e538e448fe6cebbd4cba4cf2faaff54
=> => exporting config sha256:af6781f362bb772d278fa951c68fd317288192af62833b0bfbbde4d98b62ba0f
=> => exporting attestation manifest sha256:37b4df4580d553a8240f11114a9468477e52a24249bc99a426bd982b853958c0
=> => exporting manifest list sha256:36d3d2686affb22317bf4a05f59d017eaeaf57875725634b9c6b04718444d325
=> => naming to docker.io/library/my-node-app:latest
=> => unpacking to docker.io/library/my-node-app:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ |
```

Step-38:- Run container

```
docker run -d -p 3000:3000 --name node-container my-node-app
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ docker run -d -p 3000:3000 --name node-container my-node-app
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ |
```

Step-39:- Test on Terminal

```
curl http://localhost:3000
```

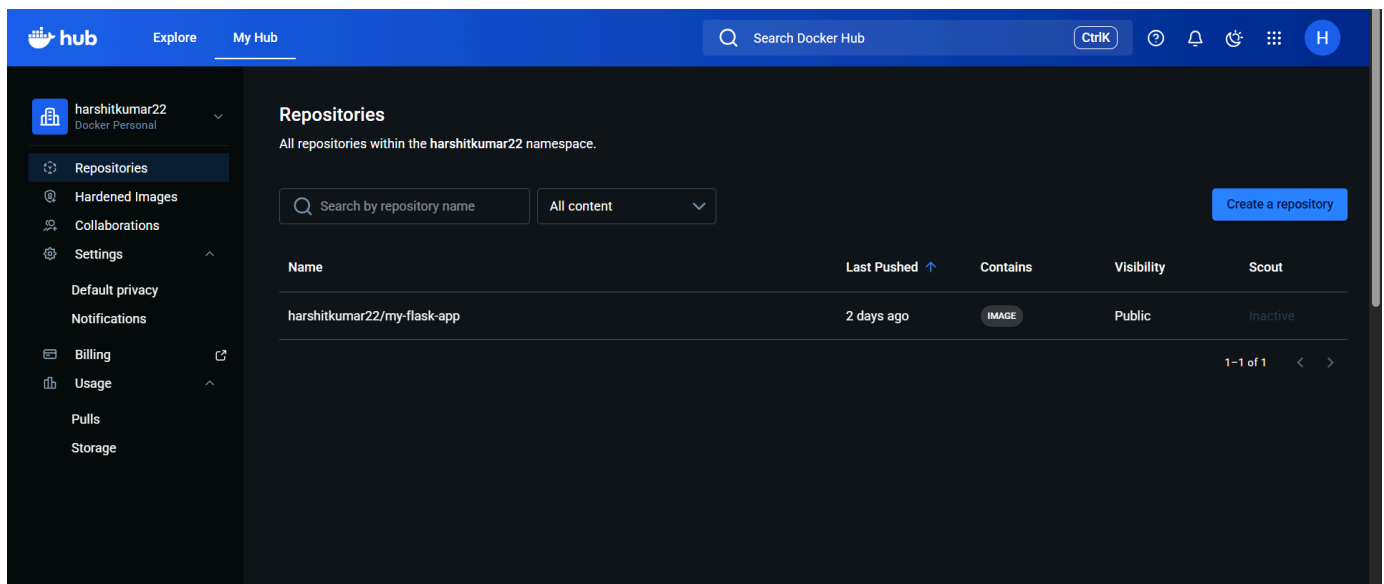
```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ curl http://localhost:3000
Hello from Node.js Docker!harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/Lab/Experiment-4/my-node-app$ |
```

Step-40:- Test on Browser

```
localhost:3000
Hello from Node.js Docker!
```

DockerHub

View Pushed Image on [DockerHub](#)



Conclusion

Essential Docker Commands Cheatsheet

Command	Purpose	Example
<code>docker build</code>	Build image	<code>docker build -t myapp .</code>
<code>docker run</code>	Run container	<code>docker run -p 3000:3000 myapp</code>
<code>docker ps</code>	List containers	<code>docker ps -a</code>
<code>docker images</code>	List images	<code>docker images</code>
<code>docker tag</code>	Tag image	<code>docker tag myapp:latest myapp:v1</code>
<code>docker login</code>	Login to Dockerhub using username and password or token	<code>echo "token" docker login -u username --password-stdin</code>
<code>docker push</code>	Push to registry	<code>docker push username/myapp</code>
<code>docker pull</code>	Pull from registry	<code>docker pull username/myapp</code>
<code>docker rm</code>	Remove container	<code>docker rm container-name</code>
<code>docker rmi</code>	Remove image	<code>docker rmi image-name</code>

Command	Purpose	Example
<code>docker logs</code>	View logs	<code>docker logs container-name</code>
<code>docker exec</code>	Execute command	<code>docker exec -it container-name bash</code>

Common Workflow Summary

Development Workflow:

```
# 1. Create Dockerfile and .dockerignore
# 2. Build image
docker build -t myapp .

# 3. Test locally
docker run -p 8080:8080 myapp

# 4. Tag for production
docker tag myapp:latest myapp:v1.0

# 5. Push to registry
docker push myapp:v1.0
```

Production Workflow:

```
# 1. Pull from registry
docker pull myapp:v1.0

# 2. Run in production
docker run -d -p 80:8080 --name prod-app myapp:v1.0

# 3. Monitor
docker logs -f prod-app
```

Key Takeaways

1. **Dockerfile** defines how to build your image
2. **.dockerignore** excludes unnecessary files (important for security and performance)
3. **Tagging** helps version and organize images
4. **Multi-stage builds** create smaller, more secure production images

5. **Docker Hub** allows sharing and distributing images

6. **Always test** images locally before publishing

Cleanup

Remove all stopped containers

```
docker container prune
```

Remove unused images

```
docker image prune
```

Remove everything unused

```
docker system prune -a
```

[< Previous](#) [Home](#) [Next >](#)

Harshit-Kumar-Containerization

Experiment 5: Docker - Volumes, Environment Variables, Monitoring & Networks

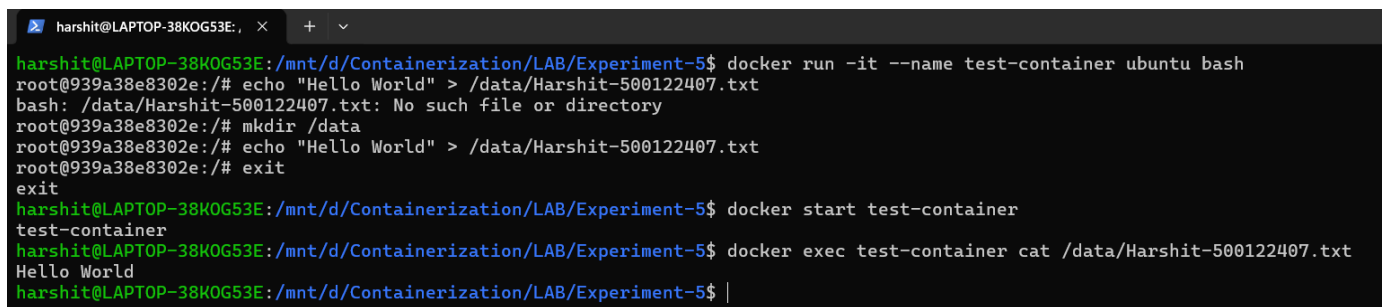
HandsOn

Step-1: Run a test container

```
# Create a container that writes data
docker run -it --name test-container ubuntu /bin/bash

# Inside container:
echo "Hello World" > /data/message.txt
cat /data/message.txt # Shows "Hello World"
exit

# Restart container
docker start test-container
docker exec test-container cat /data/message.txt
# ERROR: File doesn't exist! Data was lost.
```

A terminal window titled 'harshit@LAPTOP-38KOG53E:'. The user runs 'docker run -it --name test-container ubuntu bash'. The prompt changes to 'root@939a38e8302e:/#'. The user enters 'echo "Hello World" > /data/Harshit-500122407.txt', followed by 'bash: /data/Harshit-500122407.txt: No such file or directory', then 'mkdir /data', then 'echo "Hello World" > /data/Harshit-500122407.txt', then 'exit', and finally 'exit' at the host prompt. Then the user runs 'docker start test-container', and the prompt changes to 'test-container'. Finally, the user runs 'docker exec test-container cat /data/Harshit-500122407.txt', and the output 'Hello World' is displayed. The terminal ends with the host prompt 'harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5\$ |'.

Step-2:- Create Anonymous Volumes

```
# Create anonymous volume (auto-generated name)
docker run -d -v /app/data --name web1 nginx

# Check volume
docker volume ls
# Shows: anonymous volume with random hash

# Inspect container to see volume mount
```

docker inspect web1 | grep -A 5 Mounts

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d -v /app/data --name web1 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
448ea5cac5d5: Pull complete
4a038fd18db1: Pull complete
84e114c2bb36: Pull complete
7b5d674621c2: Pull complete
88d1d984b765: Pull complete
5435b2dcdf5c: Pull complete
054715a6bffa: Pull complete
3eff0a97d435: Download complete
cc0cf959117b: Download complete
Digest: sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
ad0f99a146371ebf2046156df71643d6239dfd1c6deaa282989e91edf55993e5
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker volume ls
DRIVER      VOLUME NAME
local       6b4eab63d4f0deea08ab2168746074c6157c8d155c142941adc917eb81dc3cf2
local       26feb_mysql_data
local       26feb_wp_content
local       89c1e0cbb07833b376150098557229b3784fef38442e7c01d11008b2d89ffbe7
local       1088cdae2a34a2c1e180e66dc5ed0cb6426e10faa1e53ebc4bdaf02f35776ce5
local       964748bb3c30fbac8573d4a67dd24c4f4dd0530d9ed27abbd555f2734cc3db04
local       HarshitVolume
local       a334b7b3ce44032963db0d818a03b7ff81e2ba5ed9687fca3867f73fdeceda47
local       a539e11647af44f1247d13dc5abb2c6b5c315a05c705e3b35db0c4f238bc324e
local       assignment-1_pgdata
local       b7286e4626db91564f6ac6af9bcf8ed097a893744def09a5d98c7045de7dc433
local       ibm-session_jenkins_home
local       k3d-mycluster-images
local       myvolume
local       portainer_data
local       test
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker inspect web1 | grep -A 5 Mounts
"Mounts": [
  {
    "Type": "volume",
    "Name": "89c1e0cbb07833b376150098557229b3784fef38442e7c01d11008b2d89ffbe7",
    "Source": "/var/lib/docker/volumes/89c1e0cbb07833b376150098557229b3784fef38442e7c01d11008b2d89ffbe7/_data",
    "Destination": "/app/data",
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ |
```

Step-3:- Create Named Volumes

Create named volume

docker volume create mydata

Use named volume

docker run -d -v mydata:/app/data --name web2 nginx

List volumes

docker volume ls

Shows: mydata

Inspect volume

docker volume inspect mydata

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker volume create mydata
mydata
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d -v mydata:/app/data --name web2 nginx
717c0d48c420cb1f98b9d7ec703a839b5ea1a1a68ebe6e0ab68a2f302f00ab47
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker volume ls
DRIVER      VOLUME NAME
local       6b4eab63d4f0deea08ab2168746074c6157c8d155c142941adc917eb81dc3cf2
local       26feb_mysql_data
local       26feb_wp_content
local       89c1e0cbb07833b376150098557229b3784fef38442e7c01d11008b2d89ffbe7
local       1088cdade2a34a2c1e180e66dc5ed0cb6426e10faa1e53ebc4bdaf02f35776ce5
local       964748bb3c30fbac8573d4a67dd24c4f4dd0530d9ed27abbd555f2734cc3db04
local       HarshitVolume
local       a334b7b3ce44032963db0d818a03b7ff81e2ba5ed9687fca3867f73fdeceda47
local       a539e11647af44f1247d13dc5abb2c6b5c315a05c705e3b35db0c4f238bc324e
local       assignment-1_pgdata
local       b7286e4626db91564f6ac6af9bcf8ed097a893744def09a5d98c7045de7dc433
local       ibm-session-jenkins_home
local       k3d-mycluster-images
local       mydata
local       myvolume
local       portainer_data
local       test
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker volume inspect mydata
[
  {
    "CreatedAt": "2026-04-17T14:24:25Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/mydata/_data",
    "Name": "mydata",
    "Options": null,
    "Scope": "local"
  }
]
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$

```

Step-4: Bind Mounts

```

# Create directory on host
mkdir ~/myapp-data

# Mount host directory to container
docker run -d -v ~/myapp-data:/app/data --name web3 nginx

# Add file on host
echo "From Host" > ~/myapp-data/host-file.txt

# Check in container
docker exec web3 cat /app/data/host-file.txt

# Shows: From Host

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ mkdir ~/myapp-data
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d -v ~/myapp-data:/app/data --name web3 nginx
b4a9df61ffa26fe22232add4c3177e24bfb408c95f288ef51ca874485e65d99a
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ echo "From Host" > ~/myapp-data/host-file.txt
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker exec web3 cat /app/data/host-file.txt
From Host
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$

```

Step-5:- Database with Persistent Storage

```

# MySQL with named volume
docker run -d \
  --name mysql-db \

```

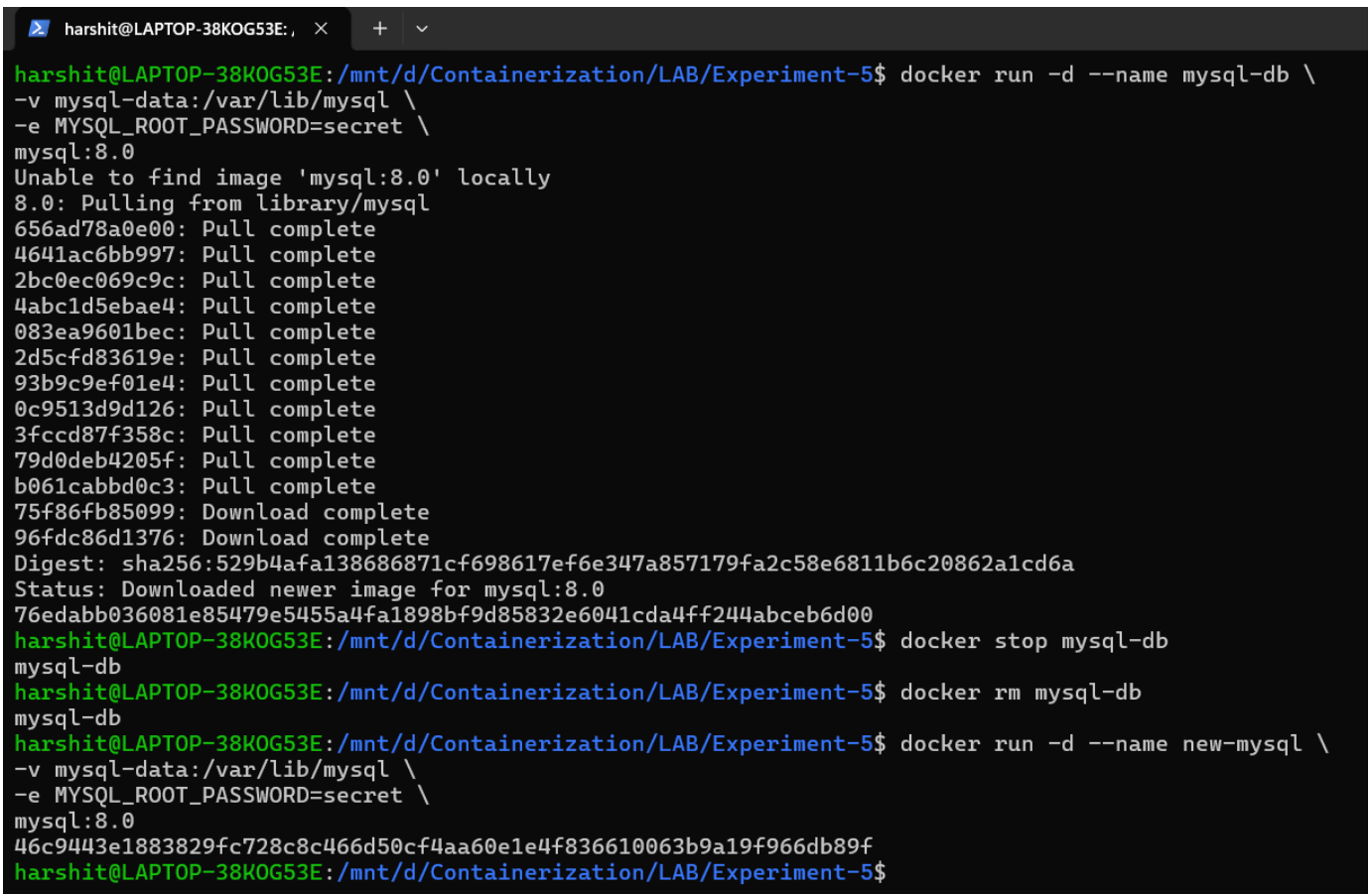
```

-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0

# Check data persists
docker stop mysql-db
docker rm mysql-db

# New container with same volume
docker run -d \
  --name new-mysql \
  -v mysql-data:/var/lib/mysql \
  -e MYSQL_ROOT_PASSWORD=secret \
  mysql:8.0
# Data is preserved!

```



```

harshit@LAPTOP-38KOG53E: ~$ docker run -d --name mysql-db \
-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
656ad78a0e00: Pull complete
4641ac6bb997: Pull complete
2bc0ec069c9c: Pull complete
4abc1d5ebae4: Pull complete
083ea9601bec: Pull complete
2d5cfd83619e: Pull complete
93b9c9ef01e4: Pull complete
0c9513d9d126: Pull complete
3fccd87f358c: Pull complete
79d0deb4205f: Pull complete
b061cabbdc3: Pull complete
75f86fb85099: Download complete
96fdc86d1376: Download complete
Digest: sha256:529b4afa138686871cf698617ef6e347a857179fa2c58e6811b6c20862a1cd6a
Status: Downloaded newer image for mysql:8.0
76edabb036081e85479e5455a4fa1898bf9d85832e6041cda4ff244abceb6d00
harshit@LAPTOP-38KOG53E: ~$ docker stop mysql-db
mysql-db
harshit@LAPTOP-38KOG53E: ~$ docker rm mysql-db
mysql-db
harshit@LAPTOP-38KOG53E: ~$ docker run -d --name new-mysql \
-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0
46c9443e1883829fc728c8c466d50cf4aa60e1e4f836610063b9a19f966db89f
harshit@LAPTOP-38KOG53E: ~$

```

Step-6: Environment Variable

```

docker run -d \
  --name app1 \
  -e DATABASE_URL="postgres://user:pass@db:5432/mydb" \
  -e DEBUG="true" \
  -p 3000:3000 \
  my-node-app

```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d --name app1 \
-e DATABASE_URL="postgres://user:pass@host:5432/db" \
-e DEBUG="true" \
-p 3000:80 nginx
306b6c0a89c1f00ffb51b354018587f34e6e6e568161e52816ec3279ddf362e1
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$
```

Step-7:- List .env Variables

```
echo "DATABASE_HOST=localhost" > .env
echo "DATABASE_PORT=5432" >> .env
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ echo "DATABASE_HOST=localhost" > .env
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ echo "DATABASE_PORT=5432" >> .env
```

Step-8: Run Container with .env

```
docker run -d \
--env-file .env \
--name app2 \
my-app
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d --env-file .env --name app2 nginx
87d1d2e93d298034c213c60bb46f03ff0b40648b76157ecb9f533e2e3405a285
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$
```

Step-9:- Dockerfile Env

```
nano Dockerfile
cat Dockerfile
```

Paste this:-

```
FROM python:3.9-slim
ENV PORT=5000
ENV DEBUG=false
CMD ["sleep", "3600"]
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ nano Dockerfile
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ cat Dockerfile
FROM python:3.9-slim
ENV PORT=5000
ENV DEBUG=false
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$
```

Step-10:- Build Flask App with the Dockerfile

```
docker build -t flask-app .
docker run -d --name flask-app -p 5000:5000 flask-app
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker build -t flask-app .
[+] Building 2.8s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 88B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/1] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acd1e13cf1731b1b
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:35769b8958194e1f25c254e835986fadcead31cf85fe0692390db3ebd94de536
=> => exporting config sha256:cbe583556e930e145836d3297c57eaf0790d00a4b85f1762cf335822ae93a428
=> => exporting attestation manifest sha256:ff6a7f84b0217496ac04d9203c449f0691e3b97fa3db43b92e249ca2cd4f7059
=> => exporting manifest list sha256:5a01b7bcdacd1bd1fcdc7602d47358e89d71d330ed794c5e5270dc24a0bd4f0
=> => naming to docker.io/library/flask-app:latest
=> => unpacking to docker.io/library/flask-app:latest
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d --name flask-app \
-p 5000:5000 \
-e DEBUG=true \
flask-app
2e4d7dc711dd49658482fd13fc21a84d842266b28df76b9db9ce1b59bb4993d0
```

Step-11:- Monitoring

```
docker stats --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
5b5ddbd5eea9	flask-app	0.00%	496KiB / 7.581GiB	0.01%	872B / 126B	123kB / 0B	1
87d1d2e93d29	app2	0.00%	10.41MiB / 7.581GiB	0.13%	1.08kB / 126B	0B / 12.3kB	13
306b6c0a89c1	app1	0.00%	10.5MiB / 7.581GiB	0.14%	1.21kB / 126B	0B / 12.3kB	13
b4a9df61ffa2	web3	0.00%	10.46MiB / 7.581GiB	0.13%	1.59kB / 126B	0B / 12.3kB	13
717c0d48c420	web2	0.00%	10.37MiB / 7.581GiB	0.13%	1.71kB / 126B	0B / 12.3kB	13
ad0f99a14637	web1	0.00%	10.5MiB / 7.581GiB	0.14%	1.84kB / 126B	0B / 12.3kB	13
939a38e8302e	test-container	0.00%	928KiB / 7.581GiB	0.01%	2.26kB / 126B	0B / 0B	1
1e1af5410829	node_backend	0.00%	16.59MiB / 7.581GiB	0.21%	1.45kB / 1.14kB	55.7MB / 0B	11
5576c99e1f84	postgres_db	0.00%	32.57MiB / 7.581GiB	0.42%	1.32kB / 1.31kB	27.7MB / 303kB	6
d8018398f51c	k3d-mycluster-serverlb	0.00%	28.06MiB / 7.581GiB	0.36%	370kB / 366kB	22.4MB / 16.4kB	23
5e303da50cdb	k3d-mycluster-server-0	17.07%	671.6MiB / 7.581GiB	8.65%	516kB / 391kB	462MB / 40MB	193

Step-12: Monitor

```
docker stats --no-stream
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker stats --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
5b5ddbd5eea9	flask-app	0.00%	496KiB / 7.581GiB	0.01%	872B / 126B	123kB / 0B	1
87d1d2e93d29	app2	0.00%	10.41MiB / 7.581GiB	0.13%	1.08kB / 126B	0B / 12.3kB	13
306b6c0a89c1	app1	0.00%	10.5MiB / 7.581GiB	0.14%	1.21kB / 126B	0B / 12.3kB	13
b4a9df61ffa2	web3	0.00%	10.46MiB / 7.581GiB	0.13%	1.59kB / 126B	0B / 12.3kB	13
717c0d48c420	web2	0.00%	10.37MiB / 7.581GiB	0.13%	1.71kB / 126B	0B / 12.3kB	13
ad0f99a14637	web1	0.00%	10.5MiB / 7.581GiB	0.14%	1.84kB / 126B	0B / 12.3kB	13
939a38e8302e	test-container	0.00%	928KiB / 7.581GiB	0.01%	2.26kB / 126B	0B / 0B	1
1e1af5410829	node_backend	0.00%	16.59MiB / 7.581GiB	0.21%	1.45kB / 1.14kB	55.7MB / 0B	11
5576c99e1f84	postgres_db	6.02%	32.58MiB / 7.581GiB	0.42%	1.32kB / 1.31kB	27.7MB / 303kB	6
d8018398f51c	k3d-mycluster-serverlb	0.00%	28.06MiB / 7.581GiB	0.36%	371kB / 368kB	22.4MB / 16.4kB	23
5e303da50cdb	k3d-mycluster-server-0	13.88%	670.5MiB / 7.581GiB	8.64%	518kB / 392kB	462MB / 40.4MB	193

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$
```

Step-13:- Get Stats via Top

```
docker top flask-app
```



```
harshit@LAPTOP-38KOG53E: ~$ docker top flask-app
UID          PID          PPID         C           STIME        TTY          TIME         CMD
root         8792         8768         0           14:40        ?            00:00:00     sleep 3600
harshit@LAPTOP-38KOG53E: ~$
```

Step-14:- Inspect Container details

docker inspect flask-app

```
harshit@LAPTOP-38KOG53E: ~$ docker inspect flask-app
[
  {
    "Id": "5b5ddbd5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bf1d6",
    "Created": "2026-04-17T14:40:25.90846018Z",
    "Path": "sleep",
    "Args": [
      "3600"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 8792,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-04-17T14:40:26.169286351Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:5a01b7bcdacd1bd1fcdc7602d47358e89d71d330ed794c5e5270dc24a0bdf4f0",
    "ResolvConfPath": "/var/lib/docker/containers/5b5ddbd5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bf1d6/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/5b5ddbd5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bf1d6/hostname",
    "HostsPath": "/var/lib/docker/containers/5b5ddbd5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bf1d6/hosts",
    "LogPath": "/var/lib/docker/containers/5b5ddbd5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bf1d6/5b5ddbd5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bf1d6-json.log",
    "Name": "/flask-app",
    "RestartCount": 0,
    "Driver": "overlayfs",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",

```

Step-15: Inspect Container Status Only

docker inspect --format='{{.State.Status}}' flask-app

```
harshit@LAPTOP-38KOG53E: ~$ docker inspect --format='{{.State.Status}}' flask-app
running
harshit@LAPTOP-38KOG53E: ~$
```

Step-16:- View Events

docker events

```
harshit@LAPTOP-38KOG53E: ~$ docker events
2026-04-17T14:46:52.582016800Z container exec_create: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=8e758199b125a5d921f911253d7974cc438503b3b8bcb9a2d39c0134cd94a97, image=assignment-1-database, name=postgres_db)
2026-04-17T14:46:52.582635800Z container exec_start: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=8e758199b125a5d921f911253d7974cc438503b3b8bcb9a2d39c0134cd94a97, image=assignment-1-database, name=postgres_db)
2026-04-17T14:46:52.726667000Z container exec_die 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=8e758199b125a5d921f911253d7974cc438503b3b8bcb9a2d39c0134cd94a97, exitCode=0, image=assignment-1-database, name=postgres_db)
```

Step-17:- View Conatiner Events Only

```
docker events --filter 'type=container'
```

```
harshit@LAPTOP-38KOG53E: ~$ docker events --filter 'type=container'
2026-04-17T14:47:12.864352300Z container exec_create: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=a54b514bfcf789947f43dd57288903644af1646a182e15e8b29a4f1be264ae27, image=assignment-1-database, name=postgres_db)
2026-04-17T14:47:12.864743800Z container exec_start: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=a54b514bfcf789947f43dd57288903644af1646a182e15e8b29a4f1be264ae27, image=assignment-1-database, name=postgres_db)
2026-04-17T14:47:12.983155700Z container exec_die 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=a54b514bfcf789947f43dd57288903644af1646a182e15e8b29a4f1be264ae27, exitCode=0, image=assignment-1-database, name=postgres_db)
```

Step-18:- Create monitor.sh

```
nano monitor.sh
cat monitor.sh
```

Paste This:-

```
#!/bin/bash

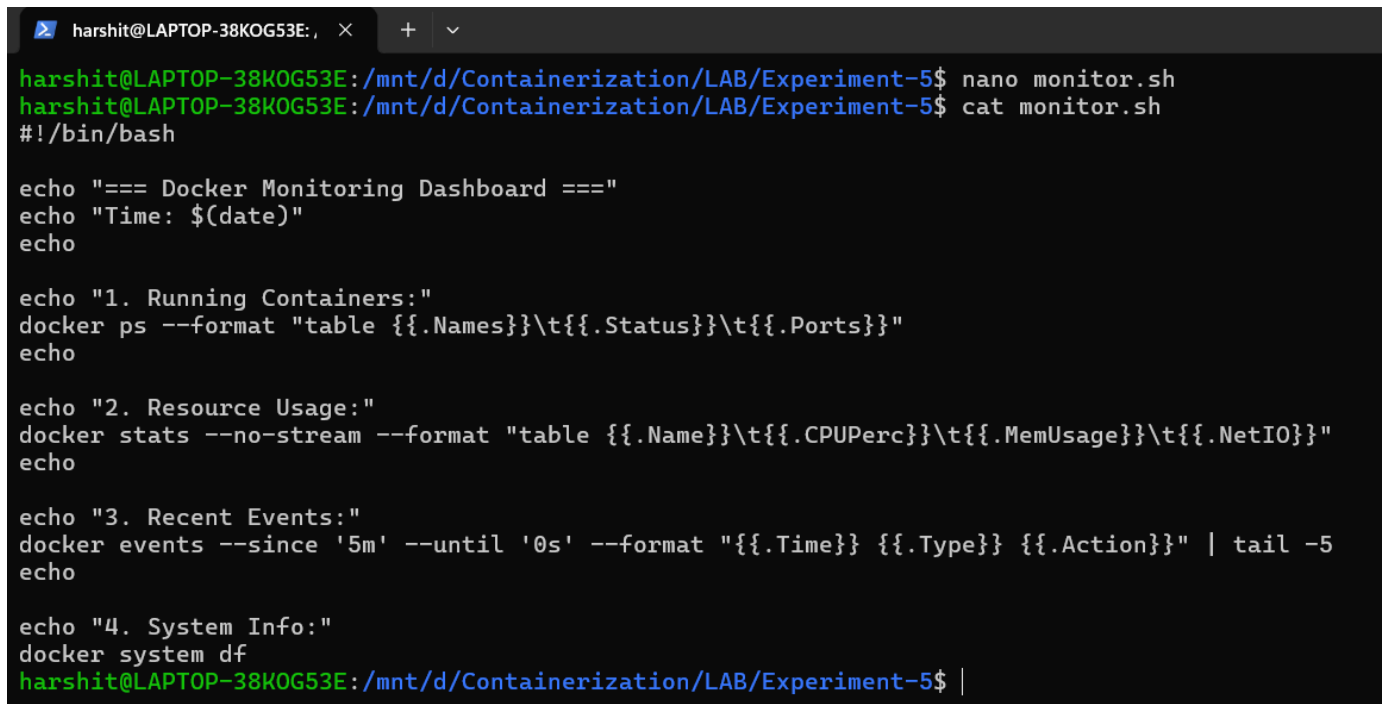
echo "=== Docker Monitoring Dashboard ==="
echo "Time: $(date)"
echo

echo "1. Running Containers:"
docker ps --format "table \t\t"
echo

echo "2. Resource Usage:"
docker stats --no-stream --format "table \t\t\t\t"
echo
```

```
echo "3. Recent Events:"
docker events --since '5m' --until '0s' --format " " | tail -5
echo

echo "4. System Info:"
docker system df
```

A terminal window titled 'harshit@LAPTOP-38KOG53E:' shows the user creating a file named 'monitor.sh' using 'nano monitor.sh'. The file's content is displayed, showing a script that prints a dashboard header, current time, and four sections of Docker-related information: running containers, resource usage, recent events, and system disk usage. The script uses 'docker ps', 'docker stats', 'docker events', and 'docker system df' commands with various flags and formatting. The terminal shows the user pressing Ctrl+X to exit nano, then running 'cat monitor.sh' to view the script's content, and finally running 'bash monitor.sh' to execute it.

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ nano monitor.sh
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ cat monitor.sh
#!/bin/bash

echo "=== Docker Monitoring Dashboard ==="
echo "Time: $(date)"
echo

echo "1. Running Containers:"
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
echo

echo "2. Resource Usage:"
docker stats --no-stream --format "table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}\t{{.NetIO}}"
echo

echo "3. Recent Events:"
docker events --since '5m' --until '0s' --format "{{.Time}} {{.Type}} {{.Action}}" | tail -5
echo

echo "4. System Info:"
docker system df
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ |
```

Step-19:- bash monitor.sh

```
bash monitor.sh
```

```

harshit@LAPTOP-38KOG53E: , × + ▾
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ bash monitor.sh
=== Docker Monitoring Dashboard ===
Time: Fri Apr 17 14:49:42 UTC 2026

1. Running Containers:
NAMES                STATUS                PORTS
flask-app             Up 9 minutes         0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
app2                  Up 18 minutes        80/tcp
app1                  Up 20 minutes        80/tcp
web3                  Up 24 minutes        80/tcp
web2                  Up 24 minutes        80/tcp
web1                  Up 27 minutes        80/tcp
test-container        Up 29 minutes
node_backend          Up 35 minutes        0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
postgres_db           Up 35 minutes (healthy)
k3d-mycluster-serverlb Up 35 minutes        0.0.0.0:42325->6443/tcp
k3d-mycluster-server-0 Up 35 minutes

2. Resource Usage:
NAME                CPU %    MEM USAGE / LIMIT    NET I/O
flask-app           0.00%    496KiB / 7.581GiB     872B / 126B
app2                 0.00%    10.41MiB / 7.581GiB   1.08kB / 126B
app1                 0.00%    10.5MiB / 7.581GiB    1.21kB / 126B
web3                 0.00%    10.46MiB / 7.581GiB   1.59kB / 126B
web2                 0.00%    10.37MiB / 7.581GiB   1.71kB / 126B
web1                 0.00%    10.5MiB / 7.581GiB    1.84kB / 126B
test-container       0.00%    928KiB / 7.581GiB     2.26kB / 126B
node_backend         0.00%    16.59MiB / 7.581GiB   1.45kB / 1.14kB
postgres_db          0.00%    32.6MiB / 7.581GiB    1.32kB / 1.31kB
k3d-mycluster-serverlb 0.00%    28.06MiB / 7.581GiB   428kB / 425kB
k3d-mycluster-server-0 17.83%    673.3MiB / 7.581GiB   575kB / 449kB

3. Recent Events:
1776437374 container exec_start: /bin/sh -c pg_isready -U admin
1776437375 container exec_die
1776437385 container exec_create: /bin/sh -c pg_isready -U admin
1776437385 container exec_start: /bin/sh -c pg_isready -U admin
1776437385 container exec_die

4. System Info:
TYPE                TOTAL    ACTIVE    SIZE    RECLAIMABLE

```

Step-20:- System Info

docker system df

```

4. System Info:
TYPE                TOTAL    ACTIVE    SIZE    RECLAIMABLE
Images              11       9        4.652GB  1.251GB (26%)
Containers          13      11        6.074MB  5.034MB (82%)
Local Volumes       18      10        1.579GB  526.1kB (0%)
Build Cache         103       0        1.25GB   1.25GB
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$

```

Step-21:- List Networks

docker networks ls

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
96de3e6643cc        assignment-1_app_net bridge              local
4ad7bd51fdb9         bridge              bridge              local
23e543289f53         docker_gwbridge     bridge              local
9bb2c2124ca6         host                host                local
4a4109b7c2bd         ibm-session_default bridge              local
ydkad0orrrnos        ingress             overlay             swarm
521d8e59c054         ipvlan_net          ipvlan              local
cb24d34877ba         k3d-mycluster       bridge              local
7982c22f2cd0         none                null                local
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$

```

Step-22:- Remove Network

```
docker network rm my-network
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network create my-network
docker run -d --name web1 --network my-network nginx
docker run -d --name web2 --network my-network nginx
docker exec web1 curl http://web2
dae1669124f718c2cd7838c9ab97a002efb170c3f5ab74f2afed2366cfb35cd4
docker: Error response from daemon: Conflict. The container name "/web1" is already in use by container "ad0f99a146371ebf2046156df71643d6239dfd1c6deaa282989e91edf55993e5". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
docker: Error response from daemon: Conflict. The container name "/web2" is already in use by container "717c0d48c420cb1f98b9d7ec703a839b5ea1a1a68ebe6e0ab68a2f302f00ab47". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %             Dload  Upload    Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  --:--:--  0:00:07  --:--:--    0

```

Step-23: Create network

```
docker network create my-network
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network create my-network
f3f6952dbdd2ec03ef154923b8228464d532ec4c196cda26d59453a37be63758
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$

```

Step-24:- Run Containers with Bridge Network

```

docker rm -f web1 web2
docker run -d --name web1 --network my-network nginx
docker run -d --name web2 --network my-network nginx

docker exec web1 curl http://web2

```

```

harshit@LAPTOP-38KOG53E: ~$ docker network create my-network
f3f6952dbdd2ec03ef154923b8228464d532ec4c196cda26d59453a37be63758
harshit@LAPTOP-38KOG53E: ~$ docker rm -f web1 web2
web1
web2
harshit@LAPTOP-38KOG53E: ~$ docker run -d --name web1 --network my-network nginx
b79949a4e27b824ff890db42fe7bd5d26df2c2967b36425122c0fb547d6825a9
harshit@LAPTOP-38KOG53E: ~$ docker run -d --name web2 --network my-network nginx
b3b1f63f61c2fb76a938d2d840a67f3992c1031f08cc854b8b2b7f75f1b8fafb
harshit@LAPTOP-38KOG53E: ~$ docker exec web1 curl http://web2
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0    0    0    0    0    0     0      0  --:--:-- --:--:-- --:--:--    0!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
100 896 100 896 0 0 97805 0 --:--:-- --:--:-- --:--:-- 109k
harshit@LAPTOP-38KOG53E: ~$

```

Step-25:- Run nginx Container

```

docker run -d --network host nginx
curl https://localhost

```

```

harshit@LAPTOP-38KOG53E: ~$ docker run -d --network host nginx
9a81f1ea70ba28d66f9299af1918a1321791320c7fc892a5296c9535ad536e40
harshit@LAPTOP-38KOG53E: ~$ curl http://localhost
<h1>Hello from NGINX</h1>
harshit@LAPTOP-38KOG53E: ~$

```

Step-26:- Run alpine Container

```

docker run -d --network none alpine sleep 3600

```

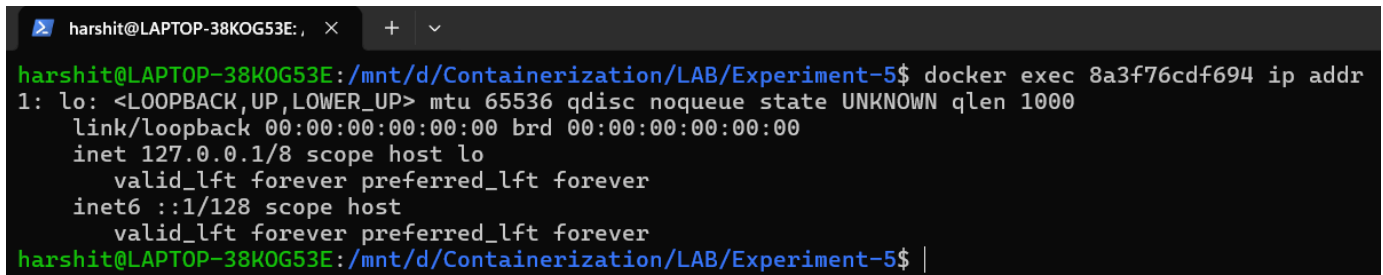
```

harshit@LAPTOP-38KOG53E: ~$ docker run -d --network none alpine sleep 3600
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
6a0ac1617861: Pull complete
4e0cb4ac63e7: Download complete
a63813f19ccb: Download complete
Digest: sha256:5b10f432ef3da1b8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11
Status: Downloaded newer image for alpine:latest
8a3f76cdf6949d9d7662e5a6cd3bac878a30221a358db418a21151d5f4049584

```

Step-27:- Get Ip Address of Container

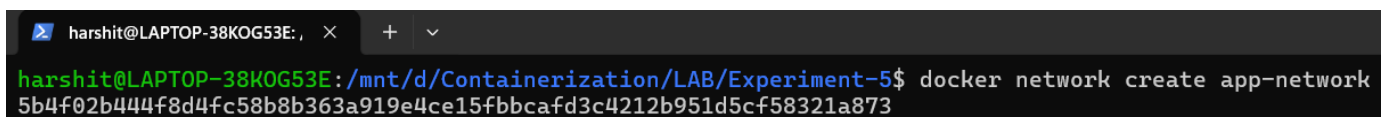
```
docker exec <Container ID> ip addr
```

A terminal window with a dark background. The prompt is 'harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5\$'. The command 'docker exec 8a3f76cdf694 ip addr' has been executed. The output shows details for the loopback interface 'lo', including its state, MTU, and IP address (127.0.0.1).

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ docker exec 8a3f76cdf694 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ |
```

Step-28:- Create a Network

```
docker network create app-network
```

A terminal window with a dark background. The prompt is 'harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5\$'. The command 'docker network create app-network' has been executed. The output is a long alphanumeric string representing the network ID.

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ docker network create app-network
5b4f02b444f8d4fc58b8b363a919e4ce15fbbcafd3c4212b951d5cf58321a873
```

Step-29:- Connect network with Container

```
docker network connect app-network web1
docker network connect app-network web2
docker network connect app-network flask-app
docker network inspect app-network
```



```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network connect app-network web1
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network connect app-network web2
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network connect app-network flask-app
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network inspect app-network
[
  {
    "Name": "app-network",
    "Id": "5b4f02b444f8d4fc58b8b363a919e4ce15fbbcafd3c4212b951d5cf58321a873",
    "Created": "2026-04-17T14:59:24.900412745Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.23.0.0/16",
          "IPRange": "",
          "Gateway": "172.23.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {},
    "Containers": {
      "5b5ddb5eea9c56fb99d3f82755eb518de9ab22021f1fb72a1a2bfc9286bfd6": {
        "Name": "flask-app",
        "EndpointID": "ed794a1d99f586f98eb25d0a7317bc5387957f24f4767356d8ac25044902a0cc",
        "MacAddress": "ee:07:0a:c0:73:b6",

```

Step-30:- curl in Container

```
docker exec web1 curl http://web2
```



```

harshit@LAPTOP-38KOG53E: ~$ docker exec web1 curl http://web2
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100    896    100    896     0     0   176k      0  --:--:-- --:--:-- --:--:--   218k
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ |

```

Step-31:- Disconnect Network

```

docker network disconnect app-network web1
docker network disconnect app-network web2
docker network disconnect app-network flask-app
docker network rm app-network

```

```

harshit@LAPTOP-38KOG53E: ~$ docker network disconnect app-network web1
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network disconnect app-network web2
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network disconnect app-network flask-app
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker network rm app-network
app-network
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ |

```

Step-32:- Create a new Network

```

docker network create my-app-network

```

```

harshit@LAPTOP-38KOG53E: ~$ docker network create myapp-network
0679302a3979d0b0d793923f305270ff181cbb8039f102b1d8848f53e9504db
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ |

```

Step-33:- Run postgres Container

```
docker run -d --name postgres \
--network myapp-network \
-e POSTGRES_PASSWORD=secret \
postgres:15
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d --name postgres \
--network myapp-network \
-e POSTGRES_PASSWORD=secret \
-v postgres-data:/var/lib/postgresql/data \
postgres:15
Unable to find image 'postgres:15' locally
15: Pulling from library/postgres
a8a0725a32a6: Pull complete
af2a6e5f285c: Pull complete
b4765ac2e7e6: Pull complete
f7049a93e93f: Pull complete
2c38b75ffca2: Pull complete
9328c154beae: Pull complete
c45b346571fc: Pull complete
5ca81a59b0a4: Pull complete
af0c3ea2fe5e: Pull complete
5c152148754e: Pull complete
71a32c7571f4: Pull complete
8ecdbd190da6: Pull complete
00f33d580dc3: Pull complete
cebcba07837b: Download complete
b9015a4f6b87: Download complete
Digest: sha256:3e43515057e113ee741fca2f621f15300be34a6d4a8dfcf2e20651288c8272f3
Status: Downloaded newer image for postgres:15
99fcd73214c9ffcc88f2eee43d9c812bd22ba88d584fda73d217a5a0d4a6080e
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ |
```

Step-34:- Run a redis Container

```
docker run -d --name redis \
--network myapp-network \
redis:7-alpine
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ docker run -d --name redis \
--network myapp-network \
-v redis-data:/data \
redis:7-alpine
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
6447bc944d85: Pull complete
1f0676652ae4: Pull complete
4f4fb700ef54: Pull complete
9572acab3f32: Pull complete
2da9501c87e4: Pull complete
51bf34b872a6: Pull complete
47836b8a3274: Pull complete
897d797d2723: Pull complete
2074d4e4f1bb: Download complete
c3c3ab8d6b01: Download complete
Digest: sha256:efc27681bf81e6ed55cb581718b7818c6c04c39b1f9aa40854edd3ea2241b6ed
Status: Downloaded newer image for redis:7-alpine
bf5a42795d7e418fd4bc2a08afd709c40e95fc69ea0f6716803b8c3dda6f73cb
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-5$ |
```

Step-35:- Run a flask app with the previous container

```
docker run -d --name flask-app \
--network myapp-network \
-p 5000:5000 \
-e DATABASE_URL="postgres://postgres:secret@postgres:5432/mydb" \
-e REDIS_URL="redis://redis:6379" \
flask-app:latest
e69d7714e36b4b3ff09adaa0b51f820019e7ef9ab59b512db6c2c20611cfff3df
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ docker run -d --name flask-app \
--network myapp-network \
-p 5000:5000 \
-e DATABASE_URL="postgres://postgres:secret@postgres:5432/mydb" \
-e REDIS_URL="redis://redis:6379" \
flask-app:latest
e69d7714e36b4b3ff09adaa0b51f820019e7ef9ab59b512db6c2c20611cfff3df
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$
```

Step-36:- List Containers

```
docker ps
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
bf5a42795d7e	redis:7-alpine	"docker-entrypoint.s..."	5 minutes ago	Up 4 minutes	6379/tcp
99fcd73214c9	postgres:15	"docker-entrypoint.s..."	5 minutes ago	Up 5 minutes	5432/tcp
8a3f76cdf694	alpine	"sleep 3600"	52 minutes ago	Up 52 minutes	
9a81f1ea70ba	clever_solomon	"/docker-entrypoint..."	53 minutes ago	Up 53 minutes	
b3b1f63f61c2	nginx	"/docker-entrypoint..."	54 minutes ago	Up 54 minutes	80/tcp
b79949a4e27b	web2	"/docker-entrypoint..."	54 minutes ago	Up 54 minutes	80/tcp
87d1d2e93d29	nginx	"/docker-entrypoint..."	About an hour ago	Up About an hour	80/tcp
306b6c0a89c1	app2	"/docker-entrypoint..."	About an hour ago	Up About an hour	80/tcp
b4a9df61ffa2	nginx	"/docker-entrypoint..."	About an hour ago	Up About an hour	80/tcp
939a38e8302e	web3	"/docker-entrypoint..."	About an hour ago	Up About an hour	80/tcp
1e1af5410829	ubuntu	"bash"	2 hours ago	Up About an hour	
5576c99e1f84	test-container	"docker-entrypoint.s..."	2 weeks ago	Up 2 hours	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
d8018398f51c	assignment-1-backend	"docker-entrypoint.s..."	2 weeks ago	Up 2 hours (healthy)	
5e303da50cdb	node_backend	"docker-entrypoint.s..."	2 weeks ago	Up 2 hours	0.0.0.0:42325->6443/tcp
e69d7714e36b	assignment-1-database	"docker-entrypoint.s..."	4 weeks ago	Up 2 hours	
	postgres_db	"bin/sh -c nginx-pr..."	4 weeks ago	Up 2 hours	
	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/k3d-entrypoint..."	4 weeks ago	Up 2 hours	
	k3d-mycluster-serverlb				
	ancher/k3s:v1.31.5-k3s1				
	k3d-mycluster-server-0				

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$
```

Step-37:- Monitor

```
docker stats --no-stream
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-5$ docker stats --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
99fcd73214c9	postgres	0.00%	31.47MiB / 7.581GiB	0.41%	1.87kB / 126B	0B / 41.6MB	6
bf5a42795d7e	redis	1.00%	3.539MiB / 7.581GiB	0.05%	914B / 126B	0B / 0B	6
e69d7714e36b	flask-app	0.00%	0B / 0B	0.00%	0B / 0B	0B / 0B	0

Step-38:- Ping inside container

```
docker exec -it flask-app bash
apt update
```

```
apt install -y iputils-ping
```

```
harshit@LAPTOP-38KOG53E: ~$ docker exec -it flask-app bash
root@4c834012398a:/# apt update
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [43.4 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [123 kB]
Fetched 10.0 MB in 4s (2387 kB/s)
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Repository 'http://deb.debian.org/debian trixie InRelease' changed its 'Version' value from '13.1' to '13.4'
root@4c834012398a:/# apt install -y iputils-ping
Installing:
  iputils-ping

Installing dependencies:
  libidn2-0 libunistring5 linux-sysctl-defaults

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 14
  Download size: 643 kB
  Space needed: 2810 kB / 1017 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 libunistring5 amd64 1.3-2 [477 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libidn2-0 amd64 2.3.8-2 [109 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 iputils-ping amd64 3:20240905-3 [51.2 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 linux-sysctl-defaults all 4.12.1 [5724 B]
Fetched 643 kB in 0s (2455 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 4.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 4.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package libunistring5:amd64.
(Reading database ... 5644 files and directories currently installed.)
Preparing to unpack .../libunistring5_1.3-2_amd64.deb ...
Unpacking libunistring5:amd64 (1.3-2) ...
```

Step-39:- Ping postgres Container

```
ping -c 2 postgres
```

```
root@4c834012398a:/# ping -c 2 postgres
PING postgres (172.23.0.2) 56(84) bytes of data.
64 bytes from postgres.myapp-network (172.23.0.2): icmp_seq=1 ttl=64 time=2.62 ms
64 bytes from postgres.myapp-network (172.23.0.2): icmp_seq=2 ttl=64 time=0.457 ms

--- postgres ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.457/1.536/2.615/1.079 ms
```

Step-40:- Ping redis Container

```
ping -c 2 redis
```

```
root@4c834012398a:/# ping -c 2 redis
PING redis (172.23.0.3) 56(84) bytes of data.
64 bytes from redis.myapp-network (172.23.0.3): icmp_seq=1 ttl=64 time=0.451 ms
64 bytes from redis.myapp-network (172.23.0.3): icmp_seq=2 ttl=64 time=0.112 ms

--- redis ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.112/0.281/0.451/0.169 ms
root@4c834012398a:/# |
```

Step-41:- Inspect Network

```
docker network inspect myapp-network
```

```
root@4c834012398a:/# ping -c 2 redis
PING redis (172.23.0.3) 56(84) bytes of data.
64 bytes from redis.myapp-network (172.23.0.3): icmp_seq=1 ttl=64 time=0.451 ms
64 bytes from redis.myapp-network (172.23.0.3): icmp_seq=2 ttl=64 time=0.112 ms

--- redis ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.112/0.281/0.451/0.169 ms
root@4c834012398a:/# |
```

Conclusion

Key Takeaways

1. **Volumes** persist data beyond container lifecycle
2. **Environment variables** configure containers dynamically
3. **Monitoring commands** help debug and optimize containers
4. **Networks** enable secure container communication
5. **Always use named volumes** for production data
6. **Custom networks** provide better isolation and DNS
7. **Monitor resource usage** to prevent issues
8. **Use .env files** for sensitive configuration

This experiment covers essential Docker features for building, configuring, and managing production-ready containerized applications.

[< Previous](#) [Home](#) [Next >](#)

Harshit-Kumar-Containerization

Experiment 6: Comparison of Docker Run and Docker Compose

Introduction

Docker Run (Imperative Approach)

The `docker run` command is used to create and start a container from an image. It requires explicit flags for:

- Port mapping (`-p`)
- Volume mounting (`-v`)
- Environment variables (`-e`)
- Network configuration (`--network`)
- Restart policies (`--restart`)
- Resource limits (`--memory` , `--cpus`)
- Container name (`--name`)

This approach is **imperative**, meaning you specify step-by-step instructions.

Docker Compose (Declarative Approach)

Docker Compose uses a YAML file (`docker-compose.yml`) to define services, networks, and volumes in a structured format.

Instead of multiple `docker run` commands, a single command is used:

```
docker compose up -d
```

Compose is **declarative**, meaning you define the desired state of the application.

Mapping: Docker Run vs Docker Compose

Docker Run Flag	Docker Compose Equivalent
<code>-p 8080:80</code>	<code>ports:</code>

Docker Run Flag	Docker Compose Equivalent
-v host:container	volumes:
-e KEY=value	environment:
--name	container_name:
--network	networks:
--restart	restart:
--memory	deploy.resources.limits.memory
--cpus	deploy.resources.limits.cpus
-d	docker compose up -d

Advantages of Docker Compose

1. Simplifies multi-container applications
2. Provides reproducibility
3. Version controllable configuration
4. Unified lifecycle management
5. Supports service scaling

Hands-On

Step-1:- Run nginx Container

```
docker run -d \  
--name lab-nginx \  
-p 8081:80 \  
-v $(pwd)/html:/usr/share/nginx/html \  
nginx:alpine  
  
docker ps
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker run -d \
--name lab-nginx \
-p 8081:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
612c0c1df4c5: Pull complete
82736a35d0e7: Pull complete
781ff50d2644: Pull complete
453da7dbc73e: Pull complete
4a8b0b2a5b19: Pull complete
aee4e54b3865: Pull complete
583599bb7d38: Pull complete
6192e1e6a438: Download complete
e8fc446e336c: Download complete
Digest: sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
Status: Downloaded newer image for nginx:alpine
6076f48cc0f97ddce83ecf30787a81064e1f09a4c8077203caf72a1b5f0beb3d
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
6076f48cc0f9   nginx:alpine   "/docker-entrypoint..." 2 seconds ago  Up 2 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80/t
cp            lab-nginx
1e1af5410829   assignment-1-backend  "docker-entrypoint.s..." 2 weeks ago   Up About a minute  0.0.0.0:3000->3000/tcp, [::]:3000->30
00/tcp       node_backend
5576c99e1f84   assignment-1-database  "docker-entrypoint.s..." 2 weeks ago   Up About a minute (healthy)
postgres_db
d8018398f51c   ghcr.io/k3d-io/k3d-proxy:5.8.3  "/bin/sh -c nginx-pr..." 4 weeks ago   Up About a minute  0.0.0.0:42325->6443/tcp
k3d-mycluster-serverlb
5e303da50cdb   rancher/k3s:v1.31.5-k3s1  "/bin/k3d-entrypoint..." 4 weeks ago   Up About a minute
k3d-mycluster-server-0
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$

```

Step-2:- Create a file index.html mounting on a container

```

nano html/index.html
cat html/index.html
docker stop lab-nginx
docker rm lab-nginx
docker run -d \
--name lab-nginx \
-p 8081:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx:alpine

```

index.html :-

```

<h1>Experiment 6 Working</h1>
<p>Nginx is running successfully</p>

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ nano html/index.html
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ cat html/index.html
<h1>Experiment 6 Working</h1>
<p>Nginx is running successfully</p>
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker stop lab-nginx
lab-nginx
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker rm lab-nginx
lab-nginx
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker run -d \
--name lab-nginx \
-p 8081:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx:alpine
d9a6e8fa3dbfb0f897c65105dd1e2167a27bfd0d177949c021cd9e6efbdd9d81
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ |

```


Step-3:- View Page on Browser



Experiment 6 Working

Nginx is running successfully

Step-4:- Create same via compose

```
nano docker-compose.yml
cat docker-compose.yml
```

docker-compose.yml :-

```
version: '3.8'

services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```

```

harshit@LAPTOP-38KOG53E: ~$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ cat docker-compose.yml
version: '3.8'

services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ |

```

Step-5:- Create Containers via Compose

```

docker-compose up -d
docker-compose ps

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker compose up -d
WARN[0000] /mnt/d/Containerization/LAB/Experiment-6/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid p
otential confusion
WARN[0000] No services to build
[+] up 2/2
✔Network experiment-6_default Created 0.1s
✔Container lab-nginx Created 0.2s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker compose ps
WARN[0000] /mnt/d/Containerization/LAB/Experiment-6/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid p
otential confusion

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
lab-nginx	nginx:alpine	"/docker-entrypoint..."	nginx	8 seconds ago	Up 8 seconds	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ |

```

Step-6:- Delete Compose Services

```

docker-compose down

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker compose down
WARN[0000] /mnt/d/Containerization/LAB/Experiment-6/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid p
otential confusion
[+] down 2/2
✔Container lab-nginx Removed 0.5s
✔Network experiment-6_default Removed 0.3s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ |

```

Step-7:- Create Network

```

docker network create wp-net

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ docker network create wp-net
e4a951b43b886074d8b9c7eed0d91c95a939219d1bf6cbd649cf50ae75e039b9
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ |

```

Step-8:- Run mysql container

```

docker run -d \
--name mysql \
--network wp-net \

```

```
-e MYSQL_ROOT_PASSWORD=secret \
-e MYSQL_DATABASE=wordpress \
mysql:5.7
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6$ docker run -d \
--name mysql \
--network wp-net \
-e MYSQL_ROOT_PASSWORD=secret \
-e MYSQL_DATABASE=wordpress \
mysql:5.7
Unable to find image 'mysql:5.7' locally
5.7: Pulling from library/mysql
df9a4d85569b: Pull complete
20e4dcae4c69: Pull complete
1c56c3d4ce74: Pull complete
90986bb8de6e: Pull complete
e9f03a1c24ce: Pull complete
68c3898c2015: Pull complete
6b95a940e7b6: Pull complete
ffc89e9dfd88: Pull complete
064b2d298fba: Pull complete
ae71319cb779: Pull complete
43d05e938198: Pull complete
9eb6a52b8441: Download complete
96889c47a44a: Download complete
Digest: sha256:4bc6bc963e6d8443453676cae56536f4b8156d78bae03c0145cbe47c2aad73bb
Status: Downloaded newer image for mysql:5.7
6b8fee5a696426d2bb79621581f4fb7d0fb5b5abfed3468fb0cc567530ff419d
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6$ |
```

Step-9:- Run wordpress container

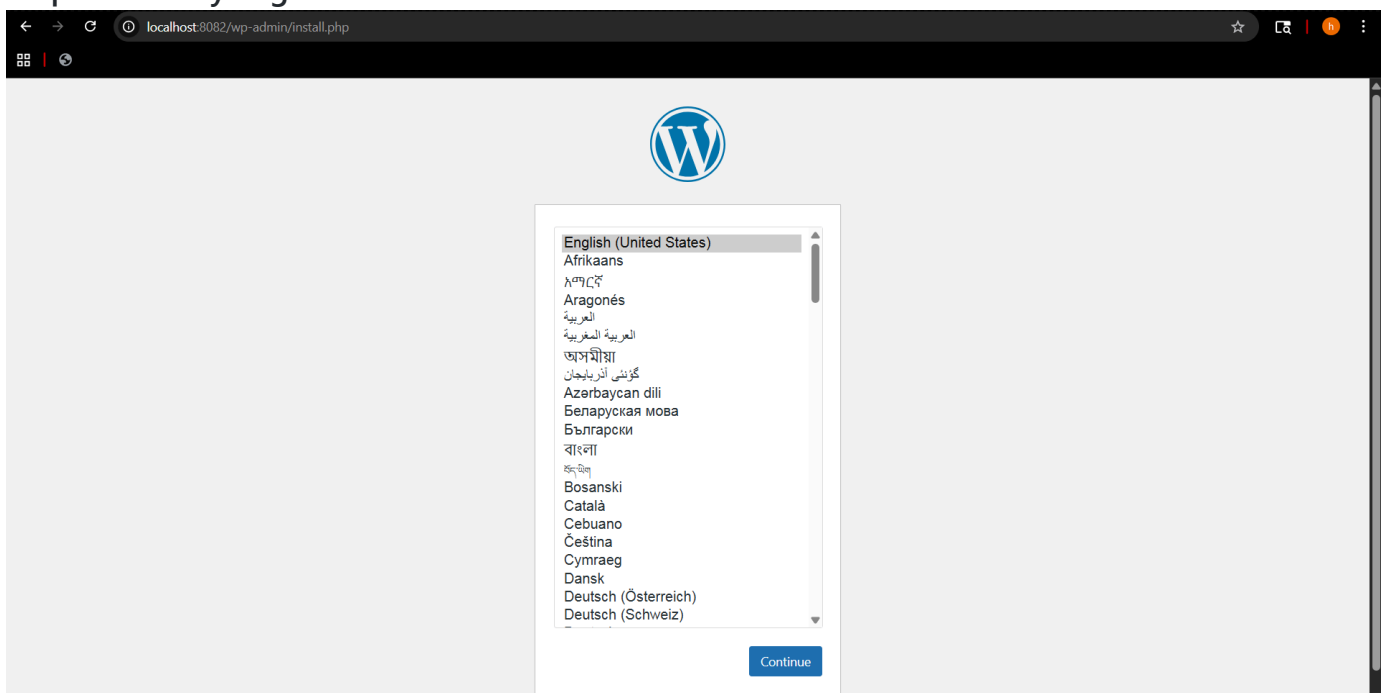
```
docker run -d \
--name wordpress \
--network wp-net \
-p 8082:80 \
-e WORDPRESS_DB_HOST=mysql \
-e WORDPRESS_DB_USER=root \
-e WORDPRESS_DB_PASSWORD=secret \
-e WORDPRESS_DB_NAME=wordpress \
wordpress:latest
```

```

harshit@LAPTOP-38KOG53E: ~$ docker run -d \
--name wordpress \
--network wp-net \
-p 8082:80 \
-e WORDPRESS_DB_HOST=mysql \
-e WORDPRESS_DB_PASSWORD=secret \
wordpress:latest
Unable to find image 'wordpress:latest' locally
latest: Pulling from library/wordpress
78a70974d738: Pull complete
d5af5dfa43d4: Pull complete
9bde0c34d1f4: Pull complete
e24121de25e6: Pull complete
871c6a1a47cc: Pull complete
de502dcf081a: Pull complete
5020203f7d33: Pull complete
7d9f8a57d57a: Pull complete
b4b1305df06a: Pull complete
79e11af4b256: Pull complete
271a7052ce75: Pull complete
9cb746f747ab: Pull complete
e052461f1b41: Pull complete
90be0470639d: Pull complete
6562f7228ec3: Pull complete
1fa182c1d458: Pull complete
3f3cb8cd0960: Pull complete
73ce216c5445: Pull complete
4f4fb700ef54: Pull complete
568912d7e28d: Pull complete
ef17106e0cde: Pull complete
0dbbd63ae63a: Pull complete
4443d9c71a47: Pull complete
05bb872ab5e8: Download complete
32a1d2f81bd2: Download complete
Digest: sha256:1a758de2f424f735a9f8ad181783a9ed29e028594869a3323903d92532b35102
Status: Downloaded newer image for wordpress:latest
f89d1455deba829640e70fd2bed7dc43b52828c5b2f2e85105bc09ba4f06f697
harshit@LAPTOP-38KOG53E: ~$

```

Step-10:- Verify Page on Browser



Step-11:- Wordpress & MySql via compose

```
mkdir wp-compose
```

```
cd wp-compose
```

```
nano docker-compose.yml
```

```
cat docker-compose.yml
```

```
services:
```

```
  mysql:
```

```
    image: mysql:5.7
```

```
    container_name: mysql
```

```
    restart: always
```

```
    environment:
```

```
      MYSQL_ROOT_PASSWORD: secret
```

```
      MYSQL_DATABASE: wordpress
```

```
      MYSQL_USER: wpuser
```

```
      MYSQL_PASSWORD: secret
```

```
    volumes:
```

```
      - mysql_data:/var/lib/mysql
```

```
  wordpress:
```

```
    image: wordpress:latest
```

```
    container_name: wordpress
```

```
    restart: always
```

```
    ports:
```

```
      - "8082:80"
```

```
    environment:
```

```
      WORDPRESS_DB_HOST: mysql
```

```
      WORDPRESS_DB_USER: wpuser
```

```
      WORDPRESS_DB_PASSWORD: secret
```

```
    depends_on:
```

```
      - mysql
```

```
volumes:
```

```
  mysql_data:
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ mkdir wp-compose
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ cd wp-compose
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ cat docker-compose.yml
version: '3.8'

services:
  mysql:
    image: mysql:5.7
    container_name: mysql
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: secret
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: secret
    volumes:
      - mysql_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress
    restart: always
    ports:
      - "8082:80"
    environment:
      WORDPRESS_DB_HOST: mysql
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: secret
    depends_on:
      - mysql

volumes:
  mysql_data:
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ |

```

Step-12:- Start Services

```

docker-compose up -d
docker-compose ps

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ docker compose up -d
WARN[0000] /mnt/d/Containerization/LAB/Experiment-6/wp-compose/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it
to avoid potential confusion
WARN[0000] No services to build
[+] up 2/2
✔Container mysql Created 0.1s
✔Container wordpress Created 0.2s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ docker compose ps
WARN[0000] /mnt/d/Containerization/LAB/Experiment-6/wp-compose/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it
to avoid potential confusion

```

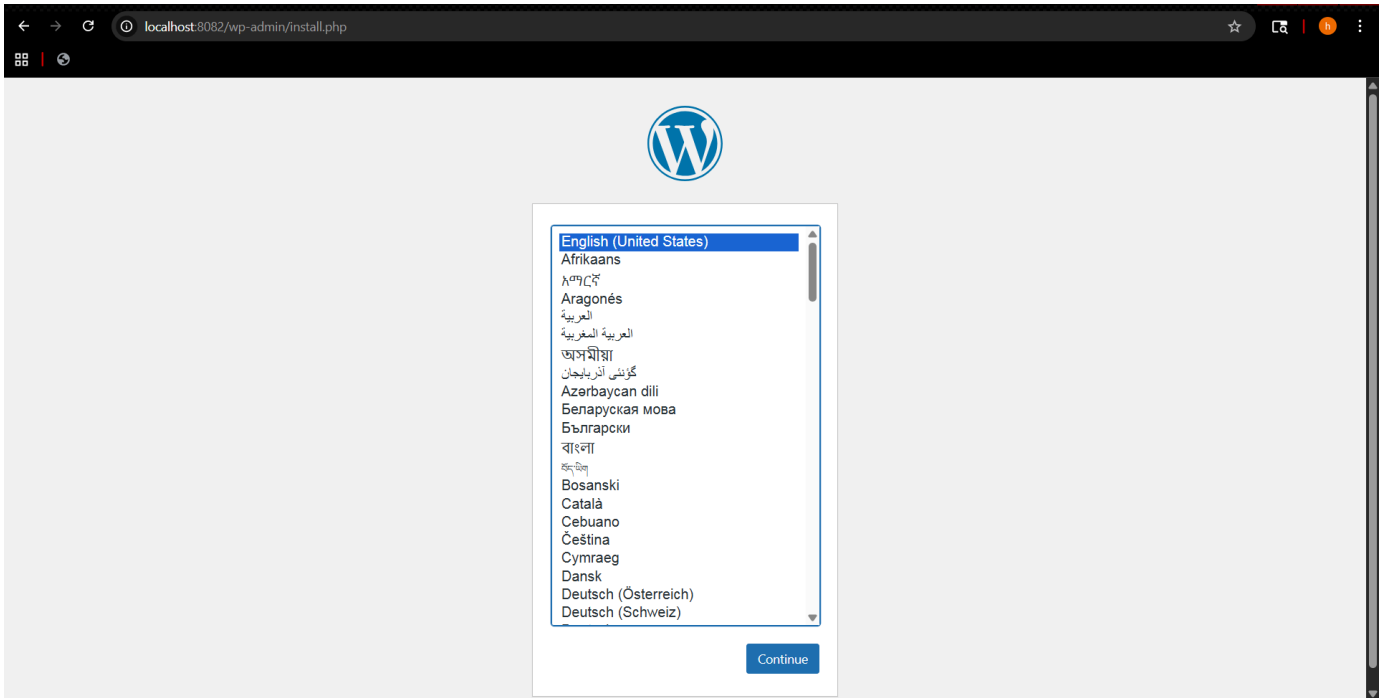
NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
mysql	mysql:5.7	"docker-entrypoint.s..."	mysql	14 seconds ago	Up 13 seconds	3306/tcp, 33060/tcp
wordpress	wordpress:latest	"docker-entrypoint.s..."	wordpress	14 seconds ago	Up 12 seconds	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ |

```

Step-13:- View On Browser



Step-14:- Create container with Node Image

```
docker run -d \
--name webapp \
-p 5000:5000 \
-e APP_ENV=production \
-e DEBUG=false \
--restart unless-stopped \
node:18-alpine
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/wp-compose$ docker run -d \
--name webapp \
-p 5000:5000 \
-e APP_ENV=production \
-e DEBUG=false \
--restart unless-stopped \
node:18-alpine
Unable to find image 'node:18-alpine' locally
18-alpine: Pulling from library/node
Digest: sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
Status: Downloaded newer image for node:18-alpine
14ef65e5155a6416c070f58c108f455a9c812835998c04e4776f693fb9aba8e7
```

Step-15:- Create Node container via Compose

```
mkdir task3-webapp
cd task3-webapp
nano docker-compose.yml
cat docker-compose.yml
```

docker-compose.yml :-

```

services:
  webapp:
    image: node:18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: "false"
    restart: unless-stopped

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ mkdir task3-webapp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6$ cd task3-webapp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task3-webapp$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task3-webapp$ cat docker-compose.yml
services:
  webapp:
    image: node:18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: "false"
    restart: unless-stopped
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task3-webapp$ |

```

Step-16:- Start Services

```

docker-compose up -d
docker-compose ps

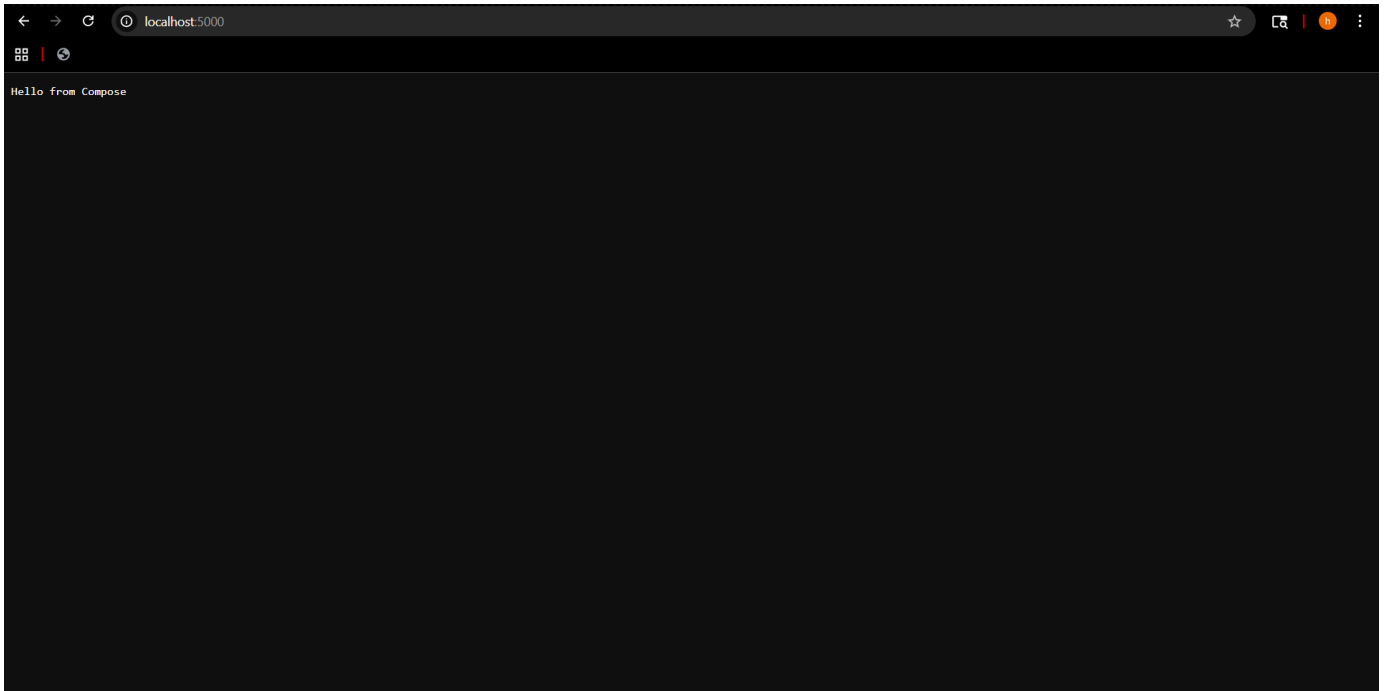
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task3-webapp$ docker compose up -d
WARN[0000] No services to build
[+] up 1/1
✔ Container webapp Created
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task3-webapp$ docker compose ps
NAME      IMAGE          COMMAND                  SERVICE    CREATED        STATUS      PORTS
webapp    node:18-alpine "docker-entrypoint.s..." webapp     11 seconds ago Restarting (0) 1 second ago
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task3-webapp$ |

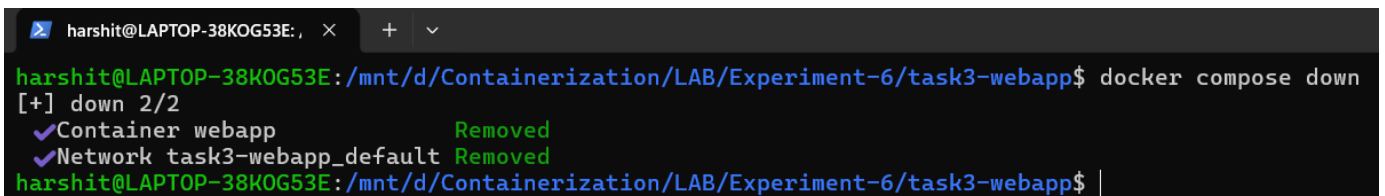
```


Step-17:- Veridy On Browser



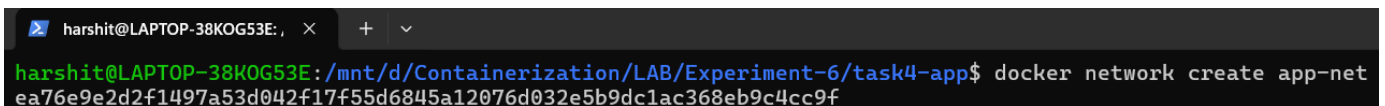
Step-18:- Stop Services

```
docker-compose down
```



Step-19:- Create Network

```
docker network create app-net
```



Step-20:- Create Python & Postgres Conatiners via Terminal

```
docker run -d \
--name postgres-db \
--network app-net \
-e POSTGRES_USER=admin \
-e POSTGRES_PASSWORD=secret \
-v pgdata:/var/lib/postgresql/data \
postgres:15
```

```
docker run -d \
```

```
--name backend \
--network app-net \
-p 8000:8000 \
-e DB_HOST=postgres-db \
-e DB_USER=admin \
-e DB_PASS=secret \
python:3.11-slim
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker run -d \
--name postgres-db \
--network app-net \
-e POSTGRES_USER=admin \
-e POSTGRES_PASSWORD=secret \
-v pgdata:/var/lib/postgresql/data \
postgres:15
8318e7206510c88db91c76f3b5c54ef6d9e19ac679337a81b94d35abdaa19e05
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker run -d \
--name backend \
--network app-net \
-p 8000:8000 \
-e DB_HOST=postgres-db \
-e DB_USER=admin \
-e DB_PASS=secret \
python:3.11-slim
Unable to find image 'python:3.11-slim' locally
3.11-slim: Pulling from library/python
0816ee351b60: Pull complete
d98f36ba2159: Pull complete
37ffa6577b04: Pull complete
79505fc62d4b: Download complete
6f70bdfdc954: Download complete
Digest: sha256:233de06753d30d120b1a3ce359d8d3be8bda78524cd8f520c99883bfe33964cf
Status: Downloaded newer image for python:3.11-slim
7ba40cced4ab546d8e1a3aeb45f71cc2987b793261092db0dd2ab7db10dfd8aa
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ |
```

Step-21:- List Containers

```
docker ps
```

```
harshit@LAPTOP-38KOG53E: x + v
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
8318e7206510	postgres:15	"docker-entrypoint.s..."	About a minute ago	Up About a minute	5432/tcp
0324beaa3d2d	wordpress:latest	"docker-entrypoint.s..."	14 minutes ago	Up 14 minutes	0.0.0.0:8082->80/tcp, [::]:8082->80/
tcp	wordpress				
e079aaf749bf	mysql:5.7	"docker-entrypoint.s..."	14 minutes ago	Up 14 minutes	3306/tcp, 33060/tcp
mysql					
1e1af5410829	assignment-1-backend	"docker-entrypoint.s..."	2 weeks ago	Up 40 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3
000/tcp	node_backend				
5576c99e1f84	assignment-1-database	"docker-entrypoint.s..."	2 weeks ago	Up 40 minutes (healthy)	
postgres_db					
d8018398f51c	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	4 weeks ago	Up 40 minutes	0.0.0.0:42325->6443/tcp
k3d-mycluster-serverlb					
5e303da50cdb	rancher/k3s:v1.31.5-k3s1	"/bin/k3d-entrypoint..."	4 weeks ago	Up 40 minutes	
k3d-mycluster-server-0					

Step-22:- List Networks

```
docker network ls
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
ea76e9e2d2f1	app-net	bridge	local
96de3e6643cc	assignment-1_app_net	bridge	local
ab8bd6e11071	bridge	bridge	local
23e543289f53	docker_gwbridge	bridge	local
b58dfd508402	experiment-10_sonarqube-lab	bridge	local
9bb2c2124ca6	host	host	local
4a4109b7c2bd	ibm-session_default	bridge	local
ydkad0orrrnos	ingress	overlay	swarm
521d8e59c054	ipvlan_net	ipvlan	local
cb24d34877ba	k3d-mycluster	bridge	local
f3f6952dbdd2	my-network	bridge	local
0679302a3979	myapp-network	bridge	local
7982c22f2cd0	none	null	local
a5d21e6e7313	wp-compose_default	bridge	local
65689d1e89f6	wp-net	bridge	local

Step-23:- List Volumes

```
docker volumes ls
```

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker volume ls
```

DRIVER	VOLUME NAME
local	6b4eab63d4f0deea08ab2168746074c6157c8d155c142941adc917eb81dc3cf2
local	26feb_mysql_data
local	26feb_wp_content
local	89c1e0cbb07833b376150098557229b3784fef38442e7c01d11008b2d89ffbe7
local	96a32a114f15734fdda9201acc3025084dba95585aff9e405f3528de840fc7b3
local	514edfec83d47d52b60e7d718baa506461e13f5d31c3fb81eea6fd65d3fb8334
local	1088cdae2a34a2c1e180e66dc5ed0cb6426e10faa1e53ebc4bdaf02f35776ce5
local	2391b8cb89172f724a30c5e494ff0fdebdaaf817d3ca3585c8149aeae675259
local	964748bb3c30fbac8573d4a67dd24c4f4dd0530d9ed27abbd555f2734cc3db04
local	18823321b3ad088e231b426ab1649322f1a52763f4288c607447566a266136ec
local	HarshitVolume
local	a1c32db277c052600f7c46ee4a27631fc54016af570b2745181bb0f19102bdda
local	a334b7b3ce44032963db0d818a03b7ff81e2ba5ed9687fca3867f73fdecda47
local	a539e11647af44f1247d13dc5abb2c6b5c315a05c705e3b35db0c4f238bc324e
local	assignment-1_pgdata
local	b7286e4626db91564f6ac6af9bcf8ed097a893744def09a5d98c7045de7dc433
local	experiment-10_sonar-data
local	experiment-10_sonar-db-data
local	experiment-10_sonar-extensions
local	ibm-session_jenkins_home
local	k3d-mycluster-images
local	mydata
local	mysql-data
local	myvolume
local	pgdata
local	portainer_data
local	postgres-data
local	redis-data
local	test
local	wp-compose_mysql_data

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ |
```

Step-24:- Stop Containers & Remove Network

```
docker stop backend postgres-db
docker rm backend postgres-db
docker network rm app-net
```

```

harshit@LAPTOP-38KOG53E: × + ▾
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker stop backend postgres-db
backend
postgres-db
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker rm backend postgres-db
backend
postgres-db
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker network rm app-net
app-net
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-6/task4-app$ |

```

Step-25:- Create same services via Compose

```

nano docker-compose.yml
cat docker-compose.yml

```

docker-compose.yml :-

```

services:
  postgres-db:
    image: postgres:15
    container_name: postgres-db
    environment:
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: secret
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks:
      - app-net

  backend:
    image: python:3.11-slim
    container_name: backend
    ports:
      - "8000:8000"
    environment:
      DB_HOST: postgres-db
      DB_USER: admin
      DB_PASS: secret
    depends_on:
      - postgres-db
    networks:
      - app-net

volumes:
  pgdata:

networks:
  app-net:

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ cat docker-compose.yml
services:
  postgres-db:
    image: postgres:15
    container_name: postgres-db
    environment:
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: secret
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks:
      - app-net

  backend:
    image: python:3.11-slim
    container_name: backend
    ports:
      - "8000:8000"
    environment:
      DB_HOST: postgres-db
      DB_USER: admin
      DB_PASS: secret
    depends_on:
      - postgres-db
    networks:
      - app-net

volumes:
  pgdata:

networks:
  app-net:
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ |

```

Step-26:- Start Services

```

docker-compose up -d
docker-compose ps

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker compose up -d
WARN[0000] No services to build
[+] up 4/4
✔Network task4-app_app-net Created
✔Volume task4-app_pgdata Created
✔Container postgres-db Created
✔Container backend Created
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
postgres-db         postgres:15          "docker-entrypoint.s..." postgres-db  6 seconds ago  Up 5 seconds  5432/tcp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ |

```

Step-27:- List Volumes

```

docker volume ls

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker volume ls
DRIVER      VOLUME NAME
local       6b4eab63d4f0deea08ab2168746074c6157c8d155c142941adc917eb81dc3cf2
local       26feb_mysql_data
local       26feb_wp_content
local       89c1e0cbb07833b376150098557229b3784fef38442e7c01d11008b2d89ffbe7
local       96a32a114f15734fdda9201acc3025084dba95585aff9e405f3528de840fc7b3
local       514edfec83d47d52b60e7d718baa506461e13f5d31c3fb81eea6fd65d3fb8334
local       1088cdae2a34a2c1e180e66dc5ed0cb6426e10faa1e53ebc4bdaf02f35776ce5
local       2391b8cb89172f724a30c5e494ff0ffebdafaf817d3ca3585c8149aeae675259
local       964748bb3c30fbac8573d4a67dd24c4f4dd0530d9ed27abbd555f2734cc3db04
local       18823321b3ad088e231b426ab1649322f1a52763f4288c607447566a266136ec
local       HarshitVolume
local       a1c32db277c052600f7c46ee4a27631fc54016af570b2745181bb0f19102bdda
local       a334b7b3ce44032963db0d818a03b7ff81e2ba5ed9687fca3867f73fdecda47
local       a539e11647af44f1247d13dc5abb2c6b5c315a05c705e3b35db0c4f238bc324e
local       assignment-1_pgdata
local       b7286e4626db91564f6ac6af9bcf8ed097a893744def09a5d98c7045de7dc433
local       experiment-10_sonar-data
local       experiment-10_sonar-db-data
local       experiment-10_sonar-extensions
local       ibm-session_jenkins_home
local       k3d-mycluster-images
local       mydata
local       mysql-data
local       myvolume
local       pgdata
local       portainer_data
local       postgres-data
local       redis-data
local       task4-app_pgdata
local       test
local       wp-compose_mysql_data
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ |

```

Step-28:- Stop Services

docker-compose down

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ docker compose down
[+] down 3/3
✔Container backend          Removed
✔Container postgres-db     Removed
✔Network task4-app_app-net Removed
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task4-app$ |

```

Step-29:- Create Node app

```

nano app.js
cat app.js

```

app.js :-

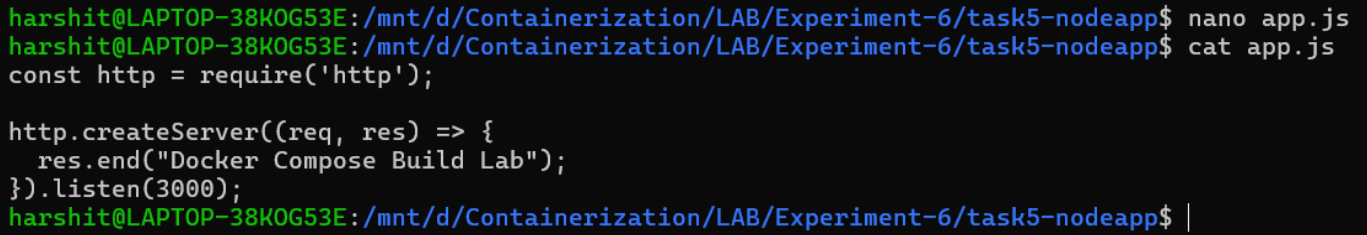
```

const http = require('http');

http.createServer((req, res) => {

```

```
res.end("Docker Compose Build Lab");
}).listen(3000);
```



A terminal window with a dark background. The prompt is `harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$`. The user runs `nano app.js`, then `cat app.js`. The output shows the content of `app.js` which is a simple HTTP server using `http.createServer` that responds with "Docker Compose Build Lab" on port 3000.

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ nano app.js
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ cat app.js
const http = require('http');

http.createServer((req, res) => {
  res.end("Docker Compose Build Lab");
}).listen(3000);
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ |
```

Step-30:- Create Dockerfile

```
nano Dockerfile
cat Dockerfile
```

Dockerfile :-

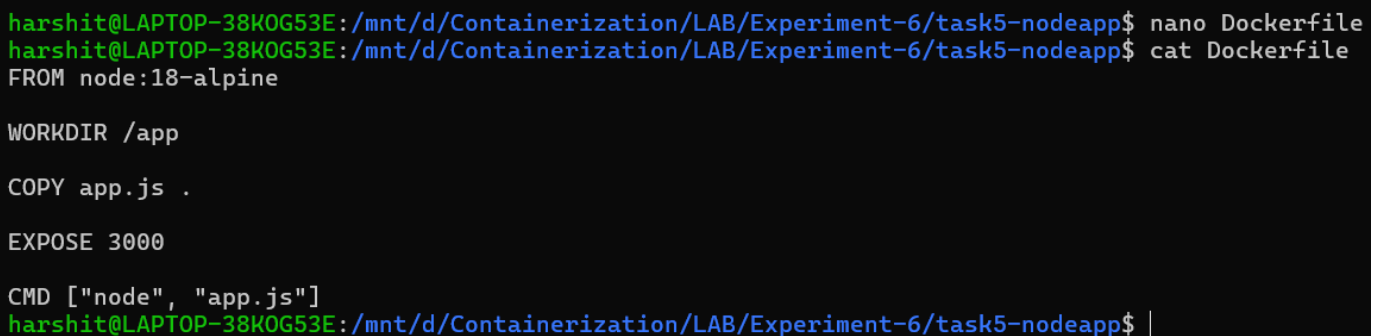
```
FROM node:18-alpine

WORKDIR /app

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
```



A terminal window with a dark background. The prompt is `harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$`. The user runs `nano Dockerfile`, then `cat Dockerfile`. The output shows the content of `Dockerfile` which defines a container image based on `node:18-alpine`, sets the working directory to `/app`, copies `app.js`, exposes port 3000, and sets the command to `node app.js`.

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ nano Dockerfile
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ cat Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ |
```

Step-31:- Create Compose File

```
nano docker-compose.yml
cat docker-compose.yml
```

docker-compose.yml :-

```

services:
  nodeapp:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: custom-node-app
    ports:
      - "3000:3000"

```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ cat docker-compose.yml
services:
  nodeapp:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: custom-node-app
    ports:
      - "3000:3000"
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ |

```

Step-32:- Start Compose Services

```

docker-compose up --build -d
docker-compose ps

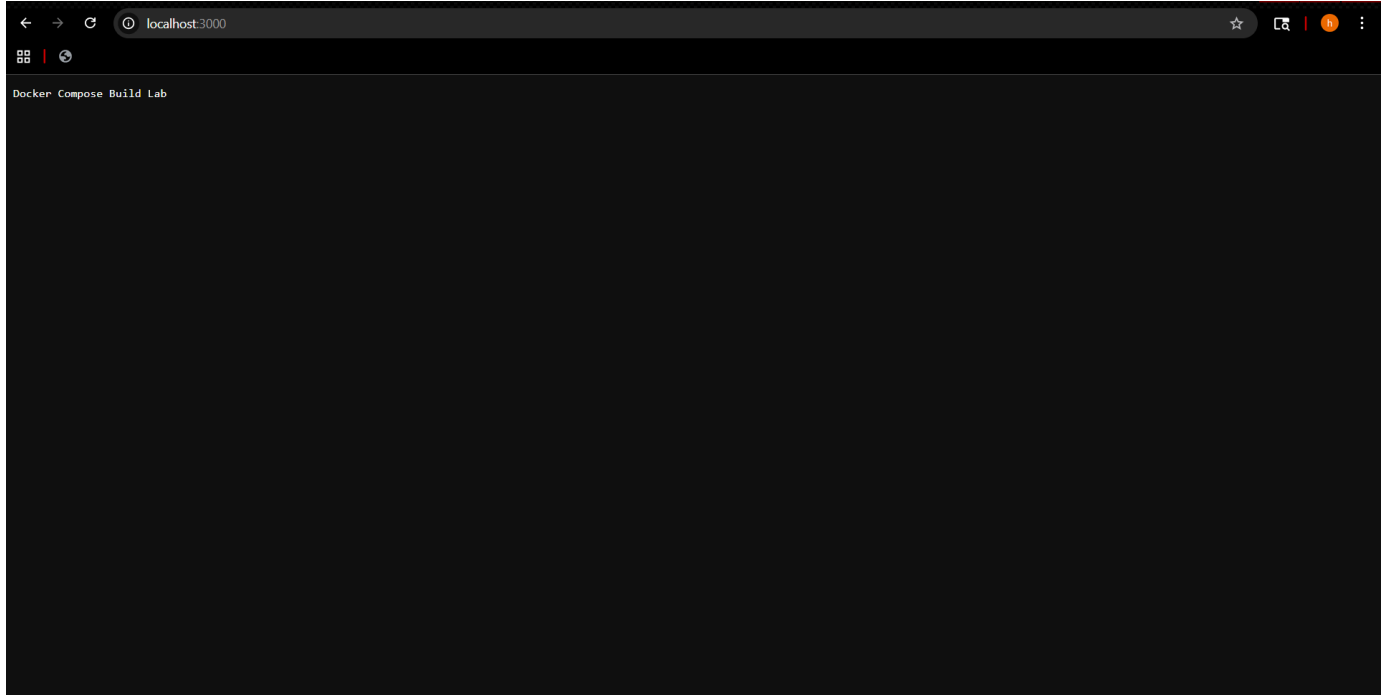
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ docker compose up --build -d
[+] Building -16.2s (10/10) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 576B 0.0s
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 123B 0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine 0.1s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d8a35cb9bc4b8ca09d9e 0.0s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d8a35cb9bc4b8ca09d9e 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 154B 0.0s
=> CACHED [2/3] WORKDIR /app 0.0s
=> [3/3] COPY app.js . 0.0s
=> => exporting to image 0.4s
=> => exporting layers 0.1s
=> => exporting manifest sha256:a5535ba34f55450c3a064942d8cd08639b36d96dd08633b191390fcc5df49c56 0.0s
=> => exporting config sha256:282992c3a276b61bce09dd6c1c093ca047199e4124b6ffa403e21f47d7feb042 0.0s
=> => exporting attestation manifest sha256:a163ac198904b61cce31c0998e917bc151948b241f33076f160b7945c9506718 0.0s
=> => exporting manifest list sha256:9c40289f51c380fa2d32f86ccfed5bac0be555e9831050ed9c9c1252b528898a 0.0s
=> => naming to docker.io/library/task5-nodeapp-nodeapp:latest 0.0s
=> => unpacking to docker.io/library/task5-nodeapp-nodeapp:latest 0.1s
=> resolving provenance for metadata file 0.0s
[+] up 3/3
✔ Image task5-nodeapp-nodeapp Built 1.9s
✔ Network task5-nodeapp_default Created 0.0s
✔ Container custom-node-app Created 0.1s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
custom-node-app task5-nodeapp-nodeapp "docker-entrypoint.s..." nodeapp 9 seconds ago Up 6 seconds 3000/tcp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-6/task5-nodeapp$ |

```


Step-33:- Verify on Browser



Conclusion

This experiment demonstrated:

- How to deploy **WordPress + MySQL** using Docker Compose.
- How containers communicate using internal networking.
- Importance of **volumes for persistence**.

[< Previous](#) [Home](#) [Next >](#)

Harshit-Kumar-Containerization

Experiment 7: CI/CD using Jenkins, GitHub and Docker Hub

Introduction

What is Jenkins?

Jenkins is a **web-based GUI automation server** used to:

- Build applications
- Test code
- Deploy software

It provides:

- Dashboard (browser-based UI)
- Plugin ecosystem (GitHub, Docker, etc.)
- Pipeline as Code using `Jenkinsfile`

What is CI/CD ??

- **Continuous Integration (CI)**: Code is automatically built and tested after each commit
- **Continuous Deployment (CD)**: Built artifacts (Docker images) are automatically delivered/deployed

Workflow Overview

Developer → GitHub → Webhook → Jenkins → Build → Docker Hub

Prerequisites

Create a repository on GitHub:

```
[my-app](https://github.com/harshitkumar22/my-app.git)
```

Project Structure

```
my-app/  
├─ app.py  
├─ requirements.txt  
├─ Dockerfile  
└─ Jenkinsfile
```

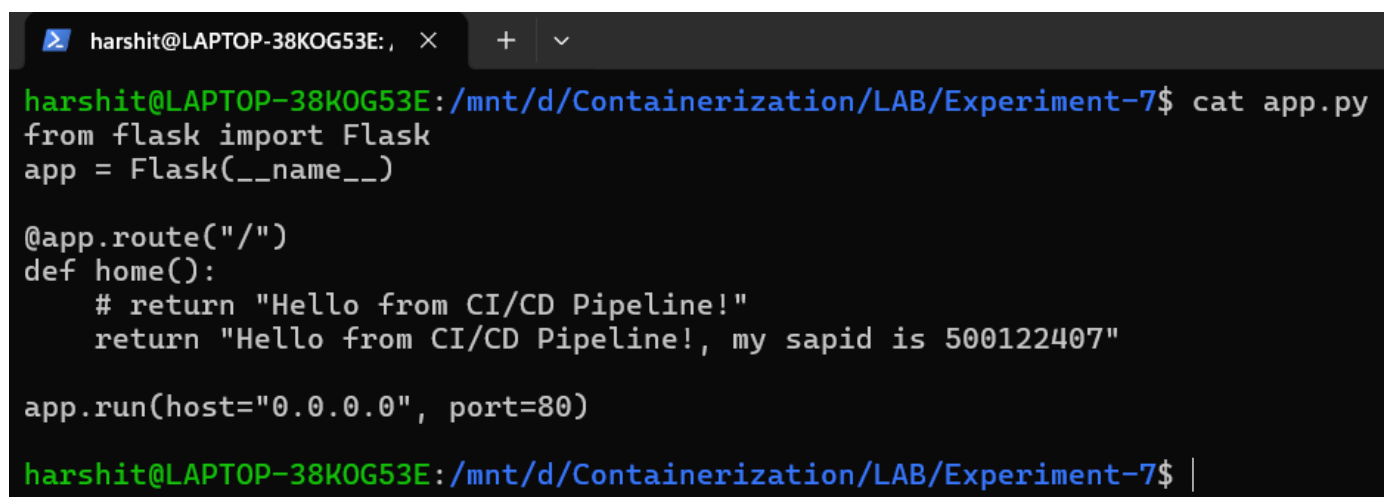
Hands-On

Step-1:- Create a `app.py`

```
nano app.py
```

Paste This:

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route("/")  
def home():  
    # return "Hello from CI/CD Pipeline!"  
    return "Hello from CI/CD Pipeline!, my sapid is 500122407"  
  
app.run(host="0.0.0.0", port=80)
```

A terminal window with a dark background. The title bar shows 'harshit@LAPTOP-38KOG53E:'. The prompt is 'harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-7\$'. The user has run 'cat app.py' and the output is displayed. The code matches the one in the previous block.

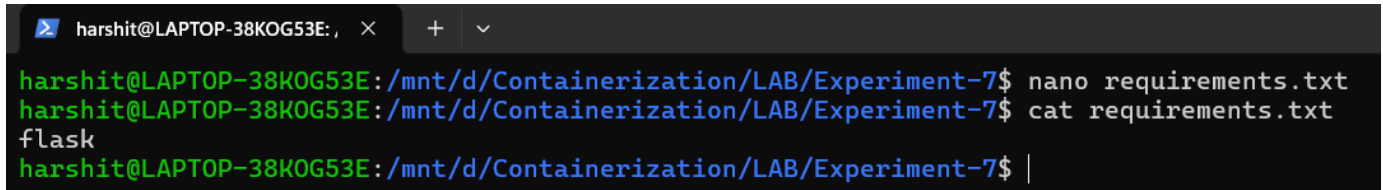
```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-7$ cat app.py  
from flask import Flask  
app = Flask(__name__)  
  
@app.route("/")  
def home():  
    # return "Hello from CI/CD Pipeline!"  
    return "Hello from CI/CD Pipeline!, my sapid is 500122407"  
  
app.run(host="0.0.0.0", port=80)  
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-7$ |
```

Step-2:- Create `requirements.txt`

```
nano requirements.txt
```

Paste this:

```
flask
```

A terminal window with a dark background. The prompt is 'harshit@LAPTOP-38KOG53E:'. The user enters 'nano requirements.txt', then 'cat requirements.txt', and finally 'flask'. The prompt is now 'harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-7\$'.

Step-3:- Create Dockerfile

```
nano Dockerfile
```

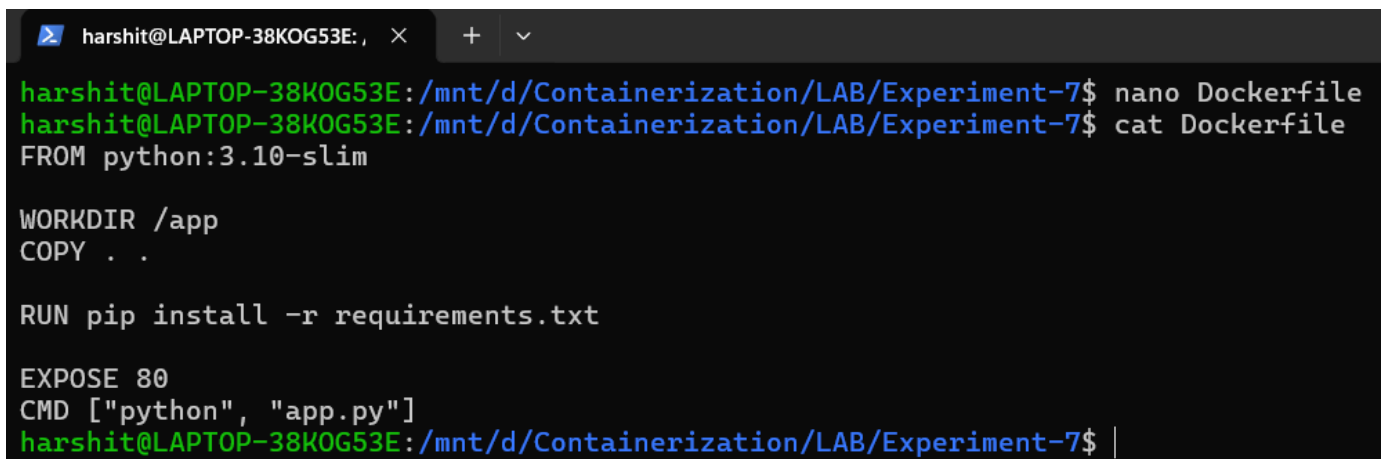
Paste this:

```
FROM python:3.10-slim

WORKDIR /app
COPY . .

RUN pip install -r requirements.txt

EXPOSE 80
CMD ["python", "app.py"]
```

A terminal window with a dark background. The prompt is 'harshit@LAPTOP-38KOG53E:'. The user enters 'nano Dockerfile', then 'cat Dockerfile'. The output shows the contents of the Dockerfile: 'FROM python:3.10-slim', 'WORKDIR /app', 'COPY . .', 'RUN pip install -r requirements.txt', 'EXPOSE 80', and 'CMD ["python", "app.py"]'. The prompt is now 'harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-7\$'.

Step-4:- Create Jenkinsfile

```
nano Jenkinsfile
```

Paste this:

```
pipeline {
    agent any
```

```
environment {
    IMAGE_NAME = "harshitkumar22/myapp"
}

stages {

    stage('Build Docker Image') {
        steps {
            sh "docker build -t ${IMAGE_NAME}:latest ."
        }
    }

    stage('Login to Docker Hub') {
        steps {
            withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKE
            sh '''
                echo "$DOCKER_TOKEN" | docker login -u harshitkumar22 --password-s
            ...
        })
    })
}

    stage('Push to Docker Hub') {
        steps {
            sh "docker push ${IMAGE_NAME}:latest"
        }
    }
}

post {
    always {
        sh "docker logout || true"
    }
}
}
```



```

harshit@LAPTOP-38KOG53E: x + v
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT/OneDrive/Desktop/Experiment-7$ nano jenkinsfile
harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT/OneDrive/Desktop/Experiment-7$ cat jenkinsfile
pipeline {
    agent pipeline {
        agent any
    }

    environment {
        IMAGE_NAME = "harshitkumar22/myapp"
    }

    stages {

        stage('Build Docker Image') {
            steps {
                s++h 'docker build -t $IMAGE_NAME:latest .'
            }
        }

        stage('Login to Docker Hub') {
            steps {
                withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
                    sh 'echo $DOCKER_TOKEN | docker login -u harshitkumar22 --password-stdin'
                }
            }
        }

        stage('Push to Docker Hub') {
            steps {
                sh 'docker push $IMAGE_NAME:latest'
            }
        }
    }
}

harshit@LAPTOP-38KOG53E:/mnt/c/Users/HARSHIT/OneDrive/Desktop/Experiment-7$ |

```

Step-5:- Create docker-compose.yml

```

version: '3.8'

services:
    jenkins:
        image: jenkins/jenkins:lts
        container_name: jenkins
        restart: always
        ports:
            - "8080:8080"
            - "50000:50000"
        volumes:
            - jenkins_home:/var/jenkins_home
            - /var/run/docker.sock:/var/run/docker.sock
        user: root

volumes:
    jenkins_home:

```

```

harshit@LAPTOP-38KOG53E: ~$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ cat docker-compose.yml
version: '3.8'

services:
  jenkins:
    image: jenkins/jenkins:lts
    container_name: jenkins
    restart: always
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    user: root

volumes:
  jenkins_home:
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ |

```

Step-6:- Start Up Compose

```
docker-compose up -d
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ docker-compose up -d
WARN[0000] /mnt/d/Containerization/LAB/Experiment-7/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid p
otential confusion
WARN[0000] No services to build
[+] up 1/1
Container jenkins Created
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
0d25962138cf	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	42 seconds ago	Up 41 seconds	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
0324beaa3d2d	wordpress:latest	"docker-entrypoint.s..."	5 days ago	Up 16 minutes	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
e079aaf749bf	mysql:5.7	"docker-entrypoint.s..."	5 days ago	Up 16 minutes	3306/tcp, 33060/tcp
5576c99e1f84	assignment-1-database	"docker-entrypoint.s..."	3 weeks ago	Up 16 minutes (healthy)	
d8018398f51c	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	5 weeks ago	Up 16 minutes	0.0.0.0:42325->6443/tcp
5e303da50cdb	rancher/k3s:v1.31.5-k3s1	"/bin/k3d-entrypoint..."	5 weeks ago	Up 16 minutes	

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ |

```

Step-7:- Exec Jenkins Contrainer to get Password

```
docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword
```

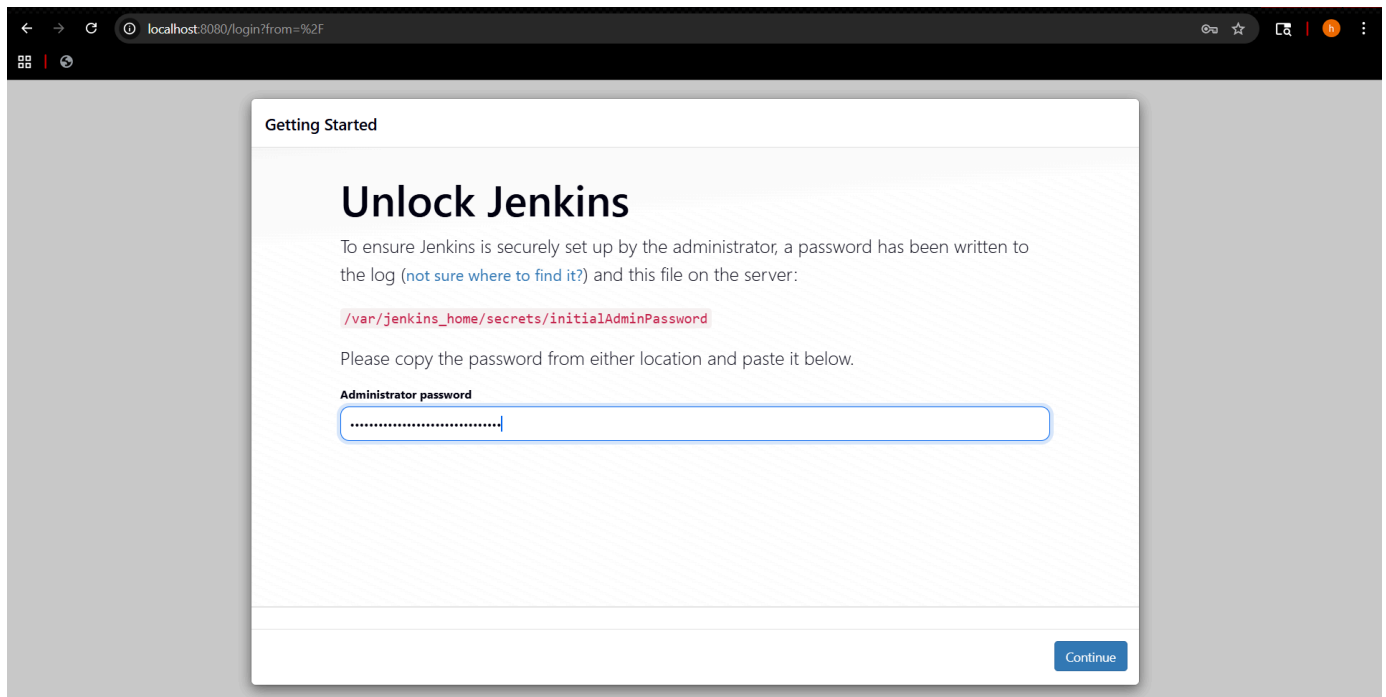
```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword
7cef1bccc38b433d8721891e6ed08c7f
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-7$ |

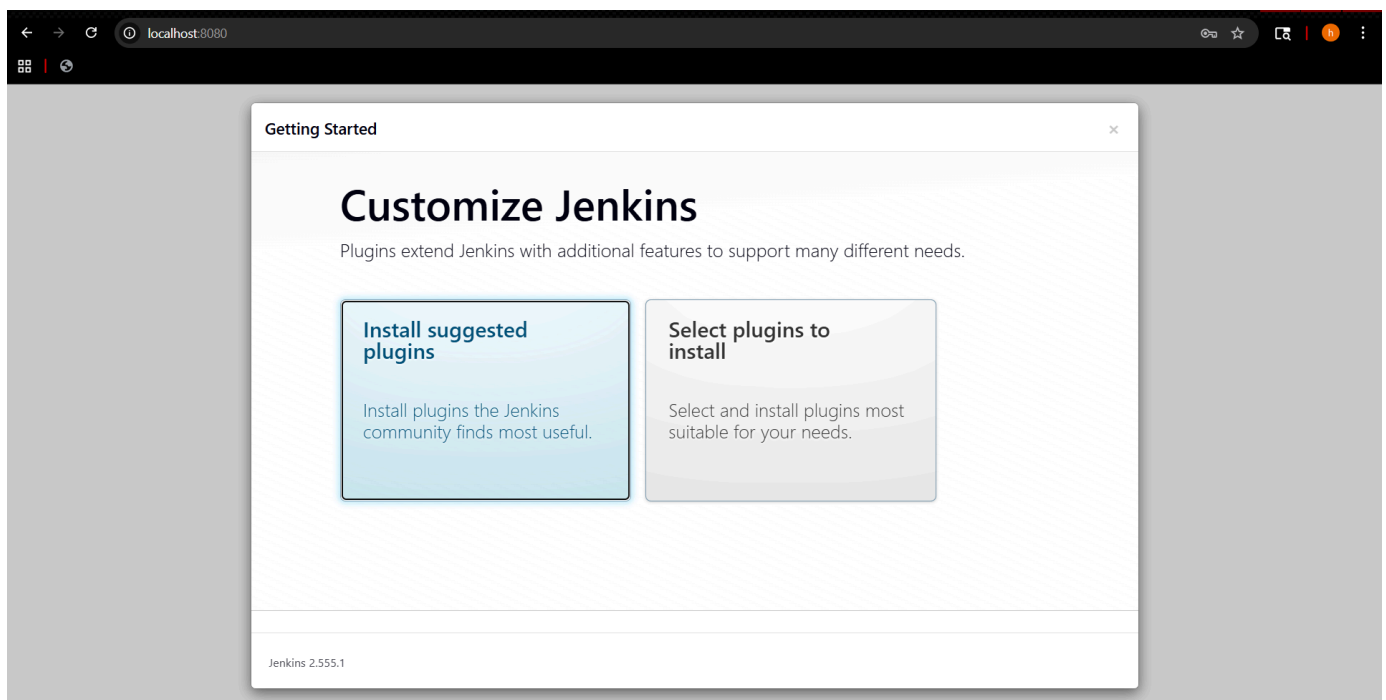
```

Step-8:- Login to Jenkins Dashboard on Browser

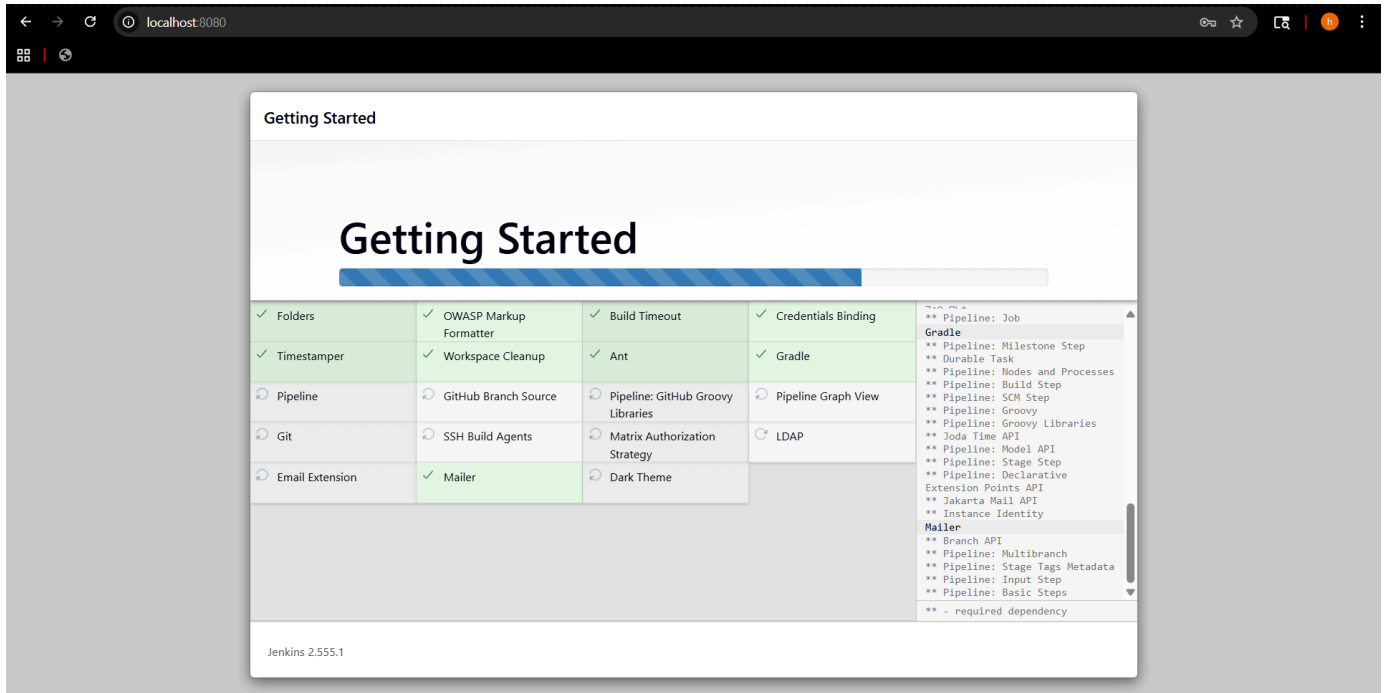
```
http://localhost:8080
```



Step-9:- Install Sugeested Plugins



Step-10:- Plugins List



The screenshot shows the Jenkins 'Getting Started' page at localhost:8080. The page title is 'Getting Started'. Below the title is a progress bar. A table lists installed plugins with green checkmarks, and a scrollable list shows the full plugin names.

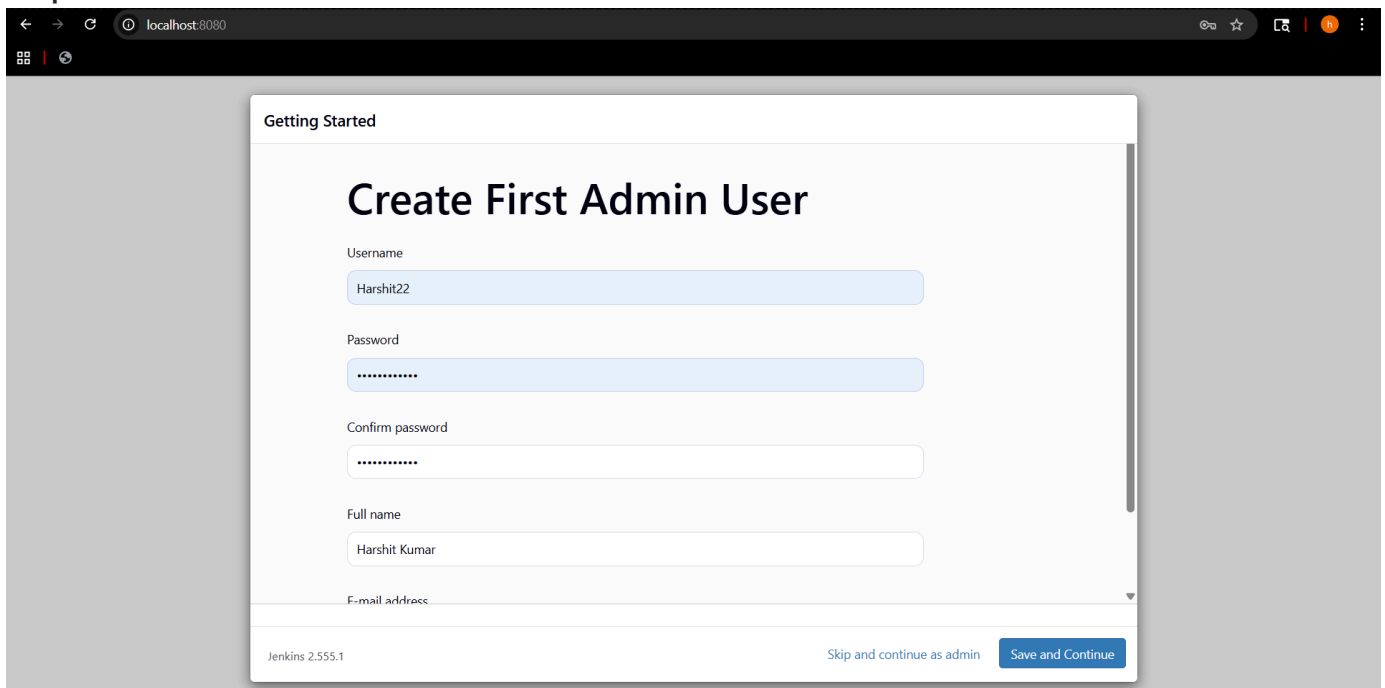
✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding
✓ Timestampers	✓ Workspace Cleanup	✓ Ant	✓ Gradle
⚙ Pipeline	⚙ GitHub Branch Source	⚙ Pipeline: GitHub Groovy Libraries	⚙ Pipeline Graph View
⚙ Git	⚙ SSH Build Agents	⚙ Matrix Authorization Strategy	⚙ LDAP
⚙ Email Extension	✓ Mailer	⚙ Dark Theme	

Plugins listed in the scrollable area:

- Gradle
 - ** Pipeline: Job
 - ** Pipeline: Milestone Step
 - ** Durable Task
 - ** Pipeline: Nodes and Processes
 - ** Pipeline: Build Step
 - ** Pipeline: SCM Step
 - ** Pipeline: Groovy
 - ** Pipeline: Groovy Libraries
 - ** Joda Time API
 - ** Pipeline: Model API
 - ** Pipeline: Stage Step
 - ** Pipeline: Declarative
- Extension Points API
- ** Jakarta Mail API
- ** Instance Identity
- Mailer**
 - ** Branch API
 - ** Pipeline: Multibranch
 - ** Pipeline: Stage Tags Metadata
 - ** Pipeline: Input Step
 - ** Pipeline: Basic Steps
- ** - required dependency

Jenkins 2.555.1

Step-11:- Create Admin User



The screenshot shows the 'Create First Admin User' form in Jenkins. The form has fields for Username, Password, Confirm password, Full name, and E-mail address. At the bottom, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'.

Username: Harshit22

Password:

Confirm password:

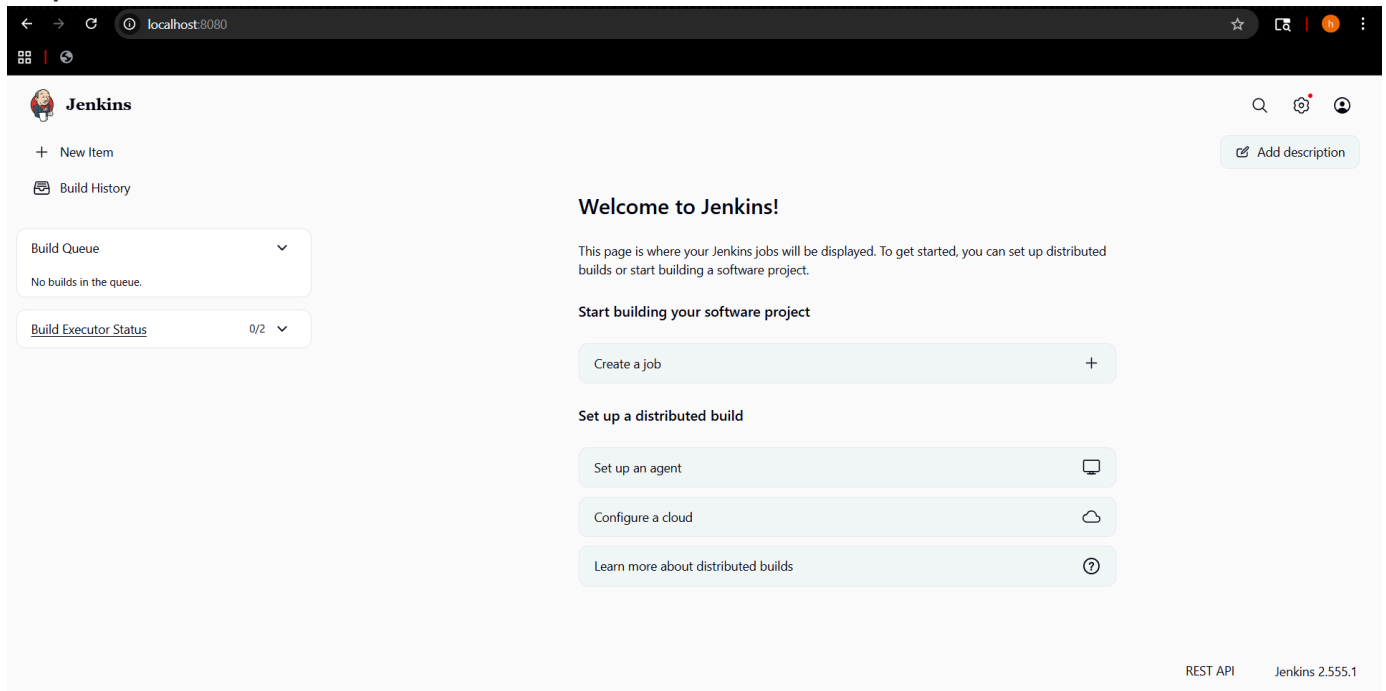
Full name: Harshit Kumar

E-mail address:

Jenkins 2.555.1

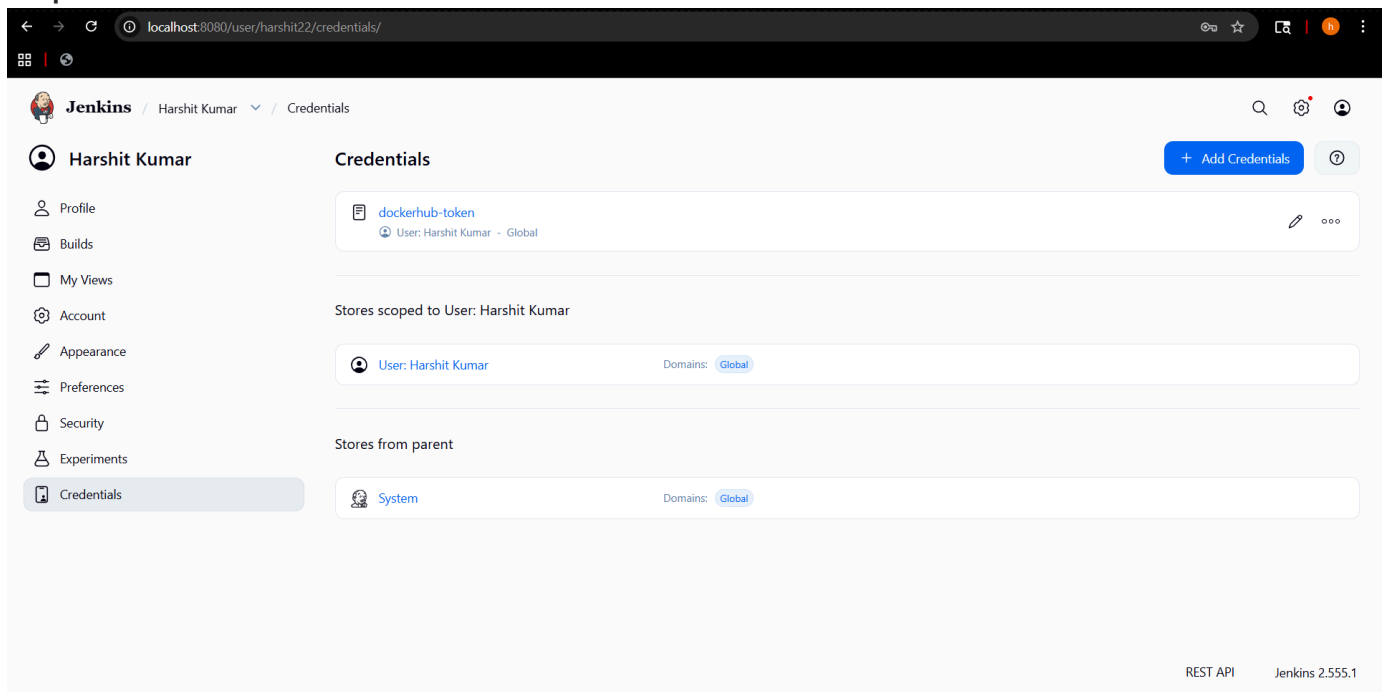
[Skip and continue as admin](#) [Save and Continue](#)

Step-12:- Admin Dashboard Interface



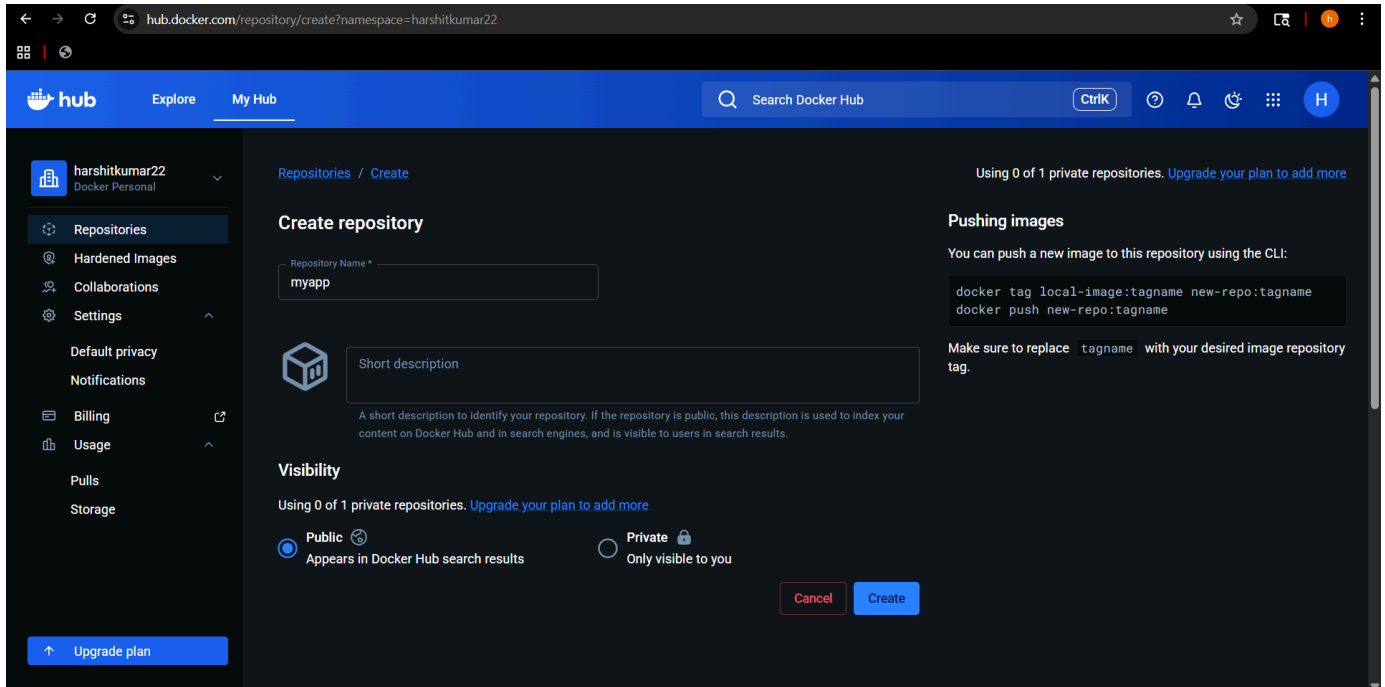
The screenshot shows the Jenkins Admin Dashboard. The top navigation bar includes the Jenkins logo, a search icon, a settings icon, and a user profile icon. Below the navigation bar, there's a sidebar with links for 'New Item', 'Build History', 'Build Queue', and 'Build Executor Status'. The main content area features a 'Welcome to Jenkins!' message, a 'Start building your software project' section with a 'Create a job' button, and a 'Set up a distributed build' section with buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. The bottom right corner displays 'REST API' and 'Jenkins 2.555.1'.

Step-13:- Add DockerHub Token in Credentials

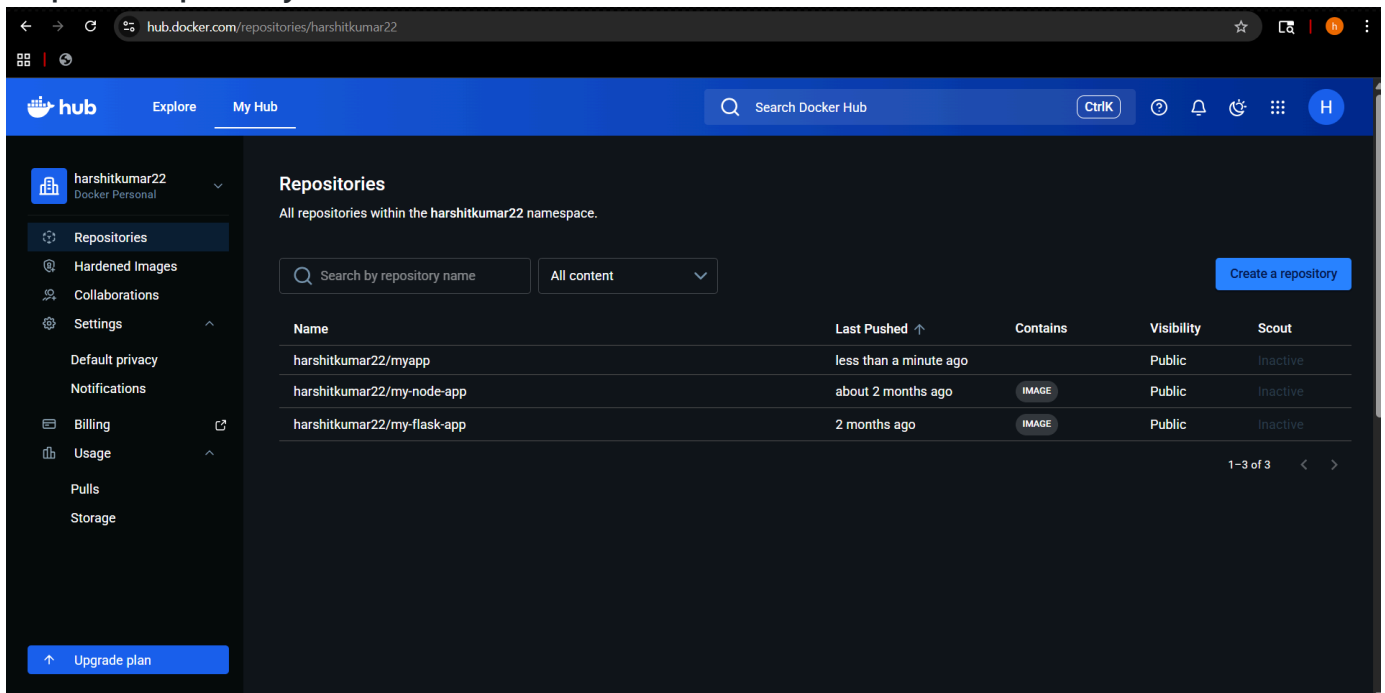


The screenshot shows the Jenkins 'Credentials' page for the user 'Harshit Kumar'. The left sidebar contains a list of user settings: Profile, Builds, My Views, Account, Appearance, Preferences, Security, Experiments, and Credentials (which is currently selected). The main content area is titled 'Credentials' and includes a '+ Add Credentials' button. Below this, there's a section for 'Stores scoped to User: Harshit Kumar' showing a 'dockerhub-token' credential. Further down, there's a section for 'Stores from parent' showing a 'System' credential. The bottom right corner displays 'REST API' and 'Jenkins 2.555.1'.

Step-14:- Create a Repository in Dockerhub



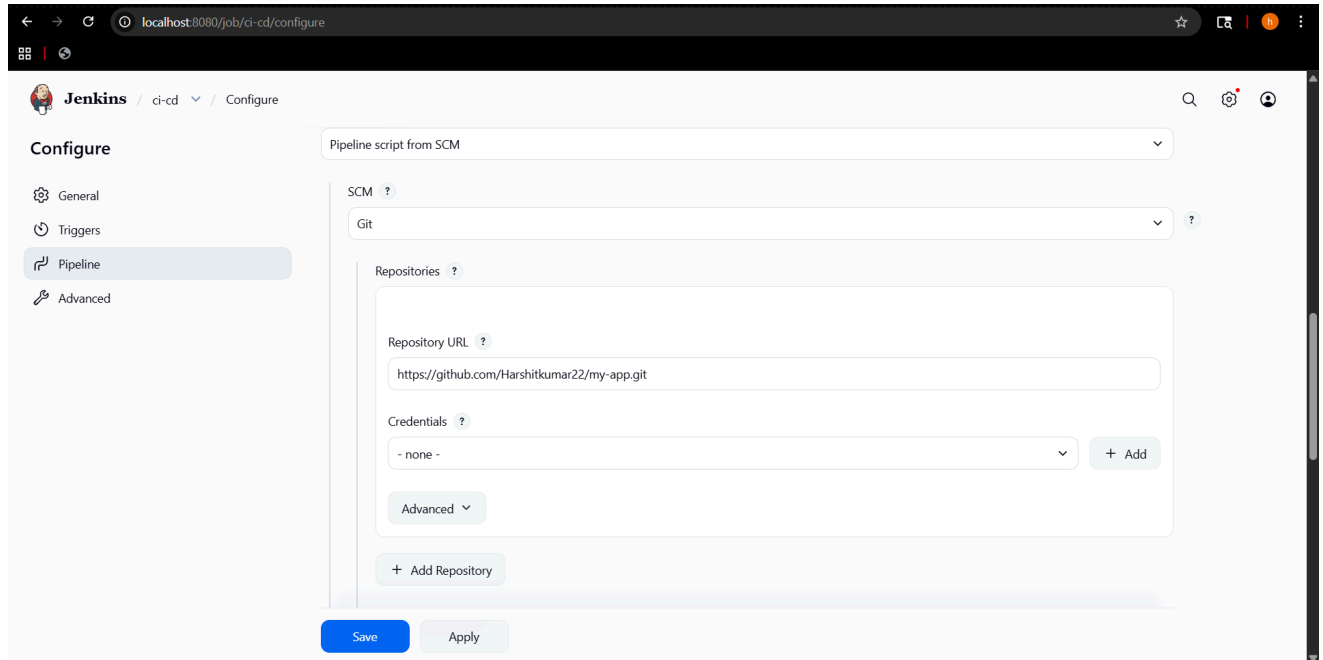
Step-15:- Repository will be listed



Step-16:- Create Pipeline Job

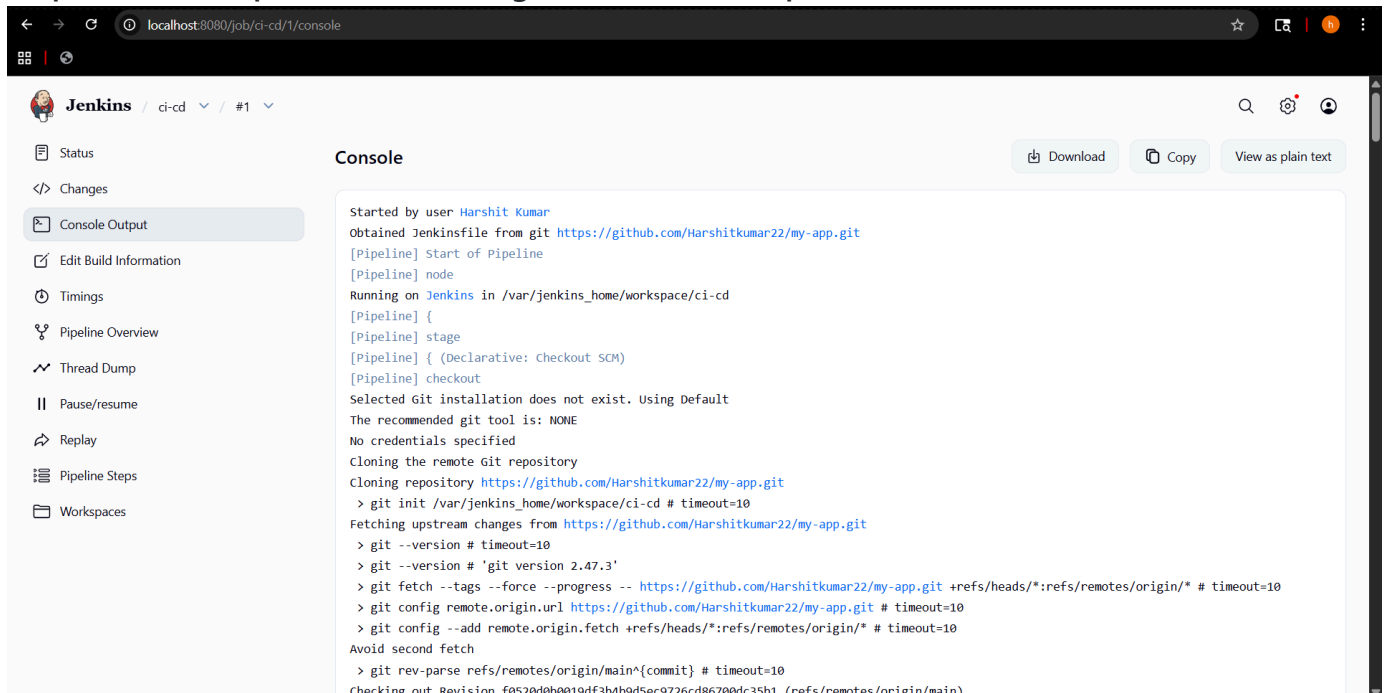
- New-Item:- Pipeline
- Name:- ci-cd-pipeline

- Definition:- Pipeline Script from SCM



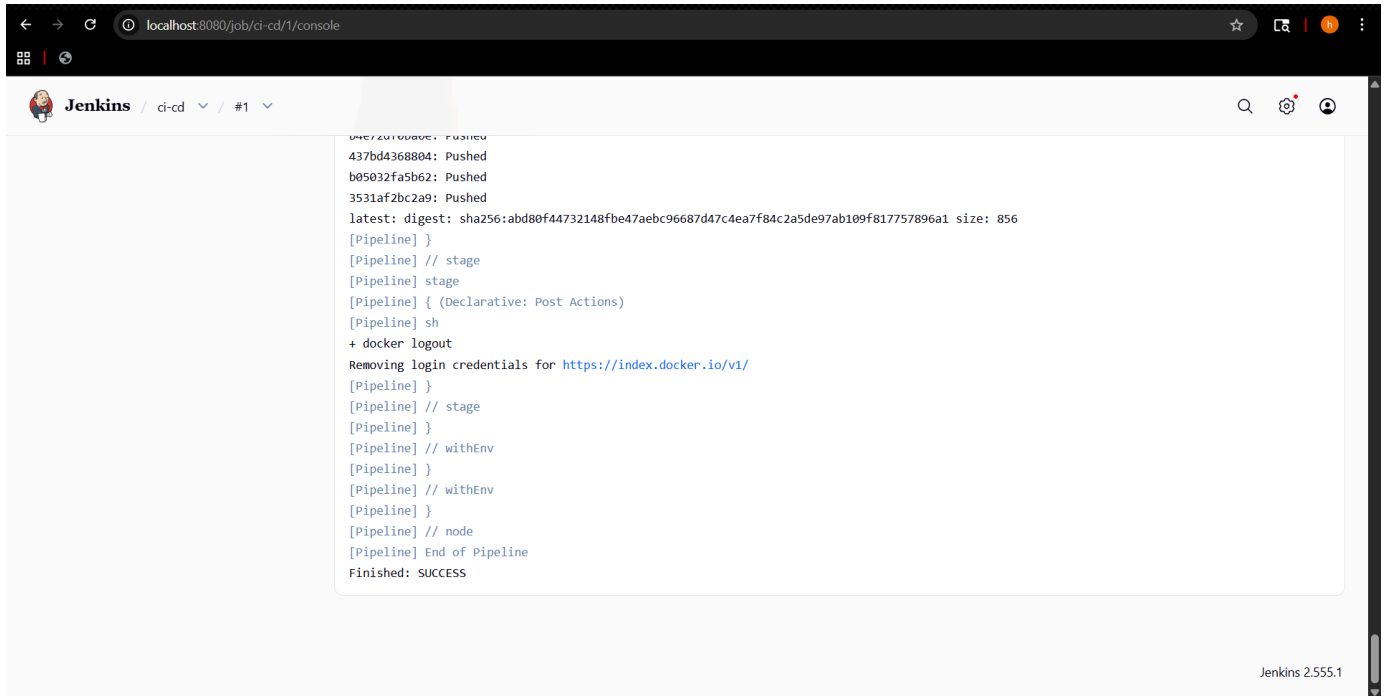
The screenshot shows the Jenkins configuration page for a pipeline script from SCM. The left sidebar has tabs for General, Triggers, Pipeline, and Advanced. The 'Pipeline' tab is selected. The main area is titled 'Configure' and shows the 'Pipeline script from SCM' configuration. The 'SCM' dropdown is set to 'Git'. The 'Repository URL' field contains 'https://github.com/Harshitkumar22/my-app.git'. The 'Credentials' dropdown is set to '- none -'. There is an 'Add' button next to the credentials dropdown. Below the repository URL field, there is an 'Advanced' dropdown and an 'Add Repository' button. At the bottom, there are 'Save' and 'Apply' buttons.

Step-17:- Build Pipeline & Track Stages in Console Output



The screenshot shows the Jenkins console output for a pipeline build. The left sidebar has tabs for Status, Changes, Console Output, Edit Build Information, Timings, Pipeline Overview, Thread Dump, Pause/resume, Replay, Pipeline Steps, and Workspaces. The 'Console Output' tab is selected. The main area shows the console output for the pipeline build. The output starts with 'Started by user Harshit Kumar' and 'Obtained Jenkinsfile from git https://github.com/Harshitkumar22/my-app.git'. It then shows the pipeline stages: 'Start of Pipeline', 'node', 'Running on Jenkins in /var/jenkins_home/workspace/ci-cd', '[Pipeline] {', '[Pipeline] stage', '[Pipeline] { (Declarative: Checkout SCM)', and '[Pipeline] checkout'. The checkout stage shows the following steps: 'Selected Git installation does not exist. Using Default', 'The recommended git tool is: NONE', 'No credentials specified', 'Cloning the remote Git repository', 'Cloning repository https://github.com/Harshitkumar22/my-app.git', 'git init /var/jenkins_home/workspace/ci-cd # timeout=10', 'Fetching upstream changes from https://github.com/Harshitkumar22/my-app.git', 'git --version # timeout=10', 'git --version # git version 2.47.3', 'git fetch --tags --force --progress -- https://github.com/Harshitkumar22/my-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10', 'git config remote.origin.url https://github.com/Harshitkumar22/my-app.git # timeout=10', 'git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10', 'Avoid second fetch', 'git rev-parse refs/remotes/origin/main^{commit} # timeout=10', and 'Checking out Revision f0520d0b0019df3b4b9d5ec9726cd86700dc35b1 (refs/remotes/origin/main)'.

Step-18:- Build Status Success will be shown with Green Tick

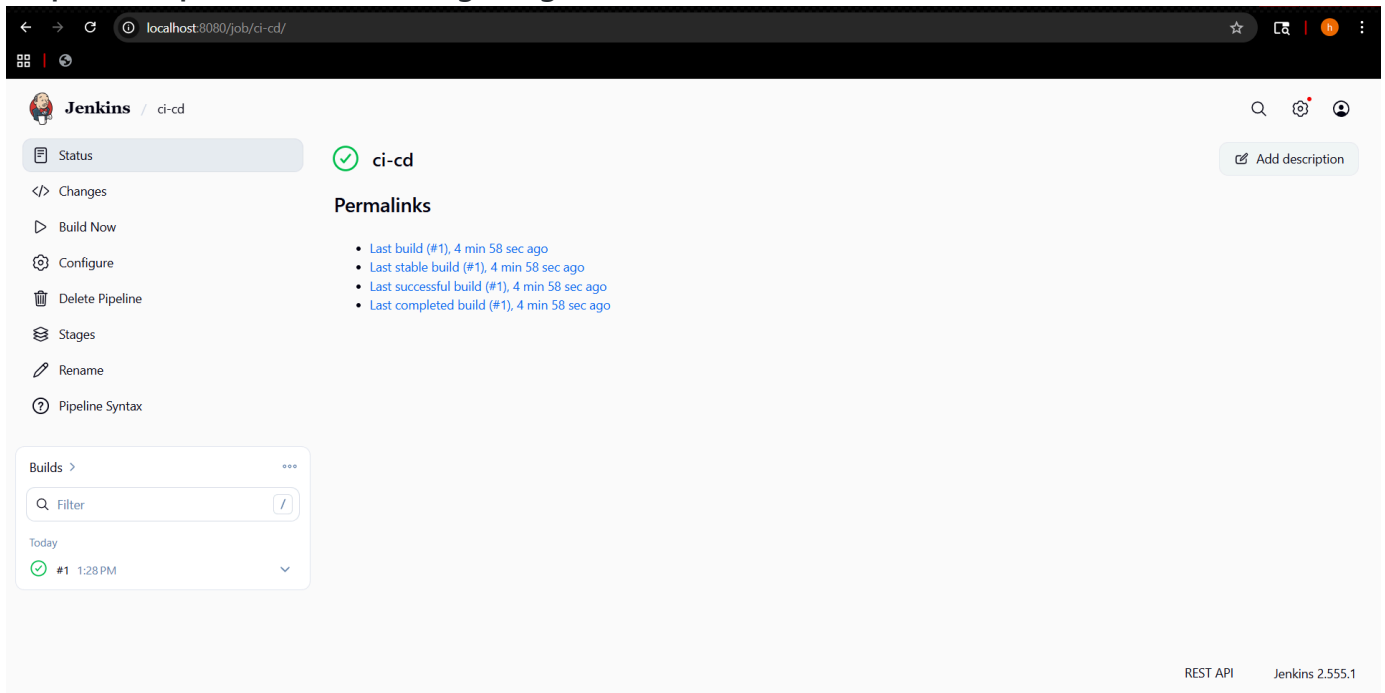


The screenshot shows the Jenkins console output for a build. The output includes the following text:

```
437bd4368804: Pushed
b05032fa5b62: Pushed
3531af2bc2a9: Pushed
latest: digest: sha256:abd80f44732148fbe47aebc96687d47c4ea7f84c2a5de97ab109f817757896a1 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] sh
+ docker logout
Removing login credentials for https://index.docker.io/v1/
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

The Jenkins interface shows the job name "ci-cd" and the build number "#1". The status is "SUCCESS".

Step-19:- Repo will be consisting Image Now !!!



The screenshot shows the Jenkins job page for "ci-cd". The status is "SUCCESS" (indicated by a green checkmark). The page includes a sidebar with navigation options: Status, Changes, Build Now, Configure, Delete Pipeline, Stages, Rename, and Pipeline Syntax. The main content area shows the job name "ci-cd" and a list of permlinks for the build. The "Builds" section shows a list of builds, with the first build (#1) being successful and completed at 1:28 PM.

Permlinks

- Last build (#1), 4 min 58 sec ago
- Last stable build (#1), 4 min 58 sec ago
- Last successful build (#1), 4 min 58 sec ago
- Last completed build (#1), 4 min 58 sec ago

Builds >

Q Filter

Today

#1 1:28 PM

[← Previous](#) [Home](#) [Next →](#)

Harshit-Kumar-Containerization

Experiment 9: Working with Ansible

Introduction

What is Ansible?

- Ansible is an open-source **automation tool** for **configuration management**, **application deployment**, and **orchestration**.
- It follows an **agentless** architecture, using SSH for Linux and WinRM for Windows.
- Uses YAML-based **playbooks** to define automation tasks.

Ansible is software that enables cross-platform automation and orchestration at scale and has become the standard choice among enterprise automation solutions.

Key Concepts

Component	Description
Control Node	Machine with Ansible installed
Managed Nodes	Target servers (no Ansible agent needed)
Inventory	Defines the list of managed nodes (EC2 instances, servers, etc.). inventory.ini
Playbooks	YAML files containing a sequence of automation steps.
Tasks	Individual actions in playbooks (e.g., installing a package).
Modules	Built-in functionality to perform tasks (e.g., yum , apt , service).
Roles	Pre-defined reusable automation scripts.

Benefits of using Ansible

- A free and open-source community project with a huge audience.
- Battle-tested over many years as the preferred tool of IT wizards.
- Easy to start and use from day one, without the need for any special coding skills.
- Simple deployment workflow without any extra agents.

- Includes some sophisticated features around modularity and reusability that come in handy as users become more proficient.
- Extensive and comprehensive official documentation that is complemented by a plethora of online material produced by its community.

To sum up, Ansible is simple yet powerful, agentless, community-powered, predictable, and secure.

Hands-On

Step-1:- Install Ansible in WSL

```
sudo apt install ansible -y
```

```
harshit@LAPTOP-38KOG53E: ~$ sudo apt install -y
[sudo] password for harshit:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 94 not upgraded.
harshit@LAPTOP-38KOG53E: ~$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud python3-lockfile python3-netaddr
  python3-ntlm-auth python3-packaging python3-passlib python3-requests-ntlm python3-resolve-lib python3-selinux python3-simplejson python3-winrm
  python3-xmltodict
Suggested packages:
  cowsay sshpass python3-trio python3-aiocli python3-h2 python3-httpx python3-httpcore python3-lockfile-doc ipython3 python3-netaddr-docs
The following NEW packages will be installed:
  ansible ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud python3-lockfile python3-netaddr
  python3-ntlm-auth python3-packaging python3-passlib python3-requests-ntlm python3-resolve-lib python3-selinux python3-simplejson python3-winrm
  python3-xmltodict
0 upgraded, 19 newly installed, 0 to remove and 94 not upgraded.
Need to get 22.0 MB of archives.
After this operation, 330 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-packaging all 24.0-1 [41.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolve-lib all 1.0.1-1 [25.7 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython all 2.6.1-1ubuntu1 [163 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 ieee-data all 20220827.1 [2113 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-netaddr all 0.8.0-2ubuntu1 [319 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core all 2.16.3-0ubuntu2 [1280 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 ansible all 9.2.0+dfsg-0ubuntu5 [16.4 MB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-argcomplete all 3.1.4-1ubuntu0.1 [33.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-jmespath all 1.0.1-1 [21.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-kerberos amd64 1.1.14-3.1build9 [21.2 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-lockfile all 1:0.12.2-3 [13.7 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-simplejson amd64 3.19.2-1build2 [54.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-libcloud all 3.4.1-5 [751 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-ntlm-auth all 1.5.0-1 [21.3 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-requests-ntlm all 1.1.0-3 [6308 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-selinux amd64 3.5-2ubuntu2.1 [165 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-xmltodict all 0.13.0-1ubuntu0.24.04.1 [14.3 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-winrm all 0.4.3-2 [31.9 kB]
```

Step-2:- Verify Installation

```
ansible --version
```

```
harshit@LAPTOP-38KOG53E: ~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/harshit/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/harshit/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar 3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
harshit@LAPTOP-38KOG53E: ~$
```

Step-3:- Ping Localhost

```
ansible localhost -m ping
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ ansible localhost -m ping
[WARNING]: No inventory was parsed, only implicit localhost is available
localhost | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |
```

Step-4:- Generate ssh Keys

```
ssh-keygen -t rsa -b 4096
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/harshit/.ssh/id_rsa):
Created directory '/home/harshit/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/harshit/.ssh/id_rsa
Your public key has been saved in /home/harshit/.ssh/id_rsa.pub
The key's fingerprint is:
SHA256:hKCvLTm/cm021XZsJRt1RNn099ZI8Lll8uM1XJGRzdc harshit@LAPTOP-38KOG53E
The key's randomart image is:
+---[RSA 4096]-----+
|      .      B%|
|      . . .   o =E|
|      . . .   =.O|
|      . o .   .+B*|
|      * S .   .oO*|
|      o o .   o *.+|
|      . . . . . O .|
|      . . . =   |
|      o.o .   |
+---[SHA256]-----+
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |
```

Step-5:- Copy keys in id_rsa

```
#Copy in current Directory
cp ~/.ssh/id_rsa.pub .
cp ~/.ssh/id_rsa .
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ cp ~/.ssh/id_rsa.pub .
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ cp ~/.ssh/id_rsa .
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |
```

Step-6:- Create Dockerfile

FROM ubuntu

RUN apt update -y

RUN apt install -y python3 python3-pip openssh-server

RUN mkdir -p /var/run/ssh

Configure SSH

RUN mkdir -p /run/ssh && \

echo 'root:password' | chpasswd && \

sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config

sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config

sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

Create .ssh directory and set proper permissions

RUN mkdir -p /root/.ssh && \

chmod 700 /root/.ssh

Copy SSH keys (note: this is not secure for production!)

COPY id_rsa /root/.ssh/id_rsa

COPY id_rsa.pub /root/.ssh/authorized_keys

Set proper permissions for keys

RUN chmod 600 /root/.ssh/id_rsa && \

chmod 644 /root/.ssh/authorized_keys

Fix for SSH Login

RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/sshd

Expose SSH port

EXPOSE 22

Start SSH service when container starts

CMD ["/usr/sbin/sshd", "-D"]

```

harshit@LAPTOP-38KOG53E: ~$ nano Dockerfile
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ cat Dockerfile
FROM ubuntu

RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/ssh

# Configure SSH
RUN mkdir -p /run/ssh && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

# Create .ssh directory and set proper permissions
RUN mkdir -p /root/.ssh && \
    chmod 700 /root/.ssh

# Copy SSH keys (note: this is not secure for production!)
COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

# Set proper permissions for keys
RUN chmod 600 /root/.ssh/id_rsa && \
    chmod 644 /root/.ssh/authorized_keys

# Fix for SSH login
RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/ssh

# Expose SSH port
EXPOSE 22

# Start SSH service when container starts
CMD ["/usr/sbin/sshd", "-D"]
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |

```

Step-7:- Build Image

```
docker build -t ubuntu-server
```

```

harshit@LAPTOP-38KOG53E: ~$ cp ~/.ssh/id_rsa.pub .
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ docker build -t ubuntu-server .
[+] Building 98.5s (15/15) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> [internal] load metadata for docker.io/library/ubuntu:latest                  0.1s
=> [internal] load .dockerignore                                                  0.0s
=> [internal] load build context                                                  0.0s
=> [1/10] FROM docker.io/library/ubuntu:latest@sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b 16.4s
=> resolve docker.io/library/ubuntu:latest@sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b 0.0s
=> [internal] load build context                                                  16.4s
=> [2/10] RUN apt update -y                                                       0.0s
=> [3/10] RUN apt install -y python3 python3-pip openssh-server                  58.5s
=> [4/10] RUN mkdir -p /var/run/ssh &&                                            0.4s
=> [5/10] RUN mkdir -p /run/ssh && echo 'root:password' | chpasswd && sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' / 0.4s
=> [6/10] RUN mkdir -p /root/.ssh && chmod 700 /root/.ssh                       0.2s
=> [7/10] COPY id_rsa /root/.ssh/id_rsa                                          0.0s
=> [8/10] COPY id_rsa.pub /root/.ssh/authorized_keys                            0.0s
=> [9/10] RUN chmod 600 /root/.ssh/id_rsa && chmod 644 /root/.ssh/authorized_keys 0.3s
=> [10/10] RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/ssh 0.2s
=> exporting to image                                                            27.5s
=> exporting layers                                                              23.5s
=> exporting manifest sha256:19ccb5f0922ce38fd9beb1849cd152734fc0c32093d3108337306a131a0eb81f 0.0s
=> exporting config sha256:4c5abb054051b896df570c54ecec9ff679e823b8ee73ddbe6e236e036550d0e 0.0s
=> exporting attestation manifest sha256:c48cdd775851c4b19e2a7d26481fb77d4c2cd205b88c4896d039fbc0c2f8bf8b 0.0s
=> exporting manifest list sha256:ba214745435d04472f5bf8c6abb0d8257e88cef4db2f4a2b32f45e18b5db7ac3 0.0s
=> naming to docker.io/library/ubuntu-server:latest                            0.0s
=> unpacking to docker.io/library/ubuntu-server:latest                         3.9s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |

```

Step-8:- Run container from build Image

```
docker run -d -p --rm 2222:22 -p 8221:8221 --name ssh-test-server ubuntu-server
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ docker run -d --rm -p 2222:22 -p 8221:8221 --name ssh-test-server ubuntu-server
19da1e0163866f5e5d06a6843e452bb420354d7a4388cafd1b8ad66974c041d
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |
```

Step-9:- Find Container IP Address

```
docker inspect -f '{{.NetworkSettings.Networks.{{.IPaddress}}}}' ssh-test-server
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' ssh-test-server
172.17.0.2
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |
```

Step-10:- Test SSH connection via Container IP

```
# with key
ssh -i ~/.ssh/id_rsa root@172.17.0.2
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQADmGLWaawKkkBN1395dh3GJQVKG5XbLR0Pm7wDsuXhadUwbPkoJeygWqhRZvUuAJPs0yISLvn9vJIehSGzUkqLk14fi0/5dRj9e5tL1oAWxRikVdLEzU1Ft
hZuKxt0yb0/a6jd9MMJV9MDLeGBxQJmusEx9iFmQYAP6U8YFP+0oG0o0GhB9dPZSUJ4c9o6Ww42K2WuX0yzEjVhDbcJiPjyuGjNH/p0aBLvN/QXbsNTvHVR3ZLiQ8twabqmbPvwunB8F9fxkqvF24lumI6rV
9mQRezU00ZyyZ3f3X6CFxgVYLOYX5qclYAfgeToGf/k08eYLZLSgaIuy61JsdhGd+3kA0YroZM9IssCu7IOe9YSy1zNzXXnXiR82dUEPimjVAswsMR2KqRZ6f07ik7vkkY5tci8wTN+KnBMPoVr1LySbt+A
rPNlygDcH2EaQJhzkyZE94XJjA0reMKsCiQy/FFU7t8EUYCYCjd6gm8v0ohujZ3hRak8g/USEUf+AoxY23wMoy3ma+9es2bkn8Bh+dcLQAwUeN2EqiDuI9e10eoB1w2KagDzsx/09o9QL/bL2lnWeeajSmNE
Crvs10aA1cNyhPLZDf2QSmVn7r+fZOR0UEWL683lv/NxBvJqL7LTrGpAL56+IvCyS3jeJW4r+yVnB8AoUqx1nS93UnHT1n0DDQ==
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ docker exec -it ssh-test-server bash
root@19da1e016386:/# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQADmGLWaawKkkBN1395dh3GJQVKG5XbLR0Pm7wDsuXhadUwbPkoJeygWqhRZvUuAJPs0yISLvn9vJIehSGzUkqLk14fi0
/5dRj9e5tL1oAWxRikVdLEzU1FtZuKxt0yb0/a6jd9MMJV9MDLeGBxQJmusEx9iFmQYAP6U8YFP+0oG0o0GhB9dPZSUJ4c9o6Ww42K2WuX0yzEjVhDbcJiPjyuGjNH/p0aBLvN/QXbsNTvHVR3ZLiQ8twab
qmbPvwunB8F9fxkqvF24lumI6rV9mQRezU00ZyyZ3f3X6CFxgVYLOYX5qclYAfgeToGf/k08eYLZLSgaIuy61JsdhGd+3kA0YroZM9IssCu7IOe9YSy1zNzXXnXiR82dUEPimjVAswsMR2KqRZ6f07ik7vkk
Y5tci8wTN+KnBMPoVr1LySbt+A rPNlygDcH2EaQJhzkyZE94XJjA0reMKsCiQy/FFU7t8EUYCYCjd6gm8v0ohujZ3hRak8g/USEUf+AoxY23wMoy3ma+9es2bkn8Bh+dcLQAwUeN2EqiDuI9e10eoB1w2Ka
gDzsx/09o9QL/bL2lnWeeajSmNECrvs10aA1cNyhPLZDf2QSmVn7r+fZOR0UEWL683lv/NxBvJqL7LTrGpAL56+IvCyS3jeJW4r+yVnB8AoUqx1nS93UnHT1n0DDQ==
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ ssh -i ~/.ssh/id_rsa root@172.17.0.2
```

Step-11:- Start Multiple Containers

```
for i in {1..4}; do
    echo -e "\n Creating server${i}\n"
    docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
    echo -e "IP of server${i} is $(docker inspect -f '{{.NetworkSettings.Networks.{{.IPaddress}}}}' server${i})"
done
```

```

harshit@LAPTOP-38KOG53E: ~$ for i in {1..4}; do
echo -e "\n Creating server${i}\n"
docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
echo -e "IP of server${i} is $(docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i})"
done

Creating server1

b85fa008c5248ef5da95af5e9048ede7b12de1f887aa6e365f82bf0dc4cbb075
IP of server1 is 172.17.0.2

Creating server2

be65bd8dba66fd62f5f9a354731779f909f76bc526dbea62ffab9120a6a386a5
IP of server2 is 172.17.0.3

Creating server3

9d1f22b7d460d85fc99357d9cd2a4f823fcf85b799a0eb37293e58f6886a9008
IP of server3 is 172.17.0.4

Creating server4

1e6d13181333170a99131c044711c43de05c614512dabfdda8179b3240f8bd4c
IP of server4 is 172.17.0.5
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |

```

Step-12:- Create inventory.ini

```

# Get container IPs
echo "[servers]" > inventory.ini
for i in {1..4}; do
    docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini
done

# Add inventory variables
cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF

```

```

harshit@LAPTOP-38KOG53E: ~$ # Get container IPs
echo "[servers]" > inventory.ini
for i in {1..4}; do
    docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini
done

# Add inventory variables
cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |

```

Step-13:- Verify inventory.ini

```
cat inventory.ini
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ cat inventory.ini
[servers]
172.17.0.2
172.17.0.3
172.17.0.4
172.17.0.5

[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |
```

Step-14:- Test Connectivity

```
ssh -i ~/.ssh/id_rsa root@172.17.0.3
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ssh -i ~/.ssh/id_rsa root@172.17.0.3
The authenticity of host '172.17.0.3 (172.17.0.3)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIk0K0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.3' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@00560ebb3alc:~# exit
logout
Connection to 172.17.0.3 closed.
```

```
**
```

Step-15:- Ansible Ping Test

```
ansible all -i inventory.ini -m ping
```

```

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible all -i inventory.ini -m ping
The authenticity of host '172.17.0.5 (172.17.0.5)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIk0K0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
The authenticity of host '172.17.0.6 (172.17.0.6)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIk0K0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
The authenticity of host '172.17.0.4 (172.17.0.4)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIk0K0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? 172.17.0.3 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
yes
172.17.0.5 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}

172.17.0.6 | UNREACHABLE! => {
  "changed": false,
  "msg": "Failed to connect to the host via ssh: Host key verification failed.",
  "unreachable": true
}
yes
172.17.0.4 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#

```

Step-16:- Create Playbook `playbook1.yml`

```

--- # it should start with three dash only
- name: Update and configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update apt packages
      apt:
        update_cache: yes
        upgrade: dist

    - name: Install required packages
      apt:
        name: ["vim", "htop", "wget"]
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on "

```

```

harshit@LAPTOP-38KOG53E: ~$ nano playbook1.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ cat playbook1.yml
--- # it should start with three dash only
- name: Update and configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update apt packages
      apt:
        update_cache: yes
        upgrade: dist

    - name: Install required packages
      apt:
        name: ["vim", "htop", "wget"]
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on {{ inventory_hostname }}"
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ |

```

Step-17:- Run Playbook

```
ansible-playbook -i inventory.ini playbook1.yml
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible-playbook -i inventory.ini playbook1.yml
PLAY [Update and configure servers] *****
TASK [Gathering Facts] *****
The authenticity of host '172.17.0.6 (172.17.0.6)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIkOK0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
 ~/.ssh/known_hosts:4: [hashed name]
 ~/.ssh/known_hosts:7: [hashed name]
 ~/.ssh/known_hosts:8: [hashed name]
 ~/.ssh/known_hosts:9: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? ok: [172.17.0.4]
ok: [172.17.0.3]
ok: [172.17.0.5]
yes
ok: [172.17.0.6]

TASK [Update apt packages] *****
changed: [172.17.0.4]
changed: [172.17.0.6]
changed: [172.17.0.5]
changed: [172.17.0.3]

TASK [Install required packages] *****
changed: [172.17.0.6]
changed: [172.17.0.5]
changed: [172.17.0.3]
changed: [172.17.0.4]

TASK [Create test file] *****
changed: [172.17.0.5]
changed: [172.17.0.3]
changed: [172.17.0.6]
changed: [172.17.0.4]

PLAY RECAP *****
172.17.0.3 : ok=4 changed=3 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
172.17.0.4 : ok=4 changed=3 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
172.17.0.5 : ok=4 changed=3 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
172.17.0.6 : ok=4 changed=3 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

```

Step-18:- Verify Changes

```
ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
172.17.0.6 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.6
172.17.0.4 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.4
172.17.0.5 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.5
172.17.0.3 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.3
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#
```

Step-19:- CleanUp

```
for i in {1..4}; do docker rm -f server${i}; done
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$ for i in {1..4}; do docker rm -f server${i}; done
server1
server2
server3
server4
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-9$
```

[< Previous](#) [Home](#) [Next >](#)

Harshit-Kumar-Containerization

Experiment 10: Working with SonarQube

Introduction

What is SonarQube?

SonarQube is an open-source platform for continuous inspection of code quality. It performs automatic reviews with static analysis to detect bugs, code smells &/ security vulnerabilities.

How SonarQube Solves the Problem:

- **Continuous Inspection:** Scans code with every commit, providing immediate feedback.
- **Quality Gates:** Defines pass/fail criteria for code quality.
- **Technical Debt Quantification:** Measures effort needed to fix issues.
- **Multi-language Support:** Supports 20+ programming languages.
- **Visual Analytics:** Dashboard showing code quality metrics and trends.

Hands-On

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
```

```
volumes:  
  - sonar-db-data:/var/lib/postgresql/data  
networks:  
  - sonarqube-lab
```

```
sonarqube:  
  image: sonarqube:lts-community  
  container_name: sonarqube  
  ports:  
    - "9000:9000"  
  environment:  
    SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube  
    SONAR_JDBC_USERNAME: sonar  
    SONAR_JDBC_PASSWORD: sonar  
  volumes:  
    - sonar-data:/opt/sonarqube/data  
    - sonar-extensions:/opt/sonarqube/extensions  
  depends_on:  
    - sonar-db  
  networks:  
    - sonarqube-lab
```

```
volumes:  
  sonar-db-data:  
  sonar-data:  
  sonar-extensions:
```

```
networks:  
  sonarqube-lab:  
    driver: bridge
```

```
harshit@LAPTOP-38KOG53E: ~$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: ~$ cat docker-compose.yml
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    depends_on:
      - sonar-db
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:

networks:
  sonarqube-lab:
```

Step-2:- Start Compose

```
docker-compose up -d
```

```

harshit@LAPTOP-38KOG53E: ~$ docker-compose up -d
WARN[0000] /mnt/d/Containerization/LAB/Experiment-10/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid
potential confusion
[+] up 28/28
[+] up 34/34arqube:lts-community Pulled 144.9s
Image sonarqube:lts-community Pulled 144.9s
  4f4fb700ef54 Pull complete 0.1s
  1df735f481ad Pull complete 61.3s
  5d5a1fad7028 Pull complete 2.4s
  60d98d907669 Pull complete 59.4s
  eb27e3a98da1 Pull complete 136.8s
  e24a8b9e652f Pull complete 60.1s
  f3929ce9ef98 Pull complete 61.2s
  c7ad1fe61e07 Pull complete 1.9s
  ce2bd68be4fc Download complete 0.0s
  600f318704dc Download complete 6.1s
Image postgres:13 Pulled 81.2s
  ae52c8e1a35 Pull complete 2.1s
  d7ecdd7702a Pull complete 67.4s
  8cb000c698bc Pull complete 2.2s
  ba168ff08d42 Pull complete 2.5s
  bad194149a82 Pull complete 73.9s
  80e8d95fd53 Pull complete 2.3s
  5f1cded65589 Pull complete 1.7s
  fcf4d9225910 Pull complete 67.5s
  7d0650bb210f Pull complete 2.0s
  947bdcac0ea3 Pull complete 7.5s
  23a39b1003e9 Pull complete 2.7s
  097f34b018c0 Pull complete 67.8s
  4d2c7a7b3dba Pull complete 67.9s
  21f7a31efcb7 Pull complete 2.0s
  c7bebc4b5ec1 Download complete 0.0s
  d932a4a7df6c Download complete 5.1s
Network experiment-10_sonarqube-lab Created 0.1s
Volume experiment-10_sonar-data Created 0.0s
Volume experiment-10_sonar-extensions Created 0.0s
Volume experiment-10_sonar-db-data Created 0.0s
Container sonar-db Created 0.9s
Container sonarqube Created 0.2s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10$

```

Step-3:- Verify SonarQube is running via Logs

`docker-compose logs -f sonarqube`

```

harshit@LAPTOP-38KOG53E: ~$ docker-compose logs -f sonarqube
WARN[0000] /mnt/d/Containerization/LAB/Experiment-10/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid
potential confusion
sonarqube | 2026.04.18 09:02:46 INFO app[[o.s.a.AppFileSystem] Cleaning or creating temp directory /opt/sonarqube/temp
sonarqube | 2026.04.18 09:02:46 INFO app[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:38617]
sonarqube | 2026.04.18 09:02:46 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [/opt/sonarqube/elasticsearch]: /opt/sonarqube/el
asticsearch/bin/elasticsearch
sonarqube | 2026.04.18 09:02:46 INFO app[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
sonarqube | 2026.04.18 09:02:48 INFO es[[o.e.n.Node] version[7.17.15], pid[27], build[default/tar/0b8ecfb4378335f4689c4223d1f115f16bef3ba/2023-11-10T22:
03:46.987399016Z], OS[Linux/6.6.87.2-microsoft-standard-WSL2/amd64], JVM[Eclipse Adoptium/OpenJDK 64-Bit Server VM/17.0.16/17.0.16+8]
sonarqube | 2026.04.18 09:02:48 INFO es[[o.e.n.Node] JVM home [/opt/java/openjdk]
sonarqube | 2026.04.18 09:02:48 INFO es[[o.e.n.Node] JVM arguments [-XX:+UseG1GC, -Djava.io.tmpdir=/opt/sonarqube/temp, -XX:ErrorFile=/opt/sonarqube/logs
/es_hs_err_pid%p.log, -Des.networkaddress.cache.ttl=60, -Des.networkaddress.cache.negative.ttl=10, -XX:+AlwaysPreTouch, -Xss1m, -Djava.awt.headless=true, -D
file.encoding=UTF-8, -Djna.nosys=true, -Djna.tmpdir=/opt/sonarqube/temp, -XX:-OmitStackTraceInFastThrow, -Dio.netty.noUnsafe=true, -Dio.netty.noKeySetOptimi
zation=true, -Dio.netty.recycler.maxCapacityPerThread=0, -Dio.netty.allocator.numDirectArenas=0, -Dlog4j.shutdownHookEnabled=false, -Dlog4j2.disable.jmx=true,
-Dlog4j2.formatMsgNoLookups=true, -Djava.locale.providers=COMPAT, -Dcom.redhat.fips=false, -Des.enforce.bootstrap.checks=true, -Xmx512m, -Xms512m, -XX:Ma
xDirectMemorySize=256m, -XX:+HeapDumpOnOutOfMemoryError, -Des.path.home=/opt/sonarqube/elasticsearch, -Des.path.conf=/opt/sonarqube/temp/conf/es, -Des.distr
ibution.flavor=default, -Des.distribution.type=tar, -Des.bundled.jdk=false]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.p.PluginsService] loaded module [analysis-common]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.p.PluginsService] loaded module [lang-painless]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.p.PluginsService] loaded module [parent-join]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.p.PluginsService] loaded module [reindex]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.p.PluginsService] loaded module [transport-netty4]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.p.PluginsService] no plugins loaded
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.e.NodeEnvironment] using [1] data paths, mounts [[/opt/sonarqube/data (/dev/sde)], net usable_space [944.3g
b], net total_space [1006.8gb], types [ext4]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.e.NodeEnvironment] heap size [512mb], compressed ordinary object pointers [true]
sonarqube | 2026.04.18 09:02:49 INFO es[[o.e.n.Node] node name [sonarqube], node ID [KjH14TexSsiIHFgSxhw83g], cluster name [sonarqube], roles [data_froze
n, master, remote_cluster_client, data, data_content, data_hot, data_warm, data_cold, ingest]
sonarqube | 2026.04.18 09:03:09 WARN es[[o.e.t.ThreadPool] timer thread slept for [16.7s/16792ms] on absolute clock which is above the warn threshold of
[5000ms]
sonarqube | 2026.04.18 09:03:09 INFO es[[o.e.t.NettyAllocator] creating NettyAllocator with the following configs: [name=unpooled, suggested_max_allocati
on_size=256kb, factors={es.unsafe.use_unpooled_allocator=null, g1gc_enabled=true, g1gc_region_size=1mb, heap_size=512mb}]
sonarqube | 2026.04.18 09:03:09 INFO es[[o.e.i.r.RecoverySettings] using rate limit [40mb] with [default=40mb, read=0b, write=0b, max=0b]
sonarqube | 2026.04.18 09:02:52 INFO es[[o.e.d.DiscoveryModule] using discovery type [zen] and seed hosts providers [settings]
sonarqube | 2026.04.18 09:02:53 WARN es[[o.e.t.ThreadPool] absolute clock went backwards by [16.3s/16375ms] while timer thread was sleeping
sonarqube | 2026.04.18 09:02:53 INFO es[[o.e.g.DanglingIndicesState] gateway.auto_import_dangling_indices is disabled, dangling indices will not be autom
atically detected or imported and must be managed manually
sonarqube | 2026.04.18 09:02:53 INFO es[[o.e.n.Node] initialized
sonarqube | 2026.04.18 09:02:53 INFO es[[o.e.n.Node] starting ...
sonarqube | 2026.04.18 09:02:53 INFO es[[o.e.t.TransportService] publish_address [127.0.0.1:38617], bound_addresses [127.0.0.1:38617]

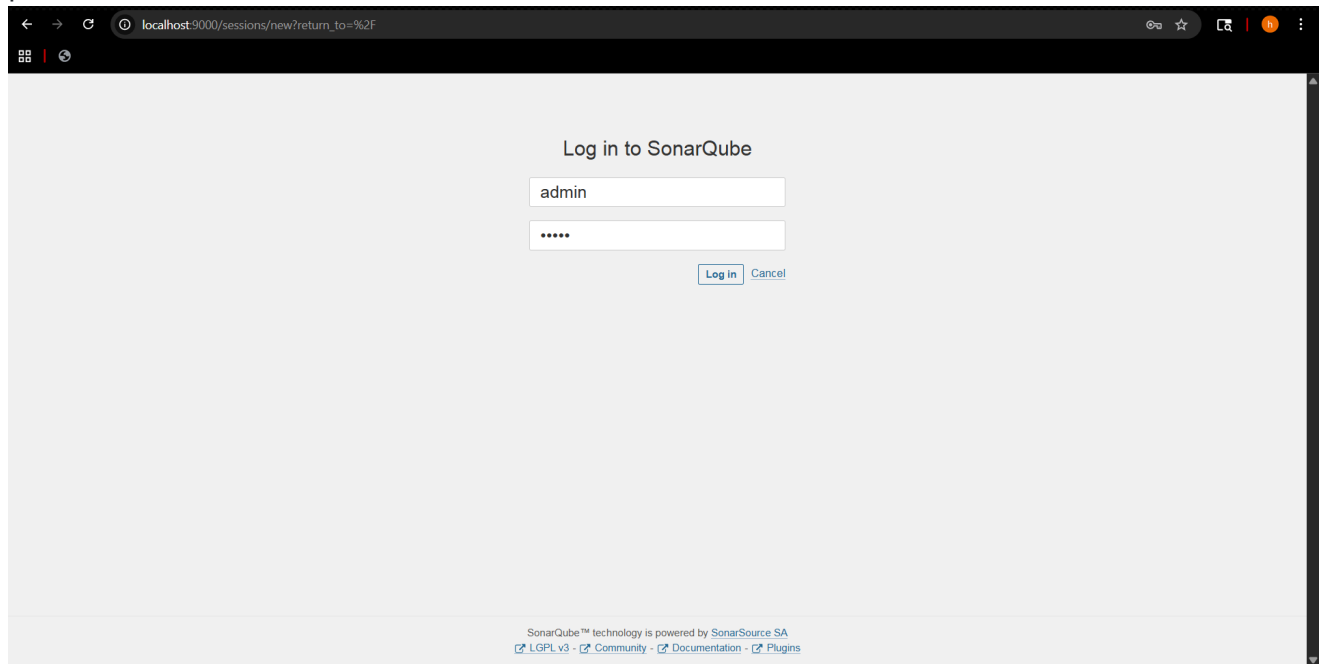
```

Step-4:- Login SonarQube

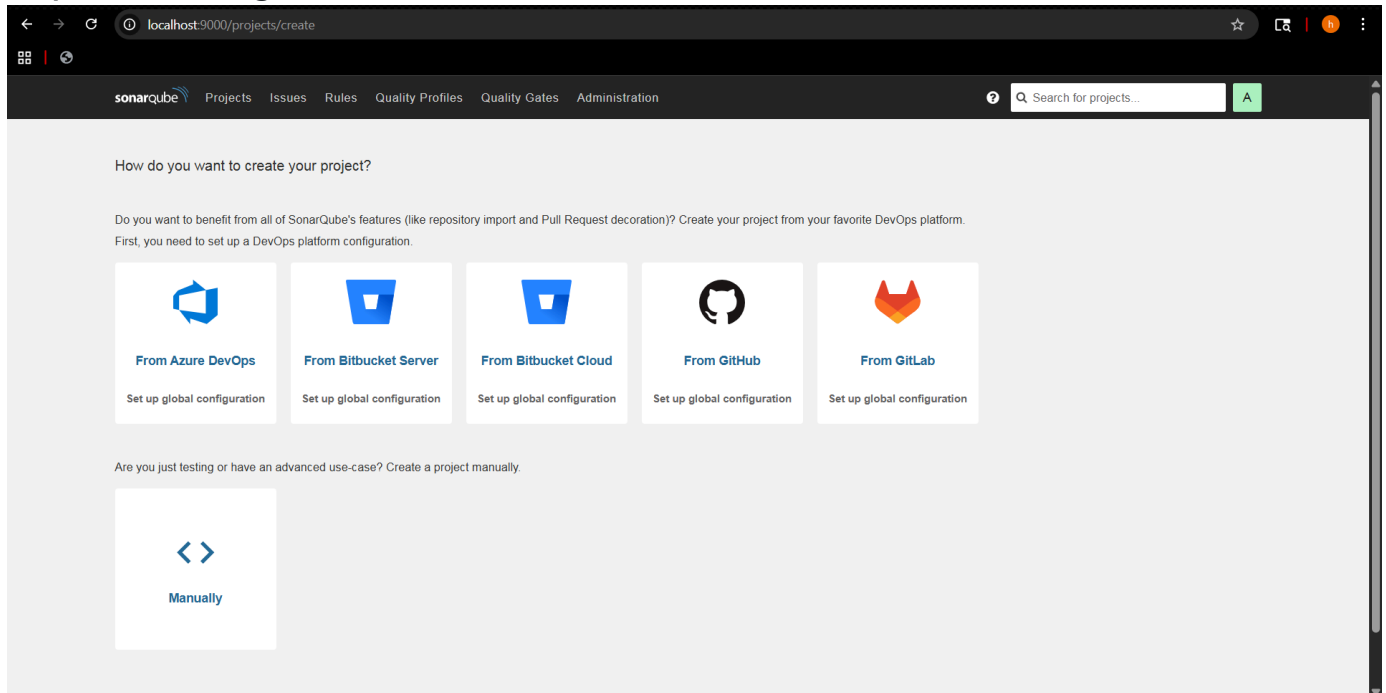
`http://localhost:9000`

- username: admin

- password: admin

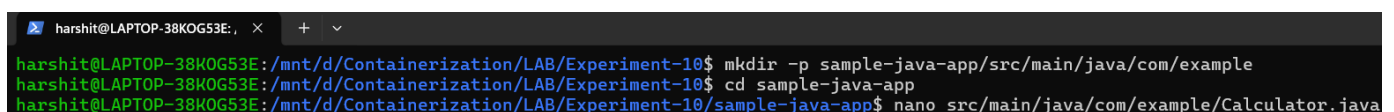


Step-5:- HomePage



Step-6:- Create Sample Java App

```
mkdir -p sample-java-app/src/main/java/com/example
cd sample-java-app
```



Step-7:- Create calculator.java

```
nano src/main/java/com/example/Calculator.java
```

```
package com.example;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }

    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public String getName(String name) {
        return name.toUpperCase();
    }

    public void riskyOperation() {
        try {
            int x = 10 / 0;
        } catch (Exception e) {
        }
    }
}
```

```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10/sample-java-app$ cat src/main/java/com/example/Calculator.java
package com.example;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }

    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public String getName(String name) {
        return name.toUpperCase();
    }

    public void riskyOperation() {

```

Step-8:- Create pom.xml

nano pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>sample-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <sonar.projectKey>sample-java-app</sonar.projectKey>
        <sonar.host.url>http://localhost:9000</sonar.host.url>
        <sonar.login>YOUR_TOKEN</sonar.login>
    </properties>

    <build>
        <plugins>
            <plugin>
                <groupId>org.sonarsource.scanner.maven</groupId>
                <artifactId>sonar-maven-plugin</artifactId>
                <version>3.9.1.2184</version>
            </plugin>
        </plugins>
    </build>

```

```
</build>
</project>
```

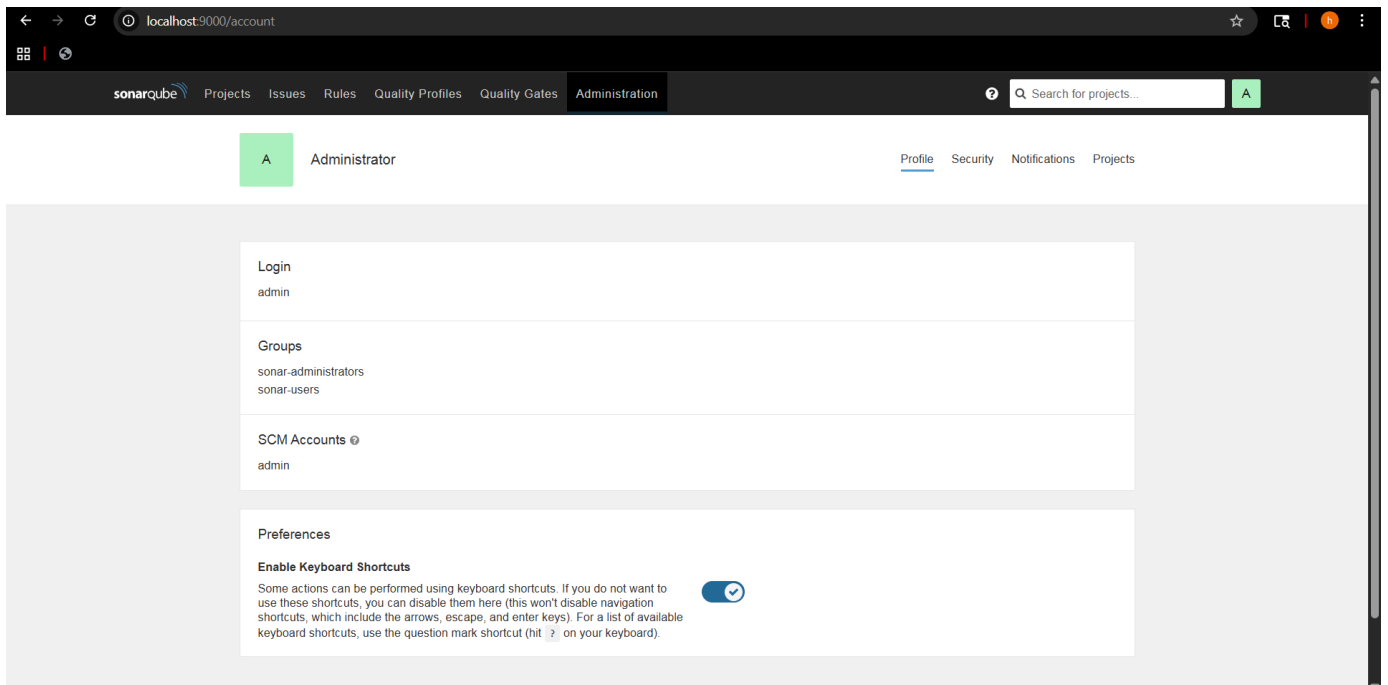
```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10/sample-java-app$ nano pom.xml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10/sample-java-app$ cat pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>sample-app</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <sonar.projectKey>sample-java-app</sonar.projectKey>
    <sonar.host.url>http://localhost:9000</sonar.host.url>
    <sonar.login>YOUR_TOKEN</sonar.login>
  </properties>

  <build>
    <plugins>
      <plugin>
        <groupId>org.sonarsource.scanner.maven</groupId>
        <artifactId>sonar-maven-plugin</artifactId>
        <version>3.9.1.2184</version>
      </plugin>
    </plugins>
  </build>
</project>
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10/sample-java-app$ |
```

Step-9:- Create Tokens Profile -> My Account -> Security -> TokenName: scanner-token -> Generate



(./Images/10ng) (./Images/11ng)

Step-10:- Run Scanner

```
mvn sonar:sonar -Dsonar.login=YOUR_TOKEN
```



```

harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10/sample-java-app$ mvn sonar:sonar -Dsonar.login=sqa_e8ed8317bb7ef57840053bee707569e29dd3942a
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/3.9.1.2184/sonar-maven-plugin-3.9.1.2184.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/3.9.1.2184/sonar-maven-plugin-3.9.1.2184.pom (12 kB at 7.5 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/parent/parent/59.0.29/parent-59.0.29.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonarsource/parent/parent/59.0.29/parent-59.0.29.pom (32 kB at 135 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/3.9.1.2184/sonar-maven-plugin-3.9.1.2184.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/3.9.1.2184/sonar-maven-plugin-3.9.1.2184.jar (50 kB at 404 kB/s)
[INFO]
[INFO] -----< com.example:sample-app >-----
[INFO] Building sample-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- sonar:3.9.1.2184:sonar (default-cli) @ sample-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-sec-dispatcher/1.4/plexus-sec-dispatcher-1.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-sec-dispatcher/1.4/plexus-sec-dispatcher-1.4.pom (3.0 kB at 31 kB/s)
)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/spice/spice-parent/12/spice-parent-12.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/spice/spice-parent/12/spice-parent-12.pom (6.8 kB at 86 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/forge/forge-parent/4/forge-parent-4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/forge/forge-parent/4/forge-parent-4.pom (8.4 kB at 90 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/1.5.5/plexus-utils-1.5.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/1.5.5/plexus-utils-1.5.5.pom (5.1 kB at 47 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.11/plexus-1.0.11.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.11/plexus-1.0.11.pom (9.0 kB at 47 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-cipher/1.4/plexus-cipher-1.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-cipher/1.4/plexus-cipher-1.4.pom (2.1 kB at 122 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.2.1/plexus-utils-3.2.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.2.1/plexus-utils-3.2.1.pom (5.3 kB at 36 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/5.1/plexus-5.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/5.1/plexus-5.1.pom (23 kB)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/api/sonar-scanner-api/2.16.2.588/sonar-scanner-api-2.16.2.588.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/api/sonar-scanner-api/2.16.2.588/sonar-scanner-api-2.16.2.588.pom (5.2 kB at 33 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/api/sonar-scanner-api-parent/2.16.2.588/sonar-scanner-api-parent-2.16.2.588.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/api/sonar-scanner-api-parent/2.16.2.588/sonar-scanner-api-parent-2.16.2.588.pom

```

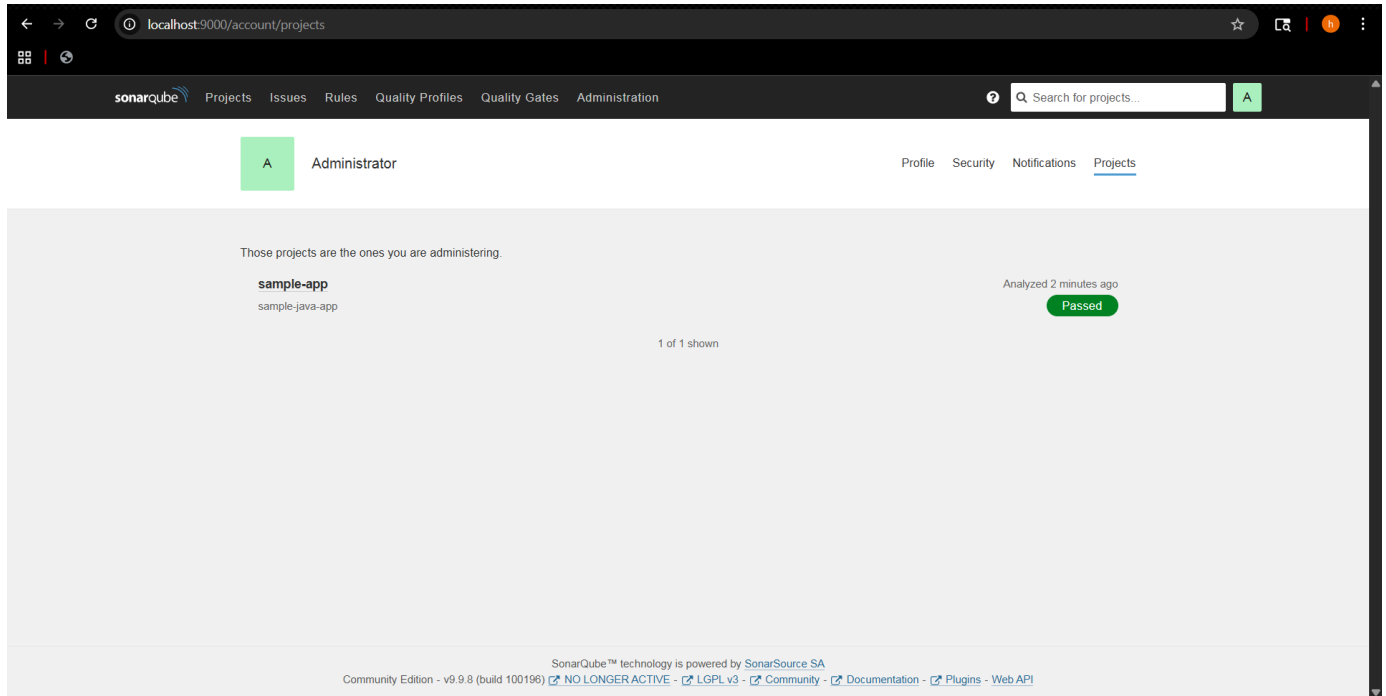
Step-11:- Scroll down to view Build Success

```

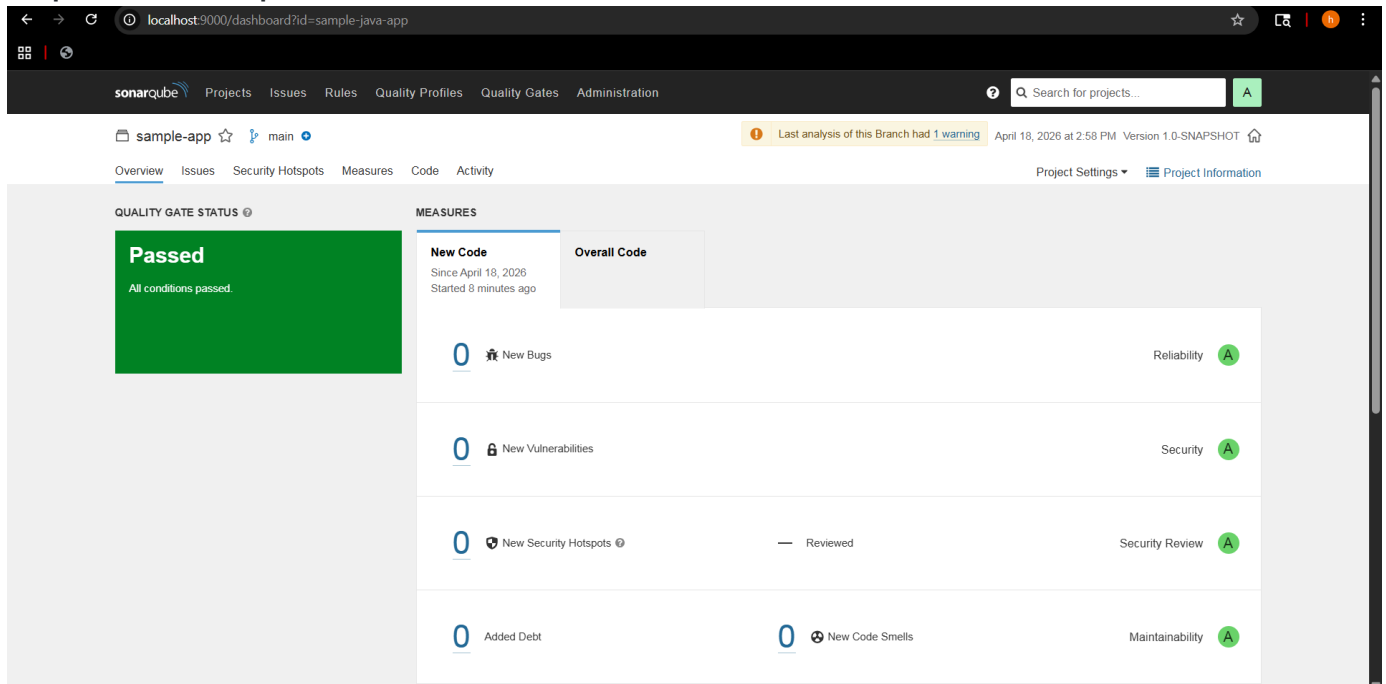
[INFO] Sensor VB.NET Project Type Information [vbnet] (done) | time=1ms
[INFO] Sensor VB.NET Analysis Log [vbnet]
[INFO] Sensor VB.NET Analysis Log [vbnet] (done) | time=35ms
[INFO] Sensor VB.NET Properties [vbnet]
[INFO] Sensor VB.NET Properties [vbnet] (done) | time=1ms
[INFO] Sensor IaC Docker Sensor [iac]
[INFO] 0 source files to be analyzed
[INFO] 0/0 source files have been analyzed
[INFO] Sensor IaC Docker Sensor [iac] (done) | time=90ms
[INFO] ----- Run sensors on project
[INFO] Sensor Analysis Warnings import [csharp]
[INFO] Sensor Analysis Warnings import [csharp] (done) | time=0ms
[INFO] Sensor Zero Coverage Sensor
[INFO] Sensor Zero Coverage Sensor (done) | time=14ms
[INFO] Sensor Java CPD Block Indexer
[INFO] Sensor Java CPD Block Indexer (done) | time=48ms
[INFO] SCM Publisher SCM provider for this project is: git
[INFO] SCM Publisher 2 source files to be analyzed
[WARNING] Thread[JGit-FileStoreAttributeReader-1,5,main]: got smaller file timestamp on /mnt/d (D:\), /mnt/d/Containerization/.git: 2026-04-18T09:22:35Z < 2026-04-18T09:22:35.417547Z. Aborting measurement at resolution PT0.582453S.
[INFO] SCM Publisher 0/2 source files have been analyzed (done) | time=2204ms
[WARNING] Missing blame information for the following files:
[WARNING] * src/main/java/com/example/Calculator.java
[WARNING] * pom.xml
[WARNING] This may lead to missing/broken features in SonarQube
[INFO] CPD Executor Calculating CPD for 1 file
[INFO] CPD Executor CPD calculation finished (done) | time=15ms
[INFO] Analysis report generated in 222ms, dir size=127.7 kB
[INFO] Analysis report compressed in 694ms, zip size=19.1 kB
[INFO] Analysis report uploaded in 354ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=sample-java-app
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://localhost:9000/api/ce/task?id=AZ2f5foGT-VwXPQldLYs
[INFO] Analysis total time: 11.033 s
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 23.127 s
[INFO] Finished at: 2026-04-18T09:22:22Z
[INFO]
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-10/sample-java-app$ |

```

Step-12:- On Browser test will be Listed



Step-13:- View Report



(./Images/18.png)

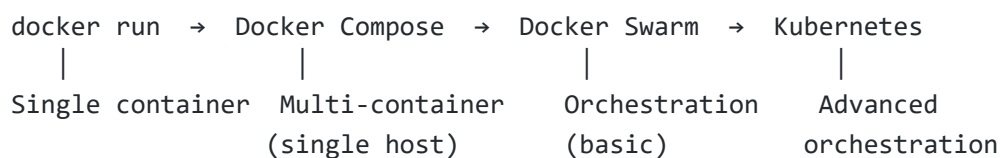
[← Previous](#) [Home](#) [Next →](#)

Harshit-Kumar-Containerization

Experiment 11: Orchestration via Docker Swarm

Introduction

The Progression Path



This experiment focuses on: Moving from Compose → Swarm

Hands-On

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

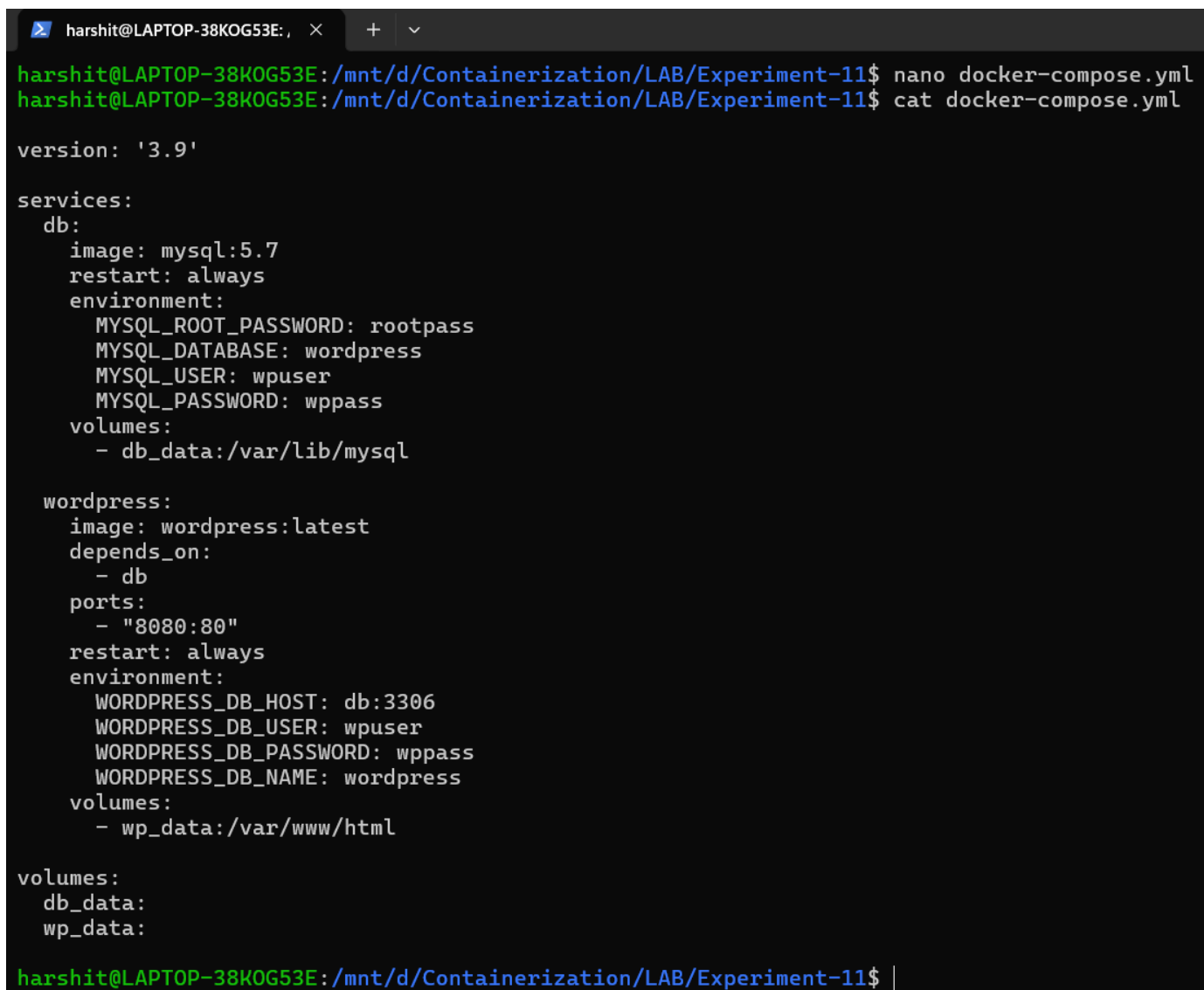
Paste This:

```
version: '3.9'

services:
  db:
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql
```

```
wordpress:
  image: wordpress:latest
  depends_on:
    - db
  ports:
    - "8080:80"
  restart: always
  environment:
    WORDPRESS_DB_HOST: db:3306
    WORDPRESS_DB_USER: wpuser
    WORDPRESS_DB_PASSWORD: wppass
    WORDPRESS_DB_NAME: wordpress
  volumes:
    - wp_data:/var/www/html
```

```
volumes:
  db_data:
  wp_data:
```



The screenshot shows a terminal window with the following content:

```
harshit@LAPTOP-38KOG53E: ~$ nano docker-compose.yml
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ cat docker-compose.yml

version: '3.9'

services:
  db:
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

The terminal prompt at the bottom is `harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$`.

Step-2:- Initialize Swarm

```
docker swarm init
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker swarm init
```

Step-3:- Verify Swarm is Active

```
docker node ls
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker node ls
ID                HOSTNAME          STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
wnqinwknjqttz8hhkcj1itb80 *  docker-desktop  Ready    Active           Leader            29.1.3
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-4:- Deploy as a stack

```
docker stack deploy -c docker-compose.yml wpstack
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_wordpress
Creating service wpstack_db
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-5:- Verify by listing Services

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker service ls
ID                NAME                MODE                REPLICAS    IMAGE                PORTS
1yuxq0obh0xt      wpstack_db           replicated          1/1          mysql:5.7
xvz7gtqzujdq      wpstack_wordpress    replicated          1/1          wordpress:latest     *:8080->80/tcp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-6:- Details for container service

```
docker service ps wpstack_wordpress
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker service ps wpstack_wordpress
ID                NAME                IMAGE                NODE                DESIRED STATE    CURRENT STATE    ERROR                PORTS
44o0apz5jdzw      wpstack_wordpress.1  wordpress:latest     docker-desktop      Running           Running 30 seconds ago
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-7:- List Containers

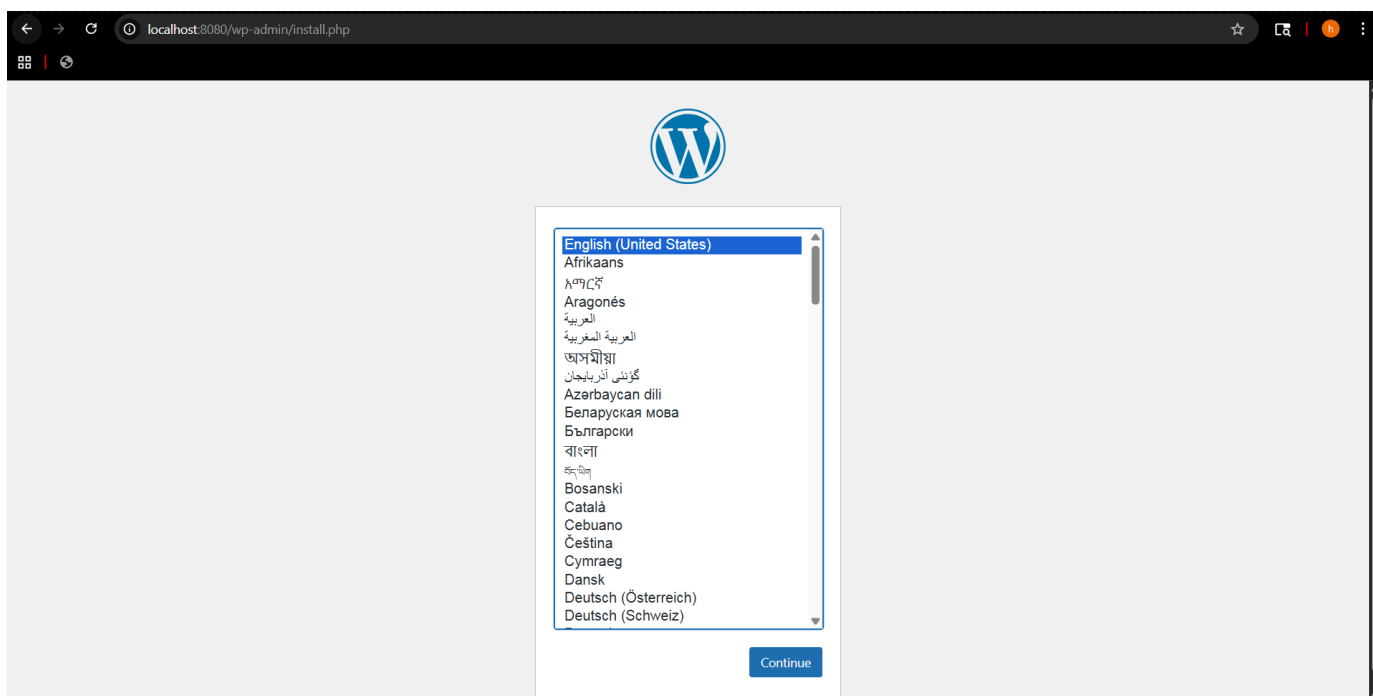
docker ps

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
f76a3939c883	wordpress:latest	"docker-entrypoint.s..."	57 seconds ago	Up 56 seconds	80/tcp
3b239bf78fb8	mysql:5.7	"docker-entrypoint.s..."	About a minute ago	Up About a minute	3306/tcp, 33060/tcp
0324beaa3d2d	wordpress:latest	"docker-entrypoint.s..."	10 days ago	Up 10 minutes	0.0.0.0:8082->80/tcp, [::]:8082->80/
e079aaf749bf	mysql:5.7	"docker-entrypoint.s..."	10 days ago	Up 10 minutes	3306/tcp, 33060/tcp
5576c99e1f84	assignment-1-database	"docker-entrypoint.s..."	3 weeks ago	Up 10 minutes (healthy)	
d8018398f51c	ghcr.io/k3d-io/k3d-proxy:5.8.3	"/bin/sh -c nginx-pr..."	5 weeks ago	Up 10 minutes	0.0.0.0:42325->6443/tcp
5e303da50cdb	mysql:5.7	"docker-entrypoint.s..."	5 weeks ago	Up 10 minutes	

Step-8:- Verify application on Containers

http://localhost:8080/



Step-9:- Scale application Containers

docker service scale wpstack_wordpress=3

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service wpstack_wordpress converged
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$
```

Step-10:- Verify Scalling

```
docker service ls
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker service ls
ID                NAME                MODE                REPLICAS    IMAGE                PORTS
1yuxq0obh0xt      wpstack_db          replicated          1/1          mysql:5.7
xvz7gtqzujdq      wpstack_wordpress    replicated          3/3          wordpress:latest     *:8080->80/tcp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-11:- List Wordpress Containers

```
docker ps | grep wordpress
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker ps | grep wordpress
15294ad9e346      wordpress:latest    "docker-entrypoint.s..."  43 seconds ago    Up 39 seconds        80/tcp
wpstack_wordpress.2.g4fqapsrzntw1hn2eaeqdp2e      wordpress:latest    "docker-entrypoint.s..."  43 seconds ago    Up 39 seconds        80/tcp
643a8ac73631      wordpress:latest    "docker-entrypoint.s..."  43 seconds ago    Up 39 seconds        80/tcp
wpstack_wordpress.3.bl1qjtuaaf240kgh208a165ifc      wordpress:latest    "docker-entrypoint.s..."  2 minutes ago     Up 2 minutes         80/tcp
f76a3939c883      wordpress:latest    "docker-entrypoint.s..."  2 minutes ago     Up 2 minutes         80/tcp
wpstack_wordpress.1.44o0apz5jdzwg2t70k9yaoetv      wordpress:latest    "docker-entrypoint.s..."  10 days ago       Up 11 minutes        0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
0324beaa3d2d      wordpress:latest    "docker-entrypoint.s..."  10 days ago       Up 11 minutes        0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-12:- Kill any Container

```
docker kill <container-id>
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker kill 1529
1529
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-13:- Watch Swarm Recreating Contrainer

```
docker service ps wpstack_wordpress
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker service ps wpstack_wordpress
ID                NAME                IMAGE                NODE                DESIRED STATE    CURRENT STATE    ERROR                PORTS
44o0apz5jdzw      wpstack_wordpress.1    wordpress:latest     docker-desktop      Running           Running 3 minutes ago
39p93zpdo5v3      wpstack_wordpress.2    wordpress:latest     docker-desktop      Running           Running 15 seconds ago
g4fqapsrzntw      \_ wpstack_wordpress.2    wordpress:latest     docker-desktop      Shutdown          Failed 21 seconds ago    "task: non-zero exit (137)"
bl1qjtuaaf240      wpstack_wordpress.3    wordpress:latest     docker-desktop      Running           Running 2 minutes ago
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ |
```

Step-14:- Verify Container running

```
docker ps | grep wordpress
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker ps | grep wordpress
91579f488cca    wordpress:latest    "docker-entrypoint.s..."    37 seconds ago    Up 32 seconds    80/tcp
wpstack_wordpress.2.39p93zpdo5v389ohyocc7yxga    "docker-entrypoint.s..."    2 minutes ago    Up 2 minutes    80/tcp
643a8ac73631    wordpress:latest    "docker-entrypoint.s..."    4 minutes ago    Up 4 minutes    80/tcp
f76a3939c883    wordpress:latest    "docker-entrypoint.s..."    10 days ago    Up 13 minutes    0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
wpstack_wordpress.1.44o0apz5jdzwg2t70k9yaoetv
0324beaa3d2d    wordpress:latest
```

Step-15:- Remove Stack

```
docker stack rm wpstack
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$
```

Step-16:- Verify Service Removal

```
docker service ls
docker ps
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-11$ docker service ps wpstack_wordpress
ID            NAME                IMAGE                NODE                DESIRED STATE    CURRENT STATE    ERROR                PORTS
44o0apz5jdzw  wpstack_wordpress.1  wordpress:latest    docker-desktop     Running          Running 3 minutes ago
39p93zpdo5v3  wpstack_wordpress.2  wordpress:latest    docker-desktop     Running          Running 15 seconds ago
g4fqapsrzntn  \_ wpstack_wordpress.2  wordpress:latest    docker-desktop     Shutdown        Failed 21 seconds ago    "task: non-zero exit (137)"
bllqjtuaaf240 wpstack_wordpress.3  wordpress:latest    docker-desktop     Running          Running 2 minutes ago
```

Conclusion

Summary

You started with	You can now do
Single container (<code>docker run</code>)	Multi-container (Compose)
Manual scaling	One-command scaling (<code>scale</code>)
Manual recovery	Automatic self-healing
Single host	Multi-host cluster ready

Final Takeaway

Compose defines the application. Swarm runs it reliably.

Quick Reference Card

Initialize Swarm

```
docker swarm init
```

Deploy stack

```
docker stack deploy -c docker-compose.yml <stack-name>
```

List services

```
docker service ls
```

Scale service

```
docker service scale <stack-name_service-name>=<replicas>
```

See service tasks

```
docker service ps <service-name>
```

Remove stack

```
docker stack rm <stack-name>
```

Leave Swarm (if needed)

```
docker swarm leave --force
```

[< Previous](#) [Home](#) [Next >](#)

Harshit-Kumar-Containerization

Experiment 12: Kubernetes

Introduction

Why Kubernetes over Docker Swarm?

Reason	Explanation
Industry standard	Most companies use Kubernetes
Powerful scheduling	Automatically decides where to run your app
Large ecosystem	Many tools and plugins available
Cloud-native support	Works on AWS, Google Cloud, Azure, etc.

Core Kubernetes Concepts (Simple Explanation)

Docker Concept	Kubernetes Equivalent	What it means
Container	Pod	A pod is a group of one or more containers. Smallest unit in K8s.
Compose service	Deployment	Describes how your app should run (e.g., 2 copies, which image to use)
Load balancing	Service	Exposes your app to the outside world or other pods
Scaling	ReplicaSet	Ensures a certain number of pod copies are always running

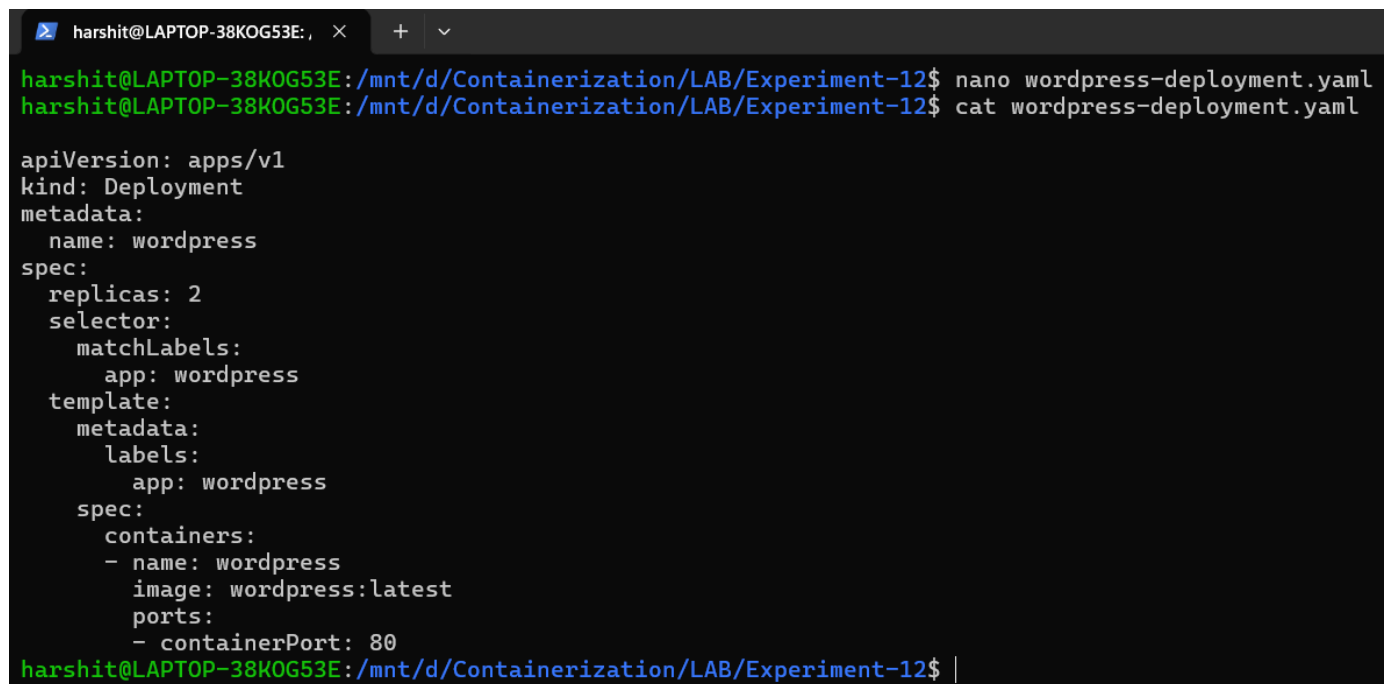
Hands-On

Step-1:- Create a `wordpress-deployment.yaml`

```
nano wordpress-deployment.yaml
```

Paste This:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress
spec:
  replicas: 2
  selector:
    matchLabels:
      app: wordpress
  template:
    metadata:
      labels:
        app: wordpress
    spec:
      containers:
      - name: wordpress
        image: wordpress:latest
        ports:
        - containerPort: 80
```

A terminal window with a dark background. The title bar shows 'harshit@LAPTOP-38KOG53E'. The prompt is 'harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-12\$'. The user enters 'nano wordpress-deployment.yaml' and then 'cat wordpress-deployment.yaml'. The output shows the YAML content of the deployment file, which is identical to the one shown in the previous block. The prompt returns to 'harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-12\$' after the cat command.

```
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-12$ nano wordpress-deployment.yaml
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-12$ cat wordpress-deployment.yaml

apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress
spec:
  replicas: 2
  selector:
    matchLabels:
      app: wordpress
  template:
    metadata:
      labels:
        app: wordpress
    spec:
      containers:
      - name: wordpress
        image: wordpress:latest
        ports:
        - containerPort: 80
harshit@LAPTOP-38KOG53E:/mnt/d/Containerization/LAB/Experiment-12$ |
```

Step-2:- Apply Deployment

```
kubectl apply -f wordpress-deployment.yaml
```

```
harshit@LAPTOP-38KOG53E: ~$ kubectl apply -f wordpress-deployment.yaml
deployment.apps/wordpress created
harshit@LAPTOP-38KOG53E: ~$
```

Step-3:- Create wordpress-service.yaml

```
nano wordpress-service.yaml
```

Paste this:-

```
apiVersion: v1
kind: Service
metadata:
  name: wordpress-service
spec:
  type: NodePort
  selector:
    app: wordpress
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

```
harshit@LAPTOP-38KOG53E: ~$ nano wordpress-service.yaml
harshit@LAPTOP-38KOG53E: ~$ cat wordpress-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: wordpress-service
spec:
  type: NodePort
  selector:
    app: wordpress
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
harshit@LAPTOP-38KOG53E: ~$
```

Step-4:- Deploy Service

```
kubectl apply -f wordpress-service.yaml
```

```
harshit@LAPTOP-38KOG53E: ~$ kubectl apply -f wordpress-service.yaml
service/wordpress-service created
harshit@LAPTOP-38KOG53E: ~$
```

Step-5:- Get pods

```
kubectl get pods
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wordpress-7d6f6db8d8-4zn2n         1/1     Running   0           64s
wordpress-7d6f6db8d8-m5ws7         1/1     Running   0           64s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ |
```

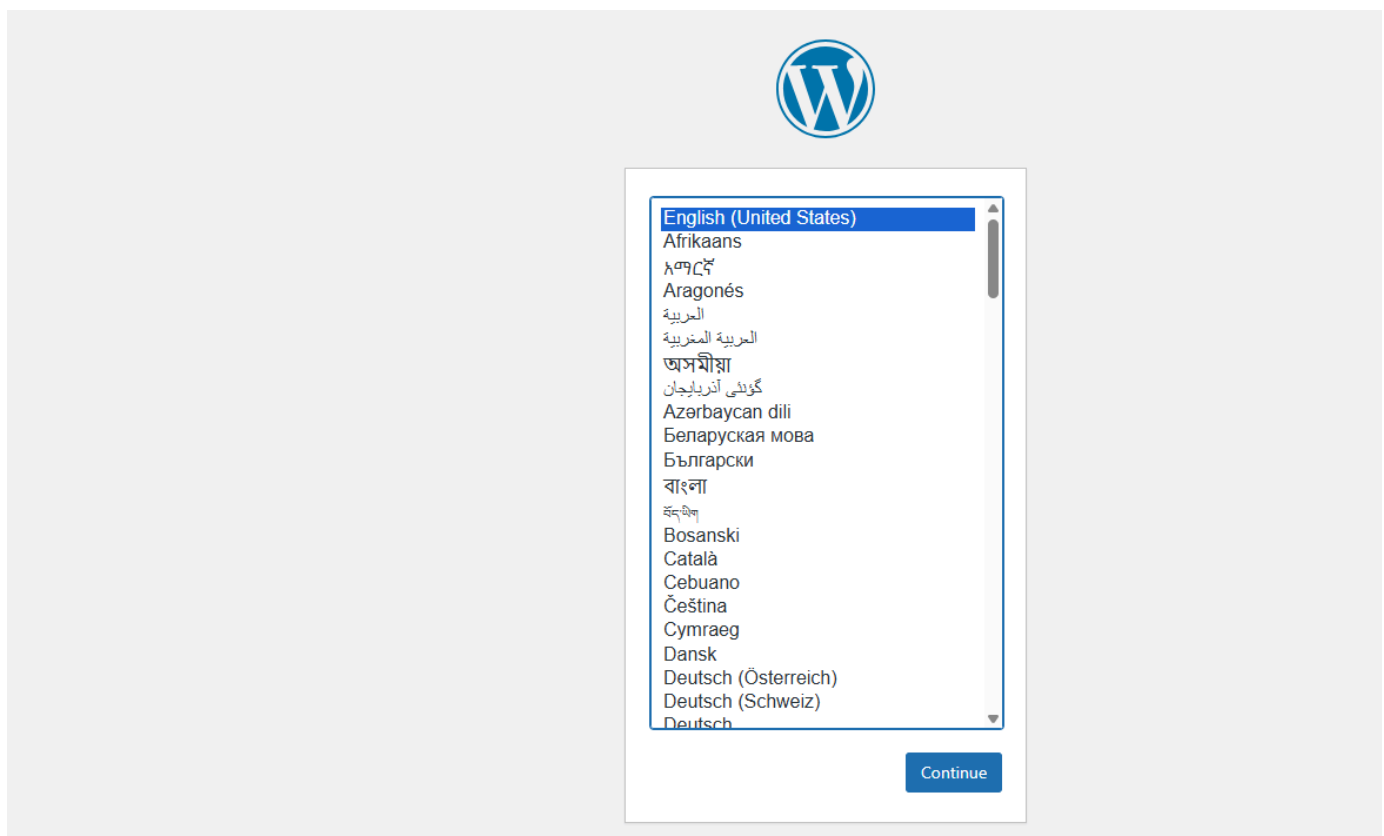
Step-6:- Check service

```
kubectl get svc
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ kubectl get svc
NAME            TYPE          CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes      ClusterIP     10.43.0.1       <none>       443/TCP          42d
web             NodePort      10.43.151.187   <none>       80:30285/TCP     42d
wordpress-service NodePort      10.43.79.203    <none>       80:30007/TCP     30s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ |
```

Step-7:- Verify On Browser

```
http://localhost:8080/
```



Step-8:- Scale Deployment

```
kubectl scale deployment wordpress --replicas=4
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ kubectl scale deployment wordpress --replicas=4
deployment.apps/wordpress scaled
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ |
```

Step-9:- Get Pods

```
kubectl get pods
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wordpress-7d6f6db8d8-2mwp7         1/1     Running   0           19s
wordpress-7d6f6db8d8-4zn2n         1/1     Running   0          3m20s
wordpress-7d6f6db8d8-m5ws7         1/1     Running   0          3m20s
wordpress-7d6f6db8d8-t6t7v         1/1     Running   0           19s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ |
```

Step-10:- Delete Pods to check Self-healing

```
kubectl delete pod <pod-name>
kubectl get pods
```

```
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ kubectl delete pod wordpress-7d6f6db8d8-2mwp7
pod "wordpress-7d6f6db8d8-2mwp7" deleted from default namespace
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wordpress-7d6f6db8d8-28m9w         1/1     Running   0           11s
wordpress-7d6f6db8d8-4zn2n         1/1     Running   0           6m3s
wordpress-7d6f6db8d8-m5ws7         1/1     Running   0           6m3s
wordpress-7d6f6db8d8-t6t7v         1/1     Running   0           3m2s
harshit@LAPTOP-38KOG53E: /mnt/d/Containerization/LAB/Experiment-12$ |
```

Insights

When to Use Which Tool

Tool	Best for
k3d	Quick learning on your laptop
Minikube	Single-node cluster testing

Tool	Best for
kubeadm	Real, production-style cluster

Summary of Commands (Cheat Sheet)

Goal	Command
Apply a YAML file	<code>kubectl apply -f file.yaml</code>
See all pods	<code>kubectl get pods</code>
See all services	<code>kubectl get svc</code>
Scale a deployment	<code>kubectl scale deployment <name> --replicas=N</code>
Delete a pod	<code>kubectl delete pod <pod-name></code>
See all nodes	<code>kubectl get nodes</code>

[← Previous](#) [🏠 Home](#)